

Geometry of Curves and Surfaces

Joe Orii

Contents

1	Curves	1
1.1	Basic Definitions	1
1.2	Curvature	4
1.3	Curves in $\mathbb{R}^3$	6
1.4	Fundamental Theorem of the Local Theory of Curves	10
1.5	Plane Curves	13
2	Surfaces	17
2.1	Definition and Examples	17
2.2	Tangent Vectors and Tangent Planes	22
2.3	Smooth Maps and Differentials	26
2.4	Normal Vectors and the Gauss Map	30
3	The First and Second Fundamental Forms	35
3.1	The Second Fundamental Form	35
3.2	Gaussian Curvature and Mean Curvature	41
3.3	Computing the Gaussian and Mean Curvatures	45
3.4	The First Fundamental Form	49
3.5	Christoffel Symbols	53
3.6	Gauss's Theorema Egregium	56
3.7	Surfaces of Constant Curvature	59
4	Area and Integration on Surfaces	63
4.1	Surface Integrals	63
4.2	Geodesics	66
4.3	A Local Version of the Gauss–Bonnet Theorem	71
4.4	Triangles and Triangulations of Surfaces	75
4.5	Gauss-Bonnet Theorem	78

1 Curves

§1.1 Basic Definitions

Definition 1.1.1. A **parametrized curve** is a smooth map $\phi: [a, b] \rightarrow \mathbb{R}^n$. It is **regular** if $\phi'(t)$ is nonzero for all t .

Remark 1.1.2 — Note that the interval for the domain can be half open, open, or $\mathbb{R}$. We just want it to be connected.

Remark 1.1.3 — In this module we will focus on regular curves in $\mathbb{R}^2$ and in $\mathbb{R}^3$. The regularity assumption means that we can think of them as describing the trajectory of some particle that is always in motion. We do not absolutely need our curves to be smooth (C^∞); three continuous derivatives (C^3) should suffice for everything we will do, but the module will be no less interesting if we require smoothness for simplicity.

Example 1.1.4

A circle parametrized by the map

$$\begin{aligned}\phi_1: [0, 2\pi] &\rightarrow \mathbb{R}^2 \\ \phi_1(t) &= (\cos(t), \sin(t))\end{aligned}$$

is a regular curve.

Example 1.1.5

The following is also another regular curve.

$$\begin{aligned}\phi_2: [0, 1] &\rightarrow \mathbb{R}^3 \\ \phi_2(t) &= (1 - t, t, t^4)\end{aligned}$$

Example 1.1.6

Here is an example of a curve that is not regular.

$$\begin{aligned}\phi_3: [-1, 1] &\rightarrow \mathbb{R}^2 \\ t &\mapsto (t^2, t^3)\end{aligned}$$

At the origin, we have $\phi'(0) = (0, 0)$.

Example 1.1.7

Here is another example of a curve that is not regular:

$$\begin{aligned}\phi_4: \mathbb{R} &\rightarrow \mathbb{R}^2 \\ t &\mapsto (1 + t^2, |t|)\end{aligned}$$

This is not differentiable at $t = 0$, since $|t|$ has no derivative there.

Remark 1.1.8 — Regularity allows us to avoid constant curves (such as $\phi(t) = (1, 2, 3)$), and to make sure that there is a well-defined tangent line to a curve ϕ at every point, of the form

$$L = \{\phi(t_0) + s \cdot \phi'(t_0) \mid s \in \mathbb{R}\}$$

at the point $\phi(t_0)$.

Remark 1.1.9 — We're interested in the image $\phi([a, b])$ of a curve, rather than the given parametrization ϕ .

Definition 1.1.10. Let $\phi: [a, b] \rightarrow \mathbb{R}^n$ be a regular curve, and take a smooth function $f: [c, d] \xrightarrow{\sim} [a, b]$ with $f'(t) \neq 0$ for all t . Then, the curve

$$\begin{aligned}\phi \circ f: [c, d] &\rightarrow \mathbb{R}^n \\ t &\mapsto \phi(f(t))\end{aligned}$$

is called a **reparametrization** of ϕ .

We need to check that a reparametrization $\phi \circ f$ of a regular curve ϕ is still regular: it is smooth since ϕ and f are, and its derivative at any t satisfies

$$(\phi \circ f)'(t) = \phi'(f(t)) \cdot f'(t).$$

Then, $\phi'(f(t)) \in \mathbb{R}^n$ is nonzero by the regularity of ϕ , and $f'(t) \neq 0$ by assumption, so $(\phi \circ f)'(t)$ is nonzero as well.

We wish to study properties of curves that are invariant under reparametrization, meaning that they do not change if we reparametrize. Recall the inner product

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i,$$

where $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$, and then associate to any vector in $\mathbb{R}^n$ its length:

$$|x| = \sqrt{\langle x, x \rangle} = \sqrt{\sum_{i=1}^n x_i^2}.$$

Definition 1.1.11. The **arclength** (or just length) of a curve $\phi: [a, b] \rightarrow \mathbb{R}^n$ is given by

$$L(\phi) = \int_a^b |\phi'(t)| dt.$$

Example 1.1.12

The circle has length as follows:

$$\begin{aligned}
 L(\phi_1) &= \int_0^{2\pi} |\phi_1'(t)| dt = \int_0^{2\pi} |(-\sin(t), \cos(t))| dt \\
 &= \int_0^{2\pi} \sqrt{\sin^2(t) + \cos^2(t)} dt \\
 &= \int_0^{2\pi} 1 dt \\
 &= 2\pi
 \end{aligned}$$

Proposition 1.1.13

The arclength of a regular curve in $\mathbb{R}^n$ is invariant under reparametrization.

Proof. Let the curve be given by $\phi: [a, b] \rightarrow \mathbb{R}^n$ and reparametrize by a smooth map $f: [c, d] \xrightarrow{\sim} [a, b]$, with $f'(t) \neq 0$ for all t . We can suppose without loss of generality that f is increasing, as the decreasing case is analogous. Now, if we write ϕ in coordinates as

$$\phi(t) = (x_1(t), \dots, x_n(t)),$$

then we can compute that its reparametrization $\psi = \phi \circ f$ has velocity

$$\begin{aligned}
 |\psi'(t)| &= \left| \frac{d}{dt}(x_1(f(t)), \dots, x_n(f(t))) \right| \\
 &= |(x_1'(f(t))f'(t), \dots, x_n'(f(t))f'(t))| \\
 &= \sqrt{(x_1'(f(t))f'(t))^2 + \dots + (x_n'(f(t))f'(t))^2} \\
 &= |f'(t)| \sqrt{(x_1'(f(t)))^2 + \dots + (x_n'(f(t)))^2} \\
 &= f'(t) \cdot |\phi'(f(t))|.
 \end{aligned}$$

With this, the length of the reparametrized curve ψ is

$$L(\psi) = \int_c^d |\psi'(t)| dt = \int_c^d |\phi'(f(t))| \cdot f'(t) dt = \int_a^b |\phi'(s)| ds,$$

where at the last step we have applied the substitution $s = f(t)$. But then the right side is $L(\phi)$ by definition, so $L(\phi) = L(\psi)$ after all. $\square$

We are interested in properties, such as length that do not change under reparametrization, so we might wish to look for a convenient choice of parametrization and stick with it from now on. One natural choice is to parametrize a curve by *arclength*.

Definition 1.1.14. A curve $\phi: [a, b] \rightarrow \mathbb{R}^n$ is **parametrized by arclength** if $|\phi'(t)| = 1$, for all t .

This is particularly useful, because if $\phi: [a, b] \rightarrow \mathbb{R}^n$ is parametrized by arclength, then we have

$$L(\phi) = \int_a^b |\phi'(t)| dt = \int_a^b 1 dt = b - a.$$

Equivalently, we have that the curve ϕ is parametrized by arclength if and only if

$$L(\phi|_{[a,t]}) = t - a, \quad \forall t \in [a, b].$$

Proposition 1.1.15

A regular curve $\phi: [a, b] \rightarrow \mathbb{R}^n$ can always be reparametrized by arclength.

Proof. Suppose we try to reparametrize it as $\psi = \phi \circ f$ for some increasing function $f: [0, L] \rightarrow [a, b]$. We want to choose f so that $|\psi'(t)| = 1$, or equivalently

$$|\phi'(f(t))| \cdot f'(t) = 1.$$

To do so, we compute the arclength of ϕ from $\phi(a)$ to $\phi(t)$ by the map

$$\begin{aligned} \ell: [a, b] &\rightarrow \mathbb{R} \\ \ell(t) &= \int_a^t |\phi'(s)| ds. \end{aligned}$$

This is strictly increasing, since

$$\ell'(t) = |\phi'(t)| > 0.$$

Therefore, ℓ is invertible, and it is moreover smooth since ϕ is regular, so we can define

$$f: [0, L] \rightarrow [a, b]$$

as its inverse: we set $f = \ell^{-1}$, meaning that $\ell(f(t)) = t$ for all $t \in [0, L]$. We differentiate this relation to get

$$\ell'(f(t))f'(t) = 1,$$

or equivalently

$$|\phi'(f(t))| \cdot f'(t) = 1.$$

This is the relation we wanted, so f provides the desired reparametrization. $\square$

§1.2 Curvature

Definition 1.2.1. Let $\phi: [a, b] \rightarrow \mathbb{R}^n$ be a curve parametrized by arclength. The quantity

$$k(t) = |\phi''(t)|$$

is the **geodesic curvature** of ϕ at the point $\phi(t)$. It is, by definition, the magnitude of the **curvature vector** $\vec{k}(t) = \phi''(t)$.

Remark 1.2.2 — We note an advantage of parametrizing curves by arclength: the curvature $k(t)$ depends only on the curve at $\phi(t)$, and not on how it is parametrized. If $\psi(t) = \phi(f(t))$ were another parametrization by arclength, then we would have

$$1 = |\psi'(t)| = |\phi'(f(t))| \cdot |f'(t)| = |f'(t)|,$$

so $f'(t) = \pm 1$. This implies that $f(t) = \pm t + C$, for some constant C ; therefore, our other parametrization is simply

$$\psi(t) = \phi(\pm t + C).$$

However, we then observe that $\psi'(t) = \pm \phi'(\pm t + C)$ and so

$$\psi''(t) = \phi''(\pm t + C) = \phi''(f(t)).$$

In particular, the curvature vector $\vec{k}(t)$ at $\psi(t) = \phi(f(t))$ is $\psi''(t) = \phi''(f(t))$, and it does not depend on which parametrization by arclength we chose. Then, neither does its magnitude, which is the geodesic curvature $k(t)$.

Proposition 1.2.3

A regular curve $\phi: [a, b] \rightarrow \mathbb{R}^n$ has curvature $k(t) \equiv 0$ if and only if ϕ is a straight line.

Proof. We take ϕ to be parametrized by arclength, and then the following are equivalent by definition:

$$k(t) \equiv 0 \iff |\phi''(t)| \equiv 0 \iff \phi''(t) \equiv 0$$

Then, $\phi''(t) = 0$ for all t if and only if $\phi'(t) = \phi'(a)$ for all t , and this in turn is equivalent to

$$\phi(t) = \phi'(a)(t - a) + \phi(a).$$

This defines a straight line. Conversely, any straight line parametrized by arclength has this form, and taking derivatives provides the result. $\square$

Proposition 1.2.4

The curvature vector $\vec{k}(t)$ is always perpendicular to the tangent line at $\phi(t)$.

Proof. Again we parametrize ϕ by arclength, so that $|\phi'(t)| = 1$ for all t . Slightly more usefully, the square

$$|\phi'(t)|^2 = \langle \phi'(t), \phi'(t) \rangle$$

is constant (this is usually easier to work with, because its definition doesn't involve any square roots) so we differentiate this to get as follows:

$$\begin{aligned} 0 &= \frac{d}{dt} \langle \phi'(t), \phi'(t) \rangle = \langle \phi''(t), \phi'(t) \rangle + \langle \phi'(t), \phi''(t) \rangle \\ &= 2 \langle \phi'(t), \phi''(t) \rangle \\ &= \langle \phi'(t), \vec{k}(t) \rangle \end{aligned}$$

The tangent line to $\phi(t)$ is defined by

$$\{\phi'(t)s + \phi(t) \mid s \in \mathbb{R}\},$$

with direction $\phi'(t)$, so $\vec{k}(t)$ must be perpendicular to it. $\square$

Example 1.2.5 (Curvature of a Circle)

Let us compute the curvature of a circle of radius $R > 0$ centered at the origin in $\mathbb{R}^2$. We should first parametrize it, so we try

$$\phi(t) = (R \cos(t), R \sin(t)), \quad t \in [0, 2\pi],$$

This is not a parametrization by arclength (for $R \neq 1$), though, because

$$|\phi'(t)| = \sqrt{(-R \sin(t))^2 + (R \cos(t))^2} = R.$$

To parametrize it by arclength, we revisit the proof of proposition 1.1.15: the curve $\phi(t)$ has arclength

$$L(t) = \int_0^t |\phi'(s)| ds = \int_0^t R ds = Rt,$$

so we reparametrize by taking $\psi(t) = \phi(f(t))$ where $f(t) = L^{-1}(t) = \frac{t}{R}$. In other words,

$$\psi(t) = \left(R \cos\left(\frac{t}{R}\right), R \sin\left(\frac{t}{R}\right) \right), \quad t \in [0, 2\pi]$$

is our desired parametrization by arclength. The curvature vector is then

$$\vec{k}(t) = \psi''(t) = \left(-\frac{1}{R} \cos\left(\frac{t}{R}\right), \frac{1}{R} \sin\left(\frac{t}{R}\right) \right) = \frac{-\psi(t)}{R^2},$$

which points toward the origin with magnitude equal to the geodesic curvature

$$k(t) = \frac{|\psi(t)|}{R^2} = \frac{R}{R^2} = \frac{1}{R}.$$

Thus our circle of radius R has curvature $\frac{1}{R}$ everywhere.

§1.3 Curves in $\mathbb{R}^3$

Let $\phi: [a, b] \rightarrow \mathbb{R}^3$ be a curve parametrized by arclength. Then the derivative $\phi'(t)$ is a unit vector for all t , and we have seen in proposition 1.2.4 that the curvature vector $\vec{k}(t) = \phi''(t)$ is perpendicular to it. In fact there is a whole plane of vectors that are perpendicular to $\phi'(t)$, but since $\vec{k}(t)$ does not depend on our chosen parametrization, it gives us a preferred choice of perpendicular vector to $\phi'(t)$, at least when it is nonzero (i.e., when $k(t) \neq 0$).

Definition 1.3.1. Let $\phi: [a, b] \rightarrow \mathbb{R}^3$ be parametrized by arclength. We define:

- the **unit tangent vector** $T(t) = \phi'(t)$
- if $\phi''(t) = T'(t) \neq 0$, then the **unit normal vector** at $\phi(t)$ is

$$N(t) = \frac{T'(t)}{|T'(t)|}$$

- if $T'(t) \neq 0$, then the **binormal vector** at $\phi(t)$ is

$$B(t) = T(t) \times N(t)$$

The triple $(T(t), N(t), B(t))$ is a positively oriented, orthonormal basis of $\mathbb{R}^3$ at $\phi(t)$, called the **Frenet frame**.

Remark 1.3.2 — Note that we can find many vectors normal to the unit tangent and unit normal vector, but the binormal vector is the *preferred choice*.

We study how the Frenet frame changes over time by computing the derivatives of its constituent vectors. First, the unit tangent vector satisfies

$$T'(t) = |T'(t)| \cdot N(t) = |\phi''(t)| \cdot N(t) = k(t)N(t)$$

by the definition of $N(t)$.

Next, since $B(t)$ is the cross product of two orthogonal unit vectors it always has unit length, so just as in the proof of proposition 1.2.4 we use $\frac{d}{dt}\langle B(t), B(t) \rangle = 0$ to see that $B'(t)$ is orthogonal to $B(t)$. We can also see that $B'(t)$ is orthogonal to $T(t)$, since

$$B'(t) = \frac{d}{dt}(T(t) \times N(t)) = \underbrace{(T'(t) \times N(t))}_{kN \times N=0} + (T(t) \times N'(t)) = T(t) \times N'(t).$$

Since $B'(t)$ is orthogonal to both $T(t)$ and $B(t)$ it must be a scalar multiple of the other element of the Frenet frame, which is $N(t)$.

Definition 1.3.3. The **torsion** $\tau(t)$, of the curve ϕ at $\phi(t)$, is the scalar such that

$$B'(t) = -\tau(t)N(t).$$

Finally, we can write $N(t) = B(t) \times T(t)$ and differentiate to get

$$\begin{aligned} N'(t) &= (B'(t) \times T(t)) + (B(t) \times T'(t)) = (-\tau(t)N(t) \times T(t)) + (B(t) \times k(t)N(t)) \\ &= \tau(t)B(t) - k(t)T(t). \end{aligned}$$

Proposition 1.3.4 (Frenet Formulas)

The Frenet frame satisfies the system of differential equations

$$\begin{aligned} T'(t) &= k(t)N(t) \\ N'(t) &= -k(t)T(t) + \tau(t)B(t) \\ B'(t) &= -\tau(t)N(t) \end{aligned}$$

or equivalently in matrix form

$$\frac{d}{dt} \begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

Remark 1.3.5 — What this proposition tells us is that the evolution of the Frenet frame over time is completely determined by the curvature k and torsion τ , together with the initial frame.

The torsion is a bit mysterious, but we can say that it measures the *failure* of the curve to be planar (i.e., to be contained in a single plane $ax + by + cz = d$). In what follows we will use the assumption $\phi''(t) \neq 0$ to make sure that the Frenet frame and hence the torsion are well-defined. Recall that the definition of $N(t)$, and hence of $B(t)$, required us to assume that $T'(t) = \phi''(t)$ is nonzero.

Proposition 1.3.6

Let $\phi: [a, b] \rightarrow \mathbb{R}^3$ be a curve parametrized by arclength, satisfying $\phi''(t) \neq 0$, for all t . Then the curve $\phi(t)$ is planar if and only if $\tau(t) = 0$ for all t .

Proof. Suppose first that $\tau(t) = 0$ for all t . Then $B' = -\tau N = 0$ says that $B(t)$ is constant, say $B(t) = \vec{c}$ for all t ; then $\vec{c}$ is a unit vector since $B(t)$ is. We compute that

$$\frac{d}{dt} \langle \phi(t), B(t) \rangle = \langle \phi'(t), B(t) \rangle + \langle \phi(t), B'(t) \rangle = \langle T(t), B(t) \rangle + \langle \phi(t), 0 \rangle = 0,$$

since T is orthogonal to B . Thus, $\langle \phi(t), B(t) \rangle = \langle \phi(t), \vec{c} \rangle$ is constant as well. We can write

$$\langle \phi(t), \vec{c} \rangle = d,$$

for some constant $d \in \mathbb{R}$, and since $\vec{c} \neq 0$, this says that $\phi(t)$ lies in a plane perpendicular to $\vec{c}$. (In fact, since it is perpendicular to $\vec{c} = B(t)$, this plane is spanned by $T(t)$ and $N(t)$.)

Conversely, suppose that ϕ lies in a plane, which we can write as $\langle \vec{x}, \vec{c} \rangle = d$, for some unit vector $\vec{c}$. We differentiate the relation $\langle \phi(t), \vec{c} \rangle = d$ to attain

$$0 = \langle \phi'(t), \vec{c} \rangle = \langle T(t), \vec{c} \rangle,$$

so $T(t)$ is perpendicular to $\vec{c}$. Then, we differentiate again to get

$$0 = \langle T'(t), \vec{c} \rangle = \langle k(t)N(t), \vec{c} \rangle,$$

which by the assumption $k(t) = \phi''(t) \neq 0$ means that $N(t)$ is also perpendicular to $\vec{c}$. Now, $\vec{c}$ is a unit vector perpendicular to both $T(t)$ and $N(t)$, so we must have $B(t) = \pm \vec{c}$. We differentiate this to get

$$0 = B'(t) = -\tau(t)N(t),$$

and since $N(t)$ is a unit vector, we must have $\tau(t) = 0$, for all t . □

Example 1.3.7

Let us compute each element of the Frenet frame, for the circle $\phi(t) = (\cos(t), \sin(t), 0)$, for $t \in [0, 2\pi]$ (note that this is parametrized by arclength.)

$$T(t) = \phi'(t) = (-\sin(t), \cos(t), 0)$$

$$N(t) = \frac{T'(t)}{|T'(t)|} = (-\cos(t), -\sin(t), 0)$$

$$B(t) = T(t) \times N(t) = (0, 0, 1)$$

Therefore, ϕ has curvature $k(t) = |T'(t)| = 1$, and since $B(t)$ is constant, the relation $B'(t) = -\tau(t)N(t)$ says that ϕ has torsion $\tau(t) = 0$. This matches our expectation from proposition 1.3.6, since ϕ lies in the plane $z = 0$.

Example 1.3.8

Define a curve $\phi: \mathbb{R} \rightarrow \mathbb{R}^3$ called a **helix** by

$$\phi(t) = \left(\cos\left(\frac{t}{\sqrt{2}}\right), \sin\left(\frac{t}{\sqrt{2}}\right), \frac{t}{\sqrt{2}} \right).$$

We check that this is parametrized by arclength:

$$|\phi'(t)| = \left| \left(-\frac{1}{\sqrt{2}} \sin\left(\frac{t}{\sqrt{2}}\right), \frac{1}{\sqrt{2}} \cos\left(\frac{t}{\sqrt{2}}\right), \frac{1}{\sqrt{2}} \right) \right| = 1.$$

We compute the Frenet frame, beginning with the unit normal:

$$T(t) = \phi'(t) = \left(-\frac{1}{\sqrt{2}} \sin\left(\frac{t}{\sqrt{2}}\right), \frac{1}{\sqrt{2}} \cos\left(\frac{t}{\sqrt{2}}\right), \frac{1}{\sqrt{2}} \right).$$

This has derivative

$$T'(t) = \left(\frac{1}{2} \cos\left(\frac{t}{\sqrt{2}}\right), -\frac{1}{2} \sin\left(\frac{t}{\sqrt{2}}\right), 0 \right),$$

with magnitude $k(t) = |T'(t)| = \frac{1}{2}$, so the principal normal vector is

$$N(t) = \frac{T'(t)}{|T'(t)|} = 2T'(t) = \left(\cos\left(\frac{t}{\sqrt{2}}\right), -\sin\left(\frac{t}{\sqrt{2}}\right), 0 \right).$$

Then, the binormal vector is given by

$$B(t) = T(t) \times N(t) = \left(\frac{1}{\sqrt{2}} \sin\left(\frac{t}{\sqrt{2}}\right), -\frac{1}{\sqrt{2}} \cos\left(\frac{t}{\sqrt{2}}\right), \frac{1}{\sqrt{2}} \right)$$

and its derivative is

$$B'(t) = \left(\frac{1}{2} \cos\left(\frac{t}{\sqrt{2}}\right), \frac{1}{2} \sin\left(\frac{t}{\sqrt{2}}\right), 0 \right) = -\frac{1}{2}N(t).$$

Then, from $B' = -\tau N$, we see that ϕ has torsion $\tau(t) = \frac{1}{2}$, for all t . Therefore, the helix ϕ is not planar.

Remark 1.3.9 — In the previous example, we determined that $k(t) = \tau(t) = \frac{1}{2}$, for all t . It is a useful exercise to check the Frenet equations

$$\frac{d}{dt} \begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

explicitly for the Frenet frame (T, N, B) of the helix.

§1.4 Fundamental Theorem of the Local Theory of Curves

Summarizing what we've seen so far, a curve $\phi: [a, b] \rightarrow \mathbb{R}^3$ parametrized by arclength that satisfies $\phi''(t) \neq 0$ for all t gives rise to the Frenet frame (T, N, B) , where

$$T = \phi'(t), N = \frac{\phi''(t)}{|\phi''(t)|}, \quad B = T \times N.$$

From this, we can extract the curvature $k = |\phi''(t)|$, which is strictly positive, and the torsion τ , which satisfies $B' = -\tau N$. These are well-defined invariants of the curve, meaning they don't depend on the parametrization. Our main result here states that in fact k and τ determine the curve ϕ , essentially uniquely.

Theorem 1.4.1 (Fundamental Theorem of the Local Theory of Curves)

Let $k, \tau: [a, b] \rightarrow \mathbb{R}$ be smooth functions, and suppose that $k(t) > 0$ for all t . Then,

- **Existence:** There is a regular curve $\phi: [a, b] \rightarrow \mathbb{R}^3$, parametrized by arclength, with curvature $k(t)$ and torsion $\tau(t)$.
- **Uniqueness:** Any other such curve $\psi: [a, b] \rightarrow \mathbb{R}^3$ differs from ϕ by a rigid motion, meaning that

$$\psi(t) = g \cdot \phi(t) + \vec{c},$$

for some fixed element $g \in SO(3)$ and vector $\vec{c} \in \mathbb{R}^3$.

Proof. We start by proving uniqueness. We first observe that rigid motions preserve arclength, curvature, and torsion. If we suppose that ϕ is parametrized by arclength and write $\tilde{\phi}(t) = g\phi(t) + \vec{c}$. Note that any $g \in SO(3)$ preserves the inner product. We have

$$\begin{aligned} |\tilde{\phi}'(t)| &= |g\phi'(t)| = \langle g'\phi(t), g\phi'(t) \rangle = (g\phi'(t))^T (g\phi'(t)) \\ &= \phi'(t)^T \underbrace{g^T g}_{=I} \phi'(t) = \langle \phi'(t), \phi'(t) \rangle = 1 \end{aligned}$$

and

$$k_{\tilde{\phi}}(t) = |\tilde{\phi}''(t)| = |g\phi''(t)| = |\phi''(t)| = k_{\phi}(t),$$

since $g \in SO(3)$ preserves lengths, so $\tilde{\phi}(t)$ is also parametrized by arclength and $\tilde{\phi}$ has the same curvature as ϕ . The Frenet frames from ϕ and $\tilde{\phi}$ are related by $T_{\tilde{\phi}}(t) = gT_{\phi}(t)$ and

$$N_{\tilde{\phi}}(t) = \frac{\tilde{\phi}''(t)}{|\tilde{\phi}''(t)|} = \frac{g\phi''(t)}{|\phi''(t)|} = gN_{\phi}(t),$$

and then

$$B_{\tilde{\phi}}(t) = T_{\tilde{\phi}}(t) \times N_{\tilde{\phi}}(t) = gT_{\phi}(t) \times gN_{\phi}(t) = gB_{\phi}(t).$$

We differentiate this to get

$$-\tau_{\tilde{\phi}}(t)N_{\tilde{\phi}}(t) = g(-\tau_{\phi}(t)N_{\phi}(t)) = -\tau_{\phi}(t)(gN_{\phi}(t)) = -\tau_{\phi}(t)N_{\tilde{\phi}}(t),$$

and then since $N_{\tilde{\phi}}(t) \neq 0$, we can conclude that $\tau_{\tilde{\phi}}(t) = -\tau_{\phi}(t)$ as well, so ϕ and $\tilde{\phi}$ have the same torsion. With this in mind, we suppose that ϕ and ψ have the same curvature k and torsion τ , and we take $g \in SO(3)$ that sends the initial Frenet frame $(T_{\psi}(a), N_{\psi}(a), B_{\psi}(a))$ of ψ to the Frenet frame $(T_{\phi}(a), N_{\phi}(a), B_{\phi}(a))$. Note that we can do this because these are both oriented orthonormal frames for $\mathbb{R}^3$. We replace $\psi(t)$ with $g \cdot \psi(t)$, so that now ϕ and ψ have the same Frenet frames at $t = a$, and then we replace $\psi(t)$ again with its translation

$$\psi(t) + (\phi(a) - \psi(a))$$

to arrange $\psi(a) = \phi(a)$ as well. Our initial hypothesis still applies, so ϕ and ψ both have curvature k and torsion τ . Now that ϕ and ψ have the same starting point and initial Frenet frames, we measure the difference between their Frenet frames for all $t \in [a, b]$, with the function

$$\begin{aligned} f(t) &= |T_{\psi}(t) - T_{\phi}(t)|^2 + |N_{\psi}(t) - N_{\phi}(t)|^2 + |B_{\psi}(t) - B_{\phi}(t)|^2 \\ &= \langle T_{\psi}(t) - T_{\phi}(t), T_{\psi}(t) - T_{\phi}(t) \rangle \\ &\quad + \langle N_{\psi}(t) - N_{\phi}(t), N_{\psi}(t) - N_{\phi}(t) \rangle \\ &\quad + \langle B_{\psi}(t) - B_{\phi}(t), B_{\psi}(t) - B_{\phi}(t) \rangle. \end{aligned}$$

We've arranged for this to satisfy $f(a) = 0$, and we compute its derivative as

$$\begin{aligned} \frac{1}{2}f'(t) &= \langle T_{\psi}(t) - T_{\phi}(t), T'_{\psi}(t) - T'_{\phi}(t) \rangle \\ &\quad + \langle N_{\psi}(t) - N_{\phi}(t), N'_{\psi}(t) - N'_{\phi}(t) \rangle \\ &\quad + \langle B_{\psi}(t) - B_{\phi}(t), B'_{\psi}(t) - B'_{\phi}(t) \rangle \\ &= k\langle T_{\psi} - T_{\phi}, N_{\psi} - N_{\phi} \rangle - k\langle N_{\psi} - N_{\phi}, T_{\psi} - T_{\phi} \rangle \\ &\quad + \tau\langle N_{\psi} - N_{\phi}, B_{\psi} - B_{\phi} \rangle - \tau\langle B_{\psi} - B_{\phi}, N_{\psi} - N_{\phi} \rangle \\ &= 0, \end{aligned}$$

so $f(t) = f(a) = 0$, for all $t \in [a, b]$. In other words, the curves ϕ and ψ have the same Frenet frames for all $t \in [a, b]$, and since the tangent vectors agree in particular, we have

$$\phi(t) = \phi(a) + \int_a^t T_{\phi}(s)ds = \psi(a) + \int_a^t T_{\psi}(s)ds = \psi(t),$$

for all $t \in [a, b]$.

Next, we move on to prove the existence part. Suppose that we've been given $k(t) > 0$ and $\tau(t)$. We can then select any positively oriented orthonormal frame (T_a, N_a, B_a) and use the existence theorem for solutions of linear differential equations to find functions

$$T, N, B: [a, b] \rightarrow \mathbb{R}^3$$

satisfying the first order differential equation with initial conditions

$$\frac{d}{dt} \begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}, \quad \begin{pmatrix} T(a) \\ N(a) \\ B(a) \end{pmatrix} = \begin{pmatrix} T_a \\ N_a \\ B_a \end{pmatrix}.$$

If this is indeed an orthonormal frame, then we can take $\phi(t) = \int_a^t T(s)ds$ to be the desired curve with (T, N, B) as its Frenet frame, so we need to check that it is orthonormal. To do this, we package the column vectors $T(t), N(t), B(t)$ into a matrix

$$M(t) = \begin{pmatrix} | & | & | \\ T(t) & N(t) & B(t) \\ | & | & | \end{pmatrix},$$

which has the property that (T, N, B) is an orthonormal basis if and only if $M^T M = I$. Notably, we've arranged for this to hold when $t = a$ (as $M(a) = I$), so it suffices to check that $\frac{d}{dt}(M^T M) = 0$, for all t . Now, the Frenet equations can be reinterpreted in terms of M by checking that

$$M' = \begin{pmatrix} | & | & | \\ T' & N' & B' \\ | & | & | \end{pmatrix} = \begin{pmatrix} | & | & | \\ T & N & B \\ | & | & | \end{pmatrix} \begin{pmatrix} 0 & -k & 0 \\ k & 0 & -\tau \\ 0 & \tau & 0 \end{pmatrix} = M \begin{pmatrix} 0 & -k & 0 \\ k & 0 & -\tau \\ 0 & \tau & 0 \end{pmatrix},$$

so we can compute that

$$\begin{aligned} \frac{d}{dt}(M^T M) &= (M')^T M + M^T M' \\ &= \left(M \begin{pmatrix} 0 & -k & 0 \\ k & 0 & -\tau \\ 0 & \tau & 0 \end{pmatrix} \right)^T M + M^T \left(M \begin{pmatrix} 0 & -k & 0 \\ k & 0 & -\tau \\ 0 & \tau & 0 \end{pmatrix} \right) \\ &= \begin{pmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} M^T M + M^T M \begin{pmatrix} 0 & -k & 0 \\ k & 0 & -\tau \\ 0 & \tau & 0 \end{pmatrix}. \end{aligned}$$

This tells us that $M^T M$ is a solution to the linear, first order differential equation

$$\frac{dA}{dt} = \begin{pmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} A - A \begin{pmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix}, \quad A(a) = I,$$

but $A(t) = I$ is also a solution, so by uniqueness we must have $M^T M = I$ for all $t \in [a, b]$. We conclude that (T, N, B) is indeed orthonormal for all t , and so $\phi(t) = \int_a^t T(s)ds$ is a curve with the desired curvature and torsion. $\square$

Remark 1.4.2 — In proving the existence, we exploited the fact that the following coefficient matrix was skew-symmetric.

$$\begin{pmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix}$$

Example 1.4.3

Let $\phi: [a, b] \rightarrow \mathbb{R}^3$ be a curve with zero torsion and constant curvature $c > 0$. Then, ϕ lies in a circle of radius $\frac{1}{c}$. Indeed, the same computation from a previous example says that the curve $\psi: [a, b] \rightarrow \mathbb{R}^3$ defined by

$$\psi(t) = \left(\frac{\cos(ct)}{c}, \frac{\sin(ct)}{c}, 0 \right)$$

has constant curvature c , and by proposition 1.3.6, it has zero torsion since it lies in the xy -plane. Thus, theorem 1.4.1 says that ϕ must be related to ψ by a rigid motion of $\mathbb{R}^3$.

§1.5 Plane Curves

For curves in the plane, we often carry out computations explicitly. For instance, let $\phi: [a, b] \rightarrow \mathbb{R}^2$ be a regular curve, described by coordinate functions $\phi(t) = (x(t), y(t))$. Then, $\phi'(t) = (x'(t), y'(t))$ is nonzero and tangent to the curve at $\phi(t)$, so the vector

$$N(t) = \frac{(-y'(t), x'(t))}{|\phi'(t)|}$$

is a unit vector orthogonal to the tangent vector $\phi'(t)$. In fact, there are two such vectors (N and $-N$), but this is a natural choice because it says that

$$T = \frac{\phi'(t)}{|\phi'(t)|}$$

and N form a oriented orthonormal basis for $\mathbb{R}^2$, meaning that

$$\det \begin{pmatrix} - & T & - \\ - & N & - \end{pmatrix} = 1.$$

Now if ϕ is parametrized by arclength, then its curvature still satisfies

$$\vec{k}(t) = \phi''(t) = k(t)N(t),$$

so we can take inner products with $N(t) = (-y'(t), x'(t))$ on both sides and apply $\langle N(t), N(t) \rangle = 1$ to get

$$k(t) = k(t) \langle N(t), N(t) \rangle = \langle \phi''(t), N(t) \rangle = x'(t)y''(t) - y'(t)x''(t).$$

In fact, it's not too much harder to determine $k(t)$ even when ϕ is not parametrized by arclength.

Proposition 1.5.1

Let $\phi: [a, b] \rightarrow \mathbb{R}^2$ be a regular curve, not necessarily parametrized by arclength. Then, its curvature at $\phi(t)$ is given by

$$k(t) = \frac{\langle \phi''(t), N(t) \rangle}{|\phi'(t)|^2}.$$

Proof. Write $\phi(t) = (x(t), y(t))$, and let $\psi(t) = \phi(f(t)) = (x(f(t)), y(f(t)))$ be a parametrization by arclength for some function $f: [0, L] \rightarrow [a, b]$. We apply the chain rule then to compute that

$$\begin{aligned}(x \circ f)' &= (x' \circ f)f', & (x \circ f)'' &= (x'' \circ f)(f')^2 + (x' \circ f)(f'') \\ (y \circ f)' &= (y' \circ f)f', & (y \circ f)'' &= (y'' \circ f)(f')^2 + (y' \circ f)(f''),\end{aligned}$$

so the above formula for the curvature of ψ becomes

$$\begin{aligned}k_\psi(t) &= (x \circ f)'(y \circ f)'' - (y \circ f)'(x \circ f)'' \\ &= [(x' \circ f)(y'' \circ f)(f')^3 + (x' \circ f)(y' \circ f)f'f''] \\ &\quad - [(y' \circ f)(x'' \circ f)(f')^3 + (y' \circ f)(x' \circ f)f'f''] \\ &= [(x' \circ f)(y'' \circ f) - (y' \circ f)(x'' \circ f)](f')^3.\end{aligned}$$

Now, since $\phi''(f(t)) = (x''(f(t)), y''(f(t)))$ and

$$N_\phi(f(t)) = N_\psi(t) = (-(y \circ f)'(t), (x \circ f)'(t)) = (-y'(f(t)), x'(f(t)))f'(t),$$

we can rewrite the curvature as

$$k_\psi(t) = \langle \phi''(f(t)), N_\phi(f(t)) \rangle \cdot f(t)^2.$$

Now recall from the proof of proposition 1.1.15 that we should take f to be the inverse of the arclength function

$$\ell(t) = \int_a^t |\phi'(s)| ds,$$

so that $\ell(f(t)) = t$, and that differentiating this last relation tells us that

$$1 = \ell'(f(t))f'(t) = |\phi'(f(t))|f'(t)$$

by the chain rule and the fundamental theorem of calculus respectively. This gives us the relation

$$f'(t) = \frac{1}{|\phi(f(t))'|},$$

so we substitute this into the above expression for $k_\psi(t)$ and set $s = f(t)$ to get

$$k_\phi(s) = k_\psi(t) = \frac{\langle \phi''(s), N_\phi(s) \rangle}{|\phi'(s)|^2}$$

□

Example 1.5.2

Let $\phi: [a, b] \rightarrow \mathbb{R}^2$ be the graph of a smooth function $f: [a, b] \rightarrow \mathbb{R}$, parametrized by $\phi(t) = (t, f(t))$. This is almost certainly not parametrized by arclength, but we can nevertheless compute its curvature using proposition 1.5.1: we have

$$\phi'(t) = (1, f'(t)), \quad \phi''(t) = (0, f''(t)), \quad N(t) = \frac{(-f'(t), 1)}{\sqrt{1 + (f'(t))^2}}$$

and so

$$k(t) = \frac{\langle \phi''(t), N(t) \rangle}{|\phi'(t)|^2} = \frac{\frac{f''(t)}{\sqrt{1 + (f'(t))^2}}}{1 + (f'(t))^2},$$

or simply

$$k(t) = \frac{f''(t)}{(1 + (f'(t))^2)^{\frac{3}{2}}}.$$

We will relate the curvature of a plane curve to the following topological quantity.

Definition 1.5.3. Let $\phi: [a, b] \rightarrow \mathbb{R}^2$ be a smooth closed curve, meaning that

$$\phi(a) = \phi(b) \quad \text{and} \quad \phi^{(k)}(a) = \phi^{(k)}(b), \forall k \geq 1.$$

If ϕ does not pass through the origin $(0, 0)$, then the **winding number** of ϕ is an integer $w(\phi) \in \mathbb{Z}$ measuring the number of counterclockwise rotations ϕ makes around the origin.

We can view our curve $\phi(t) = (x(t), y(t))$ as a complex-valued function $\phi: [a, b] \rightarrow \mathbb{R}^2 \cong \mathbb{C}$ with coordinates $z = x + iy$, and then if $\gamma = \phi([a, b])$, we know from complex analysis that

$$w(\phi) = \frac{1}{2\pi i} \oint_{\gamma} \frac{dz}{z}.$$

We expand this and attempt to simplify:

$$\begin{aligned} w(\phi) &= \frac{1}{2\pi i} \oint_{\gamma} \frac{dx + idy}{x + iy} \\ &= \frac{1}{2\pi i} \int_a^b \frac{x'(t) + iy'(t)}{x(t) + iy(t)} dt \\ &= \frac{1}{2\pi i} \int_a^b \frac{x'(t) + iy'(t)}{x(t) + iy(t)} \frac{x(t) - iy(t)}{x(t) - iy(t)} dt \\ &= \frac{1}{2\pi i} \int_a^b \frac{(xx' + yy') + i(xy' - x'y)}{x^2 + y^2} dt \\ &= \frac{1}{2\pi i} \int_a^b \frac{xx' + yy'}{x^2 + y^2} dt + \frac{1}{2\pi} \int_a^b \frac{xy' - x'y}{x^2 + y^2} dt. \end{aligned}$$

The first integral is equal to

$$\int_a^b \frac{xx' + yy'}{x^2 + y^2} dt = \frac{1}{2} \int_a^b \frac{d}{dt} \log(x^2 + y^2) dt = \frac{1}{2} (\log(|\phi(b)|^2) - \log(|\phi(a)|^2)) = 0,$$

since ϕ is a closed curve and so $\phi(a) = \phi(b)$. The winding number of ϕ is therefore given by the second integral, namely

$$w(\phi) = \frac{1}{2\pi} \int_a^b \frac{x(t)y'(t) - x'(t)y(t)}{x(t)^2 + y(t)^2} dt.$$

If $\phi: [a, b] \rightarrow \mathbb{R}^2$ is a regular closed curve parametrized by arclength, say with $\phi(t) = (x(t), y(t))$, then $\phi'(t)$ is never zero and so we can attempt to compute its winding number using this integral formula. We have $|\phi'(t)| = 1$ for all t , so

$$\begin{aligned} w(\phi') &= \frac{1}{2\pi} \int_a^b \frac{x'y'' - y'x''}{(x')^2 + (y')^2} dt \\ &= \frac{1}{2\pi} \int_a^b x'y'' - y'x'' dt \\ &= \frac{1}{2\pi} \int_a^b k(t) dt. \end{aligned}$$

In other words, up to a factor of 2π , the winding number of ϕ' is equal to the total curvature of k along ϕ . We give this quantity a name:

Definition 1.5.4. The **index** or **turning number** of a closed curve $\phi: [a, b] \rightarrow \mathbb{R}^2$ parametrized by arclength is

$$\text{Ind}(\phi) = w(T(t)),$$

i.e., the winding number of its unit tangent vector $T(t)$.

The computation above provides the following proposition.

Proposition 1.5.5

If $\phi: [a, b] \rightarrow \mathbb{R}^2$ is parametrized by arclength, then

$$\text{Ind}(\phi) = \frac{1}{2\pi} \int_a^b k(t) dt$$

The index $\text{Ind}(\phi)$ measures the total number of rotations made by the unit tangent vector as we travel along ϕ . This is a purely topological quantity: it does not change if we perturb ϕ slightly. This means that the total curvature $\int k(t) dt$ does not change either, even though the curvature function $k(t)$ might!

2 Surfaces

§2.1 Definition and Examples

We now turn our attention to the main object of study of this module.

Definition 2.1.1. A **regular surface** is a subset $S \subset \mathbb{R}^3$ such that for all points $p \in S$, there is

- an open neighborhood $V \subset \mathbb{R}^3$ of p
- an open set $U \subset \mathbb{R}^2$
- a smooth map $\phi: U \rightarrow \mathbb{R}^3$ such that:
 1. $\phi(U) = V \cap S$
 2. ϕ is a homeomorphism onto its image
 3. $\forall q \in U$, the differential $d\phi_q: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is injective.

The pair (U, ϕ) is called a **chart** or a local parametrization at p .

Remark 2.1.2 — The first two conditions on ϕ ensure that S *looks like* the plane $\mathbb{R}^2$ near every point p , while the third condition is what makes S a regular surface.

Example 2.1.3

The xy -plane $\{z = 0\} \subset \mathbb{R}^3$ is a regular surface, with chart $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $\phi(u, v) = (u, v, 0)$.

Example 2.1.4

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a smooth function. Then, the graph

$$\Gamma_f = \{(x, y, f(x, y)) \mid (x, y) \in \mathbb{R}^2\}$$

is a regular surface. We use a chart $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $\phi(u, v) = (u, v, f(u, v))$, and regularity follows because at each $q \in \mathbb{R}^2$, we have

$$d\phi_q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \frac{\partial f}{\partial u}(q) & \frac{\partial f}{\partial v}(q) \end{pmatrix},$$

whose columns are linearly independent.

Example 2.1.5

For example, the graph of $f(x, y) = \sin(\sqrt{x^2 + y^2 + 1})$ is a regular surface.

Example 2.1.6

The unit sphere $\mathbb{S}^2$ is a regular surface. We can't use a single chart to prove this, but we can define two charts

$$\begin{aligned}\phi_N: \mathbb{R}^2 &\rightarrow \mathbb{S}^2 \setminus \{(0, 0, 1)\} \\ \phi_S: \mathbb{R}^2 &\rightarrow \mathbb{S}^2 \setminus \{(0, 0, -1)\}\end{aligned}$$

by **stereographic projection** from the north and south poles. For instance, we can define $\phi_N(u, v)$ to be the unique point of $\mathbb{S}^2 \setminus \{(0, 0, 1)\}$ on the line from $(0, 0, 1)$ to $(u, v, -1)$.

Exercise 2.1.7. Check that

$$\phi_N(u, v) = \left(\frac{4u}{u^2 + v^2 + 4}, \frac{4v}{u^2 + v^2 + 4}, \frac{u^2 + v^2 - 4}{u^2 + v^2 + 4} \right).$$

To see what regularity means, we can write our chart in local coordinates as

$$\phi(u, v) = \begin{pmatrix} x(u, v) \\ y(u, v) \\ z(u, v) \end{pmatrix},$$

and then its differential has the form

$$d\phi_q = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{pmatrix}.$$

This is injective if and only if the columns $\frac{\partial \phi}{\partial u}(q)$ and $\frac{\partial \phi}{\partial v}(q)$ are linearly independent. Writing $q = (u_0, v_0)$, the curves

$$f(u) = \phi(u, v_0) \text{ and } \phi(v) = \phi(u_0, v)$$

have tangent vectors $\frac{\partial \phi}{\partial u}(q)$ and $\frac{\partial \phi}{\partial v}(q)$, respectively, at $\phi(q)$, so $d\phi_q$ is injective if and only if these tangent vectors span a plane. In other words, the surface S must have a well-defined tangent plane at each point, just like a regular curve has a tangent line at each point.

Example 2.1.8

We saw in example 2.1.3 that the xy -plane in $\mathbb{R}^3$ is a regular surface, but the map

$$\begin{aligned}\phi: \mathbb{R}^2 &\rightarrow \mathbb{R}^3 \\ (u, v) &\mapsto (u^3, v^3, 0)\end{aligned}$$

is not a chart, since $d\phi_{(0,0)} = 0$.

Example 2.1.9

The cone

$$S = \{(x, y, z) \mid x^2 + y^2 = z^2, z \geq 0\}$$

is not a regular surface. It is the graph of the function $f(x, y) = \sqrt{x^2 + y^2}$, but this is not differentiable at the origin, and it is not possible to find a smooth chart for S at the cone point $(0, 0, 0)$.

Example 2.1.10

Define $\phi: (0, 2\pi) \times (-1, 1) \times \mathbb{R}^3$ by

$$\phi(u, v) = (\sin(u), \sin(2u), v).$$

Then, ϕ is smooth and injective, and $d\phi_q$ is always injective, but ϕ is not a homeomorphism onto its image.

Proposition 2.1.12 is an important source of many examples of regular surfaces. To prove this proposition, though, will require us to construct charts at every point of $S \subset \mathbb{R}^3$ that is a regular level set of a smooth function $F: \mathbb{R}^3 \rightarrow \mathbb{R}$, because so far this is the only way we know how to prove that a surface is regular. Our key tool will be the following important result from analysis:

Theorem 2.1.11 (Inverse Function Theorem)

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be smooth on an open neighborhood V of the point $p \in \mathbb{R}^n$, and suppose that $df_p: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible. Then, there is an open neighborhood $U \subset V$ of p such that the restriction

$$f: U \rightarrow f(U)$$

is a diffeomorphism (it is smooth with a smooth inverse).

Theorem 2.1.12

Let $F: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a smooth function, and let $S \subset \mathbb{R}^3$ be a regular level set of F , meaning that

$$S = F^{-1}(c) = \{p \in \mathbb{R}^3 \mid F(p) = c\},$$

for some $c \in \mathbb{R}$, and that the gradient $\nabla F(p)$ is nonzero for all $p \in S$. Then, S is a regular surface.

Proof. Writing $S = f^{-1}(c)$, we need to construct a chart for each point $p \in S$. By assumption, $\nabla F(p) \neq 0$, so at least one of the coordinates $\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z}$ is nonzero at p . WLOG, say $\frac{\partial F}{\partial z}(p) \neq 0$.

We define a map $g: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $g(x, y, z) = (x, y, F(x, y, z))$. This sends level sets of

F into horizontal planes. The map g is smooth on all of $\mathbb{R}^3$ since F is, and the matrix

$$dg_p = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{\partial F}{\partial x}(p) & \frac{\partial F}{\partial y}(p) & \frac{\partial F}{\partial z}(p) \end{pmatrix}$$

has determinant $\frac{\partial F}{\partial z}(p) \neq 0$, so it's invertible. Therefore, we can apply the inverse function theorem to g . That is, there exists an open neighborhood $U \subset \mathbb{R}^3$ of p such that the restriction

$$g: U \rightarrow g(U)$$

is a diffeomorphism. The inverse $g^{-1}: g(U) \rightarrow U$ will be a smooth map sending planes $\{z = a\}$ to level sets $F^{-1}(a)$, so we will use this to construct our chart at p .

We define the domain for our chart by $W = \{(u, v) \in \mathbb{R}^2 \mid (u, v, c) \in g(U)\}$. This is nonempty, since if $p = (x_0, y_0, z_0)$, then $p \in U$ and $g(p) = (x_0, y_0, c) \in g(U)$, so that $(x_0, y_0) \in W$. By construction, we have that

$$W \times \{c\} = \underbrace{\{z = c\}}_{\text{plane in } \mathbb{R}^3} \cap \underbrace{g(U)}_{\text{open in } \mathbb{R}^3},$$

hence $W \times \{c\}$ is open in the plane $\{z = c\}$, and by forgetting the z -coordinate we see that W is open in $\mathbb{R}^2$. Therefore, we let $V = U \cap F^{-1}(c)$, which contains p , and define our chart by the following:

$$\begin{aligned} \phi: W &\rightarrow V \\ (u, v) &\mapsto g^{-1}(u, v, c) \end{aligned}$$

Then, ϕ is smooth since g^{-1} is, and we can check that it is a homeomorphism onto its image and that $d\phi_q$ is injective, so this satisfies all of the requirements for a chart. This construction works at every $p \in S$, so S is indeed a regular surface as promised. $\square$

Remark 2.1.13 — In the proof of theorem 2.1.12, we only needed F to be smooth near each point $p \in S$ where we wanted to construct a chart. Thus we only really need F to be smooth on an open neighborhood of S .

Example 2.1.14

Let $F(x, y, z) = x^2 + y^2 + z^2$. Then, the sphere $\mathbb{S}^2$ is the level set $F^{-1}(1)$, and we have $\nabla F = (2x, 2y, 2z) \neq 0$, for all $(x, y, z) \in \mathbb{S}^2$, so this proves that $\mathbb{S}^2$ is a regular surface without having to construct explicit charts.

Example 2.1.15

Let $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a smooth function. Then, the graph

$$\Gamma_g = \{(x, y, g(x, y))\}$$

is a level set of $F(x, y, z) = z - g(x, y)$. $S = F^{-1}(0)$, and this is regular because

$$\nabla F(x, y, z) = \left(-\frac{\partial g}{\partial x}, -\frac{\partial g}{\partial y}, 1\right) \neq 0.$$

Example 2.1.16

Let S be a torus given in cylindrical coordinates by

$$(r - 2)^2 + z^2 = 1,$$

or equivalently by taking a circle in the xz -plane of radius 1, centered at $(x, z) = (2, 0)$, and revolving it around the z -axis. Then, we can write S as a level set F^{-1} , where

$$F(x, y, z) = \left(\sqrt{x^2 + y^2} - 2 \right)^2 + z^2,$$

and we compute that

$$\nabla F(x, y, z) = \left(2(\sqrt{x^2 + y^2} - 2) \frac{x}{\sqrt{x^2 + y^2}}, 2(\sqrt{x^2 + y^2} - 2) \frac{y}{\sqrt{x^2 + y^2}}, 2z \right).$$

Therefore, if $\nabla F(x, y, z) = 0$, then $z = 0$ and either $\sqrt{x^2 + y^2} = 2$ or $x = y = 0$. However, these conditions imply that $F(x, y, z) = 0$ or $F(x, y, z) = 4$ respectively, so $\nabla F \neq 0$ on all of $S = F^{-1}(1)$, and therefore the torus S is a regular surface.

Remark 2.1.17 — In the example above, F is not smooth along the z -axis, but the torus avoids an open neighborhood of the z -axis, so the proof still works.

The following fact about surfaces will sometimes be useful: it says that every surface looks like a special case of example 2.1.4, at least locally.

Lemma 2.1.18

Let $S \subset \mathbb{R}^3$ be a regular surface, and pick a point $p \in S$. Then, there is a neighborhood $V \subset S$ of p such that V is the graph of a smooth function of the form $z = f(x, y)$, $y = g(x, z)$, or $x = h(y, z)$.

Proof. Take a chart $\phi: U \rightarrow S$ at p , and write it in coordinates as

$$\phi(u, v) = (x(u, v), y(u, v), z(u, v)).$$

Let $q = \phi^{-1}(p)$, which is a point of $U \subset \mathbb{R}^2$. Then, the derivative

$$d\phi_q = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{pmatrix}$$

has rank 2 since its columns are linearly independent. This means that one of its 2×2 minors is invertible, so we'll suppose WLOG that

$$\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

is invertible at q . We now define a map $g: U \rightarrow \mathbb{R}^2$ by

$$g(u, v) = (x(u, v), y(u, v)),$$

and dg_q is invertible, so the inverse function theorem gives us a neighborhood $V \subset U$ of q on which

$$g: V \rightarrow g(V)$$

is a diffeomorphism. Letting $W = \phi(V)$, which contains $\phi(q) = p$, we now see that W is the graph of a function $f: g(V) \rightarrow \mathbb{R}$ given by $f(x, y) = z(g^{-1}(x, y))$, i.e., as the composition

$$\begin{aligned} g(V) &\xrightarrow{g^{-1}} V \xrightarrow{z} \mathbb{R} \\ (x, y) &\longmapsto (u, v) \longmapsto z(\phi(u, v)), \end{aligned}$$

and this is smooth since both z and g^{-1} are. $\square$

Remark 2.1.19 — Note that the converse is also true. If $S \subset \mathbb{R}^3$ has the property that for every $p \in S$, there exists an open neighborhood V of p on which $S \cap V$ is the graph of a smooth function $z = f(x, y)$, $y = g(x, z)$, or $x = h(y, z)$, then S is a regular surface.

Exercise 2.1.20. Using lemma 2.1.18, show that the cone of example 2.1.9 (defined by $z = \sqrt{x^2 + y^2}$) is not a regular surface.

§2.2 Tangent Vectors and Tangent Planes

Definition 2.2.1. A **tangent vector** to S at the point $p \in S$ is a vector of the form $\alpha'(0)$, where

$$\alpha: (-\epsilon, \epsilon) \rightarrow S \subset \mathbb{R}^3$$

is a smooth curve satisfying $\alpha(0) = p$.

Definition 2.2.2. The **tangent plane** to S at p is the set $T_p S$ of all tangent vectors at p , meaning the set of all tangent vectors to curves on S that pass through p at $t = 0$.

Remark 2.2.3 — The tangent plane clearly does not depend on a choice of charts at p , because they don't appear anywhere in the definition, but this is not so useful for trying to do actual computations involving $T_p S$. When we introduce charts, this becomes easier.

Lemma 2.2.4

Let $\phi: U \rightarrow \mathbb{R}^3$ be a chart for S at p , say with $\phi(q) = p$. Then, the image

$$d\phi_q(\mathbb{R}^2) = \text{Span} \left(\frac{\partial \phi}{\partial u}(q), \frac{\partial \phi}{\partial v}(q) \right)$$

lies in $T_p S$.

Proof. The image $d\phi_q(\mathbb{R}^2)$ is spanned by the columns of

$$d\phi_q = \begin{pmatrix} | & | \\ \frac{\partial \phi}{\partial u}(q) & \frac{\partial \phi}{\partial v}(q) \\ | & | \end{pmatrix},$$

so we consider an arbitrary vector in this span, say

$$w = a \frac{\partial \phi}{\partial u}(q) + b \frac{\partial \phi}{\partial v}(q), \quad a, b \in \mathbb{R}.$$

Writing $q = (u_0, v_0)$, we consider for small $\epsilon > 0$ the smooth path $\alpha: (-\epsilon, \epsilon) \rightarrow \mathbb{R}$ defined by

$$\alpha(t) = \phi(u_0 + at, v_0 + bt).$$

Then, $\alpha(0) = \phi(u_0, v_0) = \phi(q) = p$, and by the chain rule, its velocity at time $t = 0$ is as follows:

$$\begin{aligned} \alpha'(0) &= \frac{\partial \phi}{\partial u}(u_0 + at, v_0 + bt) \cdot \frac{du}{dt} + \frac{\partial \phi}{\partial v}(u_0 + at, v_0 + bt) \cdot \frac{dv}{dt} \Big|_{t=0} \\ &= \frac{\partial \phi}{\partial u}(q) \cdot a + \frac{\partial \phi}{\partial v}(q) \cdot b \\ &= w. \end{aligned}$$

$\alpha'(0) = w \in T_p S$ since $\alpha(0) = p$ and because α is smooth. Since $w \in d\phi_q(\mathbb{R}^2)$ was arbitrary, this proves the lemma. $\square$

Lemma 2.2.5

Let $\phi: U \rightarrow \mathbb{R}^3$ be a chart for S at p . Given any smooth curve $\alpha: (-\epsilon, \epsilon) \rightarrow S$ with $\alpha(0) = p$, we can find $\epsilon' \in (0, \epsilon)$ such that $\alpha|_{(-\epsilon', \epsilon')}$ has the form

$$\alpha(t) = \phi(u_\alpha(t), v_\alpha(t)),$$

for some smooth functions $u_\alpha, v_\alpha: (-\epsilon', \epsilon') \rightarrow \mathbb{R}$.

Proof. By lemma 2.1.18, we can write S near p as a graph of the form $z = f(x, y)$, without loss of generality. The cases where it has the form $y = g(x, z)$ and $x = h(y, z)$ are essentially the same, shrinking the chart ϕ as needed to ensure that $\phi(U)$ has this form. We can therefore write

$$\phi(u, v) = (x(u, v), y(u, v), f(x(u, v), y(u, v))).$$

We let $q = \phi^{-1}(p)$, with coordinates $q = (u_0, v_0)$. Now define a function $F: U \times (-\epsilon, \epsilon) \rightarrow \mathbb{R}^3$ that parametrizes a collar neighborhood of S near p by

$$F(u, v, t) = (x(u, v), y(u, v), f(x(u, v), y(u, v)) + t).$$

Then, $F(u_0, v_0, 0) = \phi(u_0, v_0, p)$, and its derivative here is

$$dF_{(u_0, v_0, 0)} = \begin{pmatrix} \frac{\partial x}{\partial u}(q) & \frac{\partial x}{\partial v}(q) & 0 \\ \frac{\partial y}{\partial u}(q) & \frac{\partial y}{\partial v}(q) & 0 \\ \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} & \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v} & 1 \end{pmatrix} = \begin{pmatrix} | & | & 0 \\ \frac{\partial \phi}{\partial u} & \frac{\partial \phi}{\partial v} & 0 \\ | & | & 1 \end{pmatrix}.$$

The first two columns are precisely the differential $d\phi_q$, and can be factored as

$$d\phi_q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial u}(q) & \frac{\partial x}{\partial v}(q) \\ \frac{\partial y}{\partial u}(q) & \frac{\partial y}{\partial v}(q) \end{pmatrix}$$

Then since $d\phi_q$ is injective, the 2×2 factor on the right must be injective as well, hence it is invertible. This factor is also the top left 2×2 minor of $dF(u_0, v_0, 0)$, so we use expansion by minors along the third column to see that $dF(u_0, v_0, 0)$ has nonzero determinant, hence is invertible. The inverse function theorem now tells us that F is a diffeomorphism on some neighborhood of $(u_0, v_0, 0)$.

Let $G = F^{-1}$ be the inverse of F on a neighborhood of $p = F(u_0, v_0, 0)$. Then, G is smooth, hence so is the composition

$$G(x, y, f(x, y)) = (u(x, y), v(x, y), 0),$$

meaning that $u(x, y)$ and $v(x, y)$ are smooth function of x and y . Now, if we write our given curve as

$$\alpha(t) = (x(t), y(t), f(x(t), y(t)))$$

near $\alpha(0) = p$, then there is some positive $\epsilon' < \epsilon$ such that $\alpha((-\epsilon', \epsilon'))$ is contained in the domain of G . We define smooth functions $u_\alpha(t) = u(x(t), y(t))$ and $v_\alpha(t) = v(x(t), y(t))$ on the interval $(-\epsilon', \epsilon')$, and then we have

$$\alpha(t) = (\underbrace{\phi(u(x(t), y(t)))}_{=u_\alpha(t)}, \underbrace{\phi(v(x(t), y(t)))}_{=v_\alpha(t)}),$$

which is what is desired. □

We can now characterize the tangent plane $T_p S$ in terms of any chart.

Theorem 2.2.6

Let $\phi: U \rightarrow \mathbb{R}^3$ be a chart for S at p , and write $q = \phi^{-1}(p) \in U$. Then,

$$T_p S = d\phi_q(\mathbb{R}^2) = \text{Span} \left(\frac{\partial \phi}{\partial u}(q), \frac{\partial \phi}{\partial v}(q) \right).$$

Proof. The inclusion $d\phi_q(\mathbb{R}^2) \subset T_p S$ was lemma 2.2.4, so it remains to prove the reverse inclusion. By definition, any tangent vector $\vec{v} \in T_p S$ can be written as $\vec{v} = \alpha'(0)$ for some regular curve

$$\alpha: (-\epsilon, \epsilon) \rightarrow S$$

with $\alpha(0) = p$, and by lemma 2.2.5. we can write α on some smaller interval $(-\epsilon', \epsilon')$ in the form

$$\alpha(t) = \phi(u(t), v(t))$$

for smooth function u and v . Now, we have

$$\alpha'(0) = \frac{\partial \phi}{\partial u} \frac{du}{dt} + \frac{\partial \phi}{\partial v} \frac{dv}{dt} \Big|_{t=0} \in \text{Span} \left(\frac{\partial \phi}{\partial u}(q), \frac{\partial \phi}{\partial v}(q) \right),$$

so $\vec{v} = \alpha'(0)$ belongs to $d\phi_q(\mathbb{R}^2) = \text{Span} \left(\frac{\partial \phi}{\partial u}(q), \frac{\partial \phi}{\partial v}(q) \right)$. □

Remark 2.2.7 — $T_p S$ is a real two dimensional vector space because $d\phi_q(\mathbb{R}^2)$ is (note that $d\phi_q(\mathbb{R}^2)$ is injective).

Example 2.2.8

Consider the point $p = (0, 0, 1)$ on the sphere $S := \mathbb{S}^2$. Rather than use stereographic projection to define a chart at p , we view S near p as a graph, defining a chart by

$$\begin{aligned} \phi: \{(u, v) \mid u^2 + v^2 < 1\} &\rightarrow S \\ (u, v) &\mapsto (u, v, \sqrt{1 - u^2 - v^2}) \end{aligned}$$

with $\phi(0, 0) = p$. Then, theorem 2.2.6 says that $T_p S$ is spanned by the vectors

$$\begin{aligned} \frac{\partial \phi}{\partial u}(0, 0) &= \left(\begin{array}{c} 1 \\ 0 \\ \frac{-u}{\sqrt{1-u^2-v^2}} \end{array} \right) \Big|_{(u,v)=(0,0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ \frac{\partial \phi}{\partial v}(0, 0) &= \left(\begin{array}{c} 0 \\ 1 \\ \frac{-v}{\sqrt{1-u^2-v^2}} \end{array} \right) \Big|_{(u,v)=(0,0)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \end{aligned}$$

so $T_p S$ is the horizontal plane $\{z = 0\} = \text{Span} \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right)$.

For another large class of examples, we recall from proposition 2.1.12 that a regular level set of a smooth function $F: \mathbb{R}^3 \rightarrow \mathbb{R}$ is also a regular surface. It turns out that we can compute the tangent space at any point in terms of F as follows.

Proposition 2.2.9

Let $S = F^{-1}(c)$ be a regular level set of a smooth function $F: \mathbb{R}^3 \rightarrow \mathbb{R}$. For any point $p \in S$, the tangent space at p is given by

$$T_p S = \{\vec{v} \in \mathbb{R}^3 \mid \langle \vec{v}, \nabla F(p) \rangle = 0\} = (\nabla F(p))^\perp,$$

the orthogonal complement of the gradient $\nabla F(p)$.

Proof. Take any smooth path $\alpha: (-\epsilon, \epsilon) \rightarrow S$, with $\alpha(0) = p$, and write α in coordinates as

$$\alpha(t) = (x(t), y(t), z(t)).$$

Since $\alpha(t) \in S$, we know that $F(\alpha(t)) = 0$ for all t , and differentiating this last equation provides

$$0 = \frac{d(F \circ \alpha)}{dt} = \frac{\partial F}{\partial x}(\alpha(t)) \cdot x'(t) + \frac{\partial F}{\partial y}(\alpha(t)) \cdot y'(t) + \frac{\partial F}{\partial z}(\alpha(t)) \cdot z'(t).$$

Setting $t = 0$, we recognize the RHS as the inner product $\langle \nabla F(p), \alpha'(0) \rangle$, so $\alpha'(0)$ is orthogonal to $\nabla F(p)$. However, every tangent vector at p has the form $\alpha'(0)$ for some such α , so this establishes the inclusion

$$T_p S \subset (\nabla F(p))^\perp.$$

Simultaneously, theorem 2.2.6 tells us that $T_p S$ is a two-dimensional vector space, and so is the orthogonal complement to $\nabla F(p)$ since $\nabla F(p) \neq 0$, so this inclusion must be an equality. $\square$

Example 2.2.10

Let S be the paraboloid $\{z = x^2 + y^2\}$ and let $P = (1, 3, 10) \in S$. Then, we can write $S = F^{-1}(0)$, where

$$F(x, y, z) = x^2 + y^2 - z,$$

and the gradient $\nabla F = (2x, 2y, -1)$ is nonzero everywhere, so S is a regular level set of F . We compute that

$$\nabla F(p) = (2, 6, -1),$$

so the tangent plane at p is provided by

$$T_p S = (\nabla F(p))^\perp = \{(x, y, z) \mid 2x + 6y - z = 0\}.$$

§2.3 Smooth Maps and Differentials

We can use charts to identify surfaces locally with open sets of $\mathbb{R}^2$, and doing so allows us to define smooth maps and their differentials on a surface by borrowing the corresponding notion from $\mathbb{R}^2$.

Definition 2.3.1. Let $S_1, S_2 \subset \mathbb{R}^3$ be regular surfaces. A map $F: S_1 \rightarrow \mathbb{R}^3$ is **smooth** if for every chart $\phi: U \rightarrow S_1$, the composition

$$U \xrightarrow{\phi} S_1 \xrightarrow{F} \mathbb{R}^3$$

is smooth when viewed as a map from an open subset of $\mathbb{R}^2$ to $\mathbb{R}^3$. A map $F: S_1 \rightarrow S_2$ is **smooth** if it is smooth when viewed as a map $S_1 \rightarrow \mathbb{R}^3$.

Definition 2.3.2. The **differential** of a smooth map $F: S_1 \rightarrow S_2$ consists of a map

$$dF_p: T_p S_1 \rightarrow T_{F(p)} S_2$$

for each point $p \in S_1$, defined as follows. Given a tangent vector $v \in T_p S_1$, we can write $v = \alpha'(0)$ for some regular curve $\alpha: (-\epsilon, \epsilon) \rightarrow S_1$ with $\alpha(0) = p$. Then

$$\beta(t) = F(\alpha(t))$$

is a regular curve $(-\epsilon, \epsilon) \rightarrow S_2$, with $\beta(0) = F(p)$, and we define

$$dF_p(v) = \beta'(0) = (F \circ \alpha)'(0).$$

We first show that dF_p is well-defined.

Proposition 2.3.3

The value of $dF_p(v)$ does not depend on the choice of curve α .

Proof. Let $\alpha_1, \alpha_2: (-\epsilon, \epsilon) \rightarrow S_1$ be two curves with $\alpha_1(0) = \alpha_2(0) = p$ and $\alpha_1'(0) = \alpha_2'(0) = v$. Given some chart $\phi: U \rightarrow \mathbb{R}^2$ for S_1 at p , with $q = \phi^{-1}(p)$, lemma 2.2.5 says that we can write

$$\alpha_1(t) = \phi(u_1(t), v_1(t)), \quad \alpha_2(t) = \phi(u_2(t), v_2(t)),$$

for smooth functions u_1, v_1, u_2, v_2 defined on some neighborhood of $t = 0$ and satisfying $(u_1(0), v_1(0)) = (u_2(0), v_2(0)) = q$. The α'_i satisfy

$$\begin{aligned}\alpha'_1(0) &= \frac{\partial \phi}{\partial u}(q) \cdot u'_1(0) + \frac{\partial \phi}{\partial v}(q) \cdot v'_1(0) \\ \alpha'_2(0) &= \frac{\partial \phi}{\partial u}(q) \cdot u'_2(0) + \frac{\partial \phi}{\partial v}(q) \cdot v'_2(0),\end{aligned}$$

and since ϕ is a chart, the column vectors $\frac{\partial \phi}{\partial u}(q)$ and $\frac{\partial \phi}{\partial v}(q)$ are linearly independent, so the only way we can have $\alpha'_1(0) = \alpha'_2(0)$ is if $u'_1(0) = u'_2(0)$ and $v'_1(0) = v'_2(0)$. Letting $\beta_i(t) = F(\alpha_i(t))$ for $i = 1, 2$, we now compute that

$$\beta'_i(0) = \frac{d}{dt}(F(\phi(u_i(t), v_i(t))))|_{t=0} = \frac{\partial(F \circ \phi)}{\partial u}(q) \cdot u'_i(0) + \frac{\partial(F \circ \phi)}{\partial v}(q) \cdot v'_i(0).$$

However, since $u'_1(0) = u'_2(0)$ and $v'_1(0) = v'_2(0)$, this means that $\beta'_1(0) = \beta'_2(0)$, and $dF_p(v)$ was defined to either of the vectors $\beta'_i(0)$ depending on whether we took $v = \alpha'_1(0)$ or $v = \alpha'_2(0)$, so we conclude that $dF_p(v)$ doesn't depend on the choice of α after all. $\square$

In fact, the differential dF_p is not only well-defined, it carries extra useful structure.

Proposition 2.3.4

Given a smooth map $F: S_1 \rightarrow S_2$ between regular surfaces, the differential

$$dF_p: T_p S_1 \rightarrow T_{F(p)} S_2$$

is a linear map for each $p \in S$.

Proof. Take tangent vectors $w_1, w_2 \in T_p S_1$ and write $w_i = \alpha'_i(0)$ for some curves $\alpha_i: (-\epsilon, \epsilon) \rightarrow S_1$ with $\alpha_i(0) = p$. Fixing a chart $\phi: U \rightarrow S_1$ at p , and translating U if needed to arrange that $\phi^{-1}(p) = (0, 0)$, we use lemma 2.2.5 to write

$$\alpha_i(t) = \phi(u_i(t), v_i(t))$$

near $t = 0$ for each $i = 1, 2$; then $\phi^{-1}(p) = (0, 0)$ means that $u_i(0) = v_i(0) = 0$. Given constants $c_1, c_2 \in \mathbb{R}$, we define a third curve

$$\begin{aligned}\alpha_3: (-\epsilon', \epsilon') &\rightarrow S_1 \\ t &\mapsto \phi(c_1 u_1(t) + c_2 u_2(t), c_1 v_1(t) + c_2 v_2(t))\end{aligned}$$

for some small positive $\epsilon' \leq \epsilon$. We compute that $\alpha_3(0) = \phi(0, 0) = p$, and that the corresponding tangent vector in $T_p S_1$ is the following:

$$\begin{aligned}\alpha'_3(0) &= \frac{\partial \phi}{\partial u}(0, 0) \cdot (c_1 u'_1(0) + c_2 u'_2(0)) + \frac{\partial \phi}{\partial v}(0, 0) \cdot (c_1 v'_1(0) + c_2 v'_2(0)) \\ &= c_1 \left(\frac{\partial \phi}{\partial u}(0, 0) v'_1(0) + \frac{\partial \phi}{\partial u}(0, 0) v'_2(0) \right) + c_2 \left(\frac{\partial \phi}{\partial v}(0, 0) v'_1(0) + \frac{\partial \phi}{\partial v}(0, 0) v'_2(0) \right) \\ &= c_1 \alpha'_1(0) + c_2 \alpha'_2(0) \\ &= c_1 w_1 + c_2 w_2\end{aligned}$$

We can therefore apply F to α_3 to compute that:

$$\begin{aligned}
 dF_p(c_1 w_1 + c_2 w_2) &= dF_p(\alpha'_3(0)) \\
 &= \frac{d}{dt}(F(\alpha_3(t))) \big|_{t=0} \\
 &= \frac{d}{dt}((F \circ \phi)(c_1 u_1(t) + c_2 u_2(t), c_1 v_1(t) + c_2 v_2(t))) \big|_{t=0} \\
 &= \frac{\partial(F \circ \phi)}{\partial u}(0, 0) \cdot (c_1 u'_1(0) + c_2 u'_2(0)) \\
 &\quad + \frac{\partial(F \circ \phi)}{\partial v}(0, 0) \cdot (c_1 v'_1(0) + c_2 v'_2(0)) \\
 &= c_1 \left(\frac{\partial(F \circ \phi)}{\partial u}(0, 0) u'_1(0) + \frac{\partial(F \circ \phi)}{\partial v}(0, 0) v'_1(0) \right) \\
 &\quad + c_2 \left(\frac{\partial(F \circ \phi)}{\partial u}(0, 0) u'_2(0) + \frac{\partial(F \circ \phi)}{\partial v}(0, 0) v'_2(0) \right) \\
 &= c_1 (F \circ \alpha_1)'(0) + c_2 (F \circ \alpha_2)'(0) \\
 &= c_1 dF_p(w_1) + c_2 dF_p(w_2),
 \end{aligned}$$

This proves the desired linearity. $\square$

The following chain rule is implicit in the proof of proposition 2.3.4, and is generally useful for computing differentials.

Lemma 2.3.5

Let $F: S_1 \rightarrow S_2$ be a smooth map between regular surfaces, and fix a chart $\phi: U \rightarrow S_1$. Then for each $(u_0, v_0) \in U$, if $p = \phi(u_0, v_0)$ then we have the following:

$$\begin{aligned}
 dF_p \left(\frac{\partial \phi}{\partial u}(u_0, v_0) \right) &= \frac{\partial(F \circ \phi)}{\partial u}(u_0, v_0) \\
 dF_p \left(\frac{\partial \phi}{\partial v}(u_0, v_0) \right) &= \frac{\partial(F \circ \phi)}{\partial v}(u_0, v_0)
 \end{aligned}$$

Proof. We focus on the first identity, since the proof of the second is identical. We define

$$\alpha(t) = \phi(u_0 + t, v_0)$$

on some small interval $t \in (-\epsilon, \epsilon)$. Then $\alpha'(0) = \frac{\partial \phi}{\partial u}(u_0, v_0)$, and we have

$$\begin{aligned}
 dF_p(\alpha'(0)) &= (F \circ \alpha)'(0) = \frac{d}{dt}(F(\phi(u_0 + t, v_0))) \bigg|_{t=0} \\
 &= \frac{\partial(F \circ \phi)}{\partial u}(u_0, v_0) \cdot 1 + \frac{\partial(F \circ \phi)}{\partial v}(u_0, v_0) \cdot 0 \\
 &= \frac{\partial(F \circ \phi)}{\partial u}(u_0, v_0).
 \end{aligned}$$

$\square$

Finally, we can also extend the previous definitions to smooth, real-valued functions on a surface as follows.

Definition 2.3.6. If S is a regular surface, then $f: S \rightarrow \mathbb{R}$ is **smooth** if for every chart $\phi: U \rightarrow S$, the composition

$$U \xrightarrow{\phi} S \xrightarrow{f} \mathbb{R}$$

is a smooth map from $U \subset \mathbb{R}^2$ to $\mathbb{R}$. Its differential

$$df_p: T_p S \rightarrow \mathbb{R}$$

is defined as follows: given a tangent vector $v \in T_p S$, of the form $v = \alpha'(0)$ for some $\alpha: (-\epsilon, \epsilon) \rightarrow S$ with $\alpha(0) = p$, we define

$$df_p(v) = \left. \frac{d}{dt}(f(\alpha(t))) \right|_{t=0}.$$

Exercise 2.3.7. Show that $df_p: T_p S \rightarrow \mathbb{R}$ is both well-defined and linear.

Example 2.3.8

Let $S = \{x^2 + y^2 + z^2 = 1\}$ be the unit sphere, and fix $p = (0, 1, 0) \in S$. Letting $f: S \rightarrow \mathbb{R}$ be the function $f(x, y, z) = z$, we wish to compute $T_p S$ and $df_p: T_p S \rightarrow \mathbb{R}$.

Solution. We know from example 2.1.14 that S is a regular level set of $g(x, y, z) = x^2 + y^2 + z^2$, namely $S = g^{-1}(1)$. Proposition 2.2.9 therefore says that $T_p S$ is the orthogonal complement of

$$\nabla g(p) = (2x, 2y, 2z) |_{(x,y,z)=(0,1,0)} = (0, 2, 0),$$

so for example

$$T_p S = \text{Span} \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right).$$

Now, near p we can parametrize S by a chart

$$\begin{aligned} \phi: U = \{(u, v) \mid u^2 + v^2 < 1\} &\rightarrow S \\ (u, v) &\mapsto (u, \sqrt{1 - u^2 - v^2}, v), \end{aligned}$$

with $(f \circ \phi)(u, v) = v$. We apply the chain rule to $f \circ \phi$ to say that

$$\begin{aligned} df_p \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right) &= df_p \left(\frac{\partial \phi}{\partial u}(0, 0) \right) = \frac{\partial (f \circ \phi)}{\partial u}(0, 0) = 0 \\ df_p \left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) &= df_p \left(\frac{\partial \phi}{\partial v}(0, 0) \right) = \frac{\partial (f \circ \phi)}{\partial v}(0, 0) = 1. \end{aligned}$$

□

Differentials give us an analogue of the inverse function theorem for surfaces.

Proposition 2.3.9

Let $F: S_1 \rightarrow S_2$ be a smooth map between surfaces, and suppose for some $p \in S_1$ that the differential

$$dF_p: T_p S_1 \rightarrow T_{F(p)} S_2$$

is an isomorphism. Then, there is an open neighborhood $V \subset S_1$ of p such that

$$F|_V: V \rightarrow f(V)$$

is a diffeomorphism.

Proof. We take a pair of charts

$$\phi_1: U_1 \rightarrow S_1, \quad \phi_2: U_2 \rightarrow S_2$$

at p and at $F(p)$ respectively, and identify $q_1 \in U_1$ and $q_2 \in U_2$ with $\phi(q_1) = p$ and $\phi(q_2) = F(p)$. We can replace U_1 with a smaller neighborhood of q_1 as needed to ensure that $f(g(U_1)) \subset \phi_2(U_2)$. Then, the map

$$g: U_1 \xrightarrow{\phi_1} S_1 \xrightarrow{F} S_2 \xrightarrow{\phi_2^{-1}} U_2$$

is a well-defined map between open sets of $\mathbb{R}^2$, and it satisfies $g(q_1) = q_2$. the differentials $(d\phi_1)_{q_1}$ and $(d\phi_2)_{q_2}$ are invertible by the definition of a chart, and dF_p is invertible by hypothesis, so the chain rule tells us that

$$dg_{q_1} = (d(\phi_2^{-1}))_{F(p)} \circ dF_p \circ (d\phi_1)_{q_1}$$

is also invertible. We apply the inverse function theorem (theorem 2.1.11) to see that g is a diffeomorphism on a neighborhood of $q_1 \in U_1$, so the composition

$$F = \phi_2 \circ g \circ \phi_1^{-1}$$

is also a diffeomorphism from some neighborhood V of $\phi_1(q_1) = p$ onto its image. $\square$

§2.4 Normal Vectors and the Gauss Map

A regular surface $S \subset \mathbb{R}^3$ has exactly two unit normal vectors at each point $p \in S$. If we can express S as a regular level set $F^{-1}(c)$ of some smooth function $F: \mathbb{R}^3 \rightarrow \mathbb{R}$, then there is a natural way to choose a distinguished normal vector: we saw in proposition 2.2.9 that $T_p S$ is the orthogonal complement of the gradient $\nabla F(p) \neq 0$, so we can define the unit normal vector to S at p by

$$N(p) = \frac{\nabla F(p)}{|\nabla F(p)|}.$$

We may not be handed such a function F in general, but if we have a chart $\phi: U \rightarrow S$ at some point p , say with $q = \phi^{-1}(p)$, then theorem 2.2.6 says that the vectors

$$\frac{\partial \phi}{\partial u}(q), \quad \frac{\partial \phi}{\partial v}(q)$$

are linearly independent and span $T_p S$. This means that their cross product is nonzero and orthogonal to $T_p S$, so again it gives us a natural choice of normal vector: we can take

$$N(p) = \frac{\frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v}}{|\frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v}|},$$

for all $p \in \phi(U)$. This construction gives us a continuous choice of normal vector $N(p)$ locally, meaning on any single chart of S , but we want to know whether there's a global choice that works: can we extend this from a single chart to the rest of S ? The answer is sometimes no, for example when S is a Möbius strip.

Definition 2.4.1. A surface $S \subset \mathbb{R}^3$ is **orientable** if it admits a continuous choice of normal vector $N(p)$ for all $p \in S$. Since each $N(p)$ is a unit vector in $\mathbb{R}^3$, it lies on the unit sphere $\mathbb{S}^2$, and so we can view this choice as a continuous map

$$N: S \rightarrow \mathbb{S}^2$$

called the **Gauss map** of S .

Remark 2.4.2 — The Möbius strip is non-orientable, though we won't prove this.

Remark 2.4.3 — By contrast, if $S = F^{-1}(c)$ is a regular level set of some function F , then the assignment

$$N(p) = \frac{\nabla F(p)}{|\nabla F(p)|}$$

described above shows that S is orientable.

Example 2.4.4

Let $S = \mathbb{S}^2$ be the unit sphere. We can once again take $F(x, y, z) = x^2 + y^2 + z^2$ and describe S as the regular level set $F^{-1}(1)$, so it has Gauss map

$$N(x, y, z) = \frac{\nabla F}{|\nabla F|} = \frac{(2x, 2y, 2z)}{|(2x, 2y, 2z)|} = (x, y, z).$$

In other words, the Gauss map $N: S \rightarrow \mathbb{S}^2$ is the identity. We can also determine N on the northern hemisphere $\{(x, y, z) \in S \mid z > 0\}$ by using the chart

$$\begin{aligned} \phi: U = \{(u, v) \mid u^2 + v^2 < 1\} &\rightarrow \mathbb{S}^2 \\ (u, v) &\mapsto (u, \sqrt{1 - u^2 - v^2}). \end{aligned}$$

At each point in the image of ϕ , we can take

$$N(u, v, \sqrt{1 - u^2 - v^2}) = N(\phi(u, v)) = \frac{\frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v}}{|\frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v}|}$$

We compute the partial derivatives:

$$\begin{aligned} \frac{\partial \phi}{\partial u} &= \left(1, 0, \frac{-u}{\sqrt{1 - u^2 - v^2}}\right) \\ \frac{\partial \phi}{\partial v} &= \left(0, 1, \frac{-v}{\sqrt{1 - u^2 - v^2}}\right), \end{aligned}$$

whose cross product is

$$\frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} = \left(\frac{u}{\sqrt{1 - u^2 - v^2}}, \frac{v}{\sqrt{1 - u^2 - v^2}}, 1\right)$$

with length $\frac{1}{\sqrt{1 - u^2 - v^2}}$. Dividing by this length gives us

$$N(u, v, \sqrt{1 - u^2 - v^2}) = (u, v, \sqrt{1 - u^2 - v^2}),$$

so again we see that $N: S \rightarrow \mathbb{S}^2$ is the identity on the northern hemisphere, and nearly identical computations in other charts will confirm that $N = id$ on all of $\mathbb{S}^2$.

Example 2.4.5

Let $P \subset \mathbb{R}^3$ be a plane, with equation $ax + by + cz = d$. If we let $F(x, y, z) = ax + by + cz$, then $\nabla F = (a, b, c)$ is nonzero everywhere, so $P = F^{-1}(d)$ is a regular level set of F and we compute that Gauss map of P by

$$N(x, y, z) = \frac{\nabla F(x, y, z)}{|\nabla F(x, y, z)|} = \frac{(a, b, c)}{\sqrt{a^2 + b^2 + c^2}}.$$

In other words, the Gauss map $N: P \rightarrow \mathbb{S}^2$ is constant.

The Gauss map of S , by definition, is a smooth map between surfaces, so we can describe its differential, which a priori gives us a map

$$dN_p: T_p S \rightarrow T_{N(p)} \mathbb{S}^2,$$

for each $p \in S$. Now we know from example 2.4.4 that at the point $N(p) \in \mathbb{S}^2$, the vector $N(p)$ is itself orthogonal to $\mathbb{S}^2$, so the tangent plane $T_{N(p)}\mathbb{S}^2$ consists of all vectors orthogonal to $N(p)$. But since $N(p)$ is a normal vector to S at p , this is simply the tangent plane $T_p S$! In other words, we have a natural identification

$$T_{N(p)}\mathbb{S}^2 = T_p S,$$

and we can use this to view the differential of the Gauss map as a map

$$dN_p: T_p S \rightarrow T_p S.$$

Example 2.4.6

If $P \subset \mathbb{R}^3$ is a plane, then example 2.4.5 says that $N: P \rightarrow \mathbb{S}^2$ is constant, so its differential satisfies

$$dN_p(v) = 0,$$

for all p and v . In other words, the map $dN_x: T_x P \rightarrow T_x P$ is identically zero.

Example 2.4.7

Let $S = \{(x, y, z) \mid x^2 + y^2 + z^2 = r^2\}$ be a sphere of radius $r > 0$. This is a regular level set of $F(x, y, z) = x^2 + y^2 + z^2$, namely $S = F^{-1}(r^2)$, so the Gauss map is

$$N(p) = \frac{\nabla F(p)}{|\nabla F(p)|} = \frac{(2x, 2y, 2z)}{2|(x, y, z)|} = \frac{1}{r} \cdot p.$$

To compute its differential, we let $\alpha: (-\epsilon, \epsilon) \rightarrow S$ be any regular curve with $\alpha(0) = p$ and let $v = \alpha'(0)$. Then, we have

$$dN_p(v) = \left. \frac{d}{dt}(N(\alpha(t))) \right|_{t=0} = \left. \frac{d}{dt} \frac{\alpha(t)}{r} \right|_{t=0} = \frac{1}{r} \alpha'(0) = \frac{v}{r}.$$

Thus, we can describe the differential of the Gauss map as

$$dN_p = \frac{1}{r} \cdot Id: T_p S \rightarrow T_p S.$$

We recall from proposition 1.2.31 and example 1.2.5 that a line in $\mathbb{R}^2$ has curvature zero, and that a circle of radius r in $\mathbb{R}^2$ has geodesic curvature $\frac{1}{r}$. In the same sense, examples 2.4.6 and 2.4.7 suggest that dN_p is measuring some sort of curvature of planes and spheres. In the following sections we will work to make this precise, but before doing so we finish with a useful tool for computing dN_p .

Proposition 2.4.8

Let S be a regular surface, and let $\phi: U \rightarrow \mathbb{R}^3$ be a chart for S at p , with $q = \phi^{-1}(p) \in U$. Then, dN_p is completely determined by the formulas

$$dN_p \left(\frac{\partial \phi}{\partial u}(q) \right) = \frac{\partial(N \circ \phi)}{\partial u}(q), \quad dN_p \left(\frac{\partial \phi}{\partial v}(q) \right) = \frac{\partial(N \circ \phi)}{\partial v}(q).$$

Proof. We know that $\frac{\partial \phi}{\partial u}(q)$ and $\frac{\partial \phi}{\partial v}(q)$ span $T_p S$ by theorem 2.2.6, so by linearity, it suffices to compute dN_p on these vectors. The actual computation follows immediately from the chain rule (lemma 2.3.5) on $(N \circ \phi)$. $\square$

3 The First and Second Fundamental Forms

§3.1 The Second Fundamental Form

In this section we will use the derivative $dN_p: T_pS \rightarrow T_pS$ of the Gauss map to define a notion of curvature on the surface S . The following map is a crucial part of this definition.

Definition 3.1.1. Let S be a regular surface and fix a point $p \in S$. The **second fundamental form** at p is the map

$$A: T_pS \times T_pS \rightarrow \mathbb{R}$$

defined by $A(X, Y) = -\langle X, dN_p(Y) \rangle$.

Proposition 3.1.2

The second fundamental form $A: T_pS \times T_pS \rightarrow \mathbb{R}$ is a symmetric bilinear form.

Proof. We first prove that A is bilinear, using the linearity of dN_p (proposition 2.3.4) and of the inner product $\langle \cdot, \cdot \rangle$. Given tangent vectors $X, Y, Z \in T_pS$ and constants $a, b \in \mathbb{R}$, we have

$$\begin{aligned} A(X, aY + bZ) &= -\langle X, dN_p(aY + bZ) \rangle \\ &= -\langle X, a \cdot dN_p(Y) + b \cdot dN_p(Z) \rangle \\ &= -a\langle X, dN_p(Y) \rangle - b\langle X, dN_p(Z) \rangle \\ &= a \cdot A(X, Y) + b \cdot A(X, Z) \end{aligned}$$

and through similar computation, we get that $A(aX + bY, Z) = a \cdot A(X, Z) + b \cdot A(Y, Z)$. To prove that A is symmetric, we take a chart $\phi: U \rightarrow \mathbb{R}^3$ at p and let $q = \phi^{-1}(p)$. Theorem 2.2.6 says that T_pS is spanned by $\frac{\partial \phi}{\partial u}(q)$ and $\frac{\partial \phi}{\partial v}(q)$ so since we have shown that A is bilinear, we only need to check that

$$A\left(\frac{\partial \phi}{\partial u}(q), \frac{\partial \phi}{\partial v}(q)\right) = A\left(\frac{\partial \phi}{\partial v}(q), \frac{\partial \phi}{\partial u}(q)\right).$$

Using the definition of A and proposition 2.4.8, we know that

$$A\left(\frac{\partial \phi}{\partial u}(q), \frac{\partial \phi}{\partial v}(q)\right) = -\left\langle \frac{\partial \phi}{\partial u}(q), dN_p\left(\frac{\partial \phi}{\partial v}(q)\right) \right\rangle = -\left\langle \frac{\partial \phi}{\partial u}(q), \frac{\partial(N \circ \phi)}{\partial v}(q) \right\rangle.$$

Since $\frac{\partial \phi}{\partial u}(q)$ is tangent to S at p while $N(\phi(q))$ is normal to S at p , we have

$$\left\langle \frac{\partial \phi}{\partial u}(q), (N \circ \phi)(q) \right\rangle = 0,$$

and we can differentiate both sides with respect to v to obtain

$$\left\langle \frac{\partial}{\partial v} \frac{\partial \phi}{\partial u}(q), (N \circ \phi)(q) \right\rangle + \left\langle \frac{\partial \phi}{\partial u}(q), \frac{\partial(N \circ \phi)}{\partial v}(q) \right\rangle = 0,$$

so we substitute this into the above computation of A to obtain

$$A\left(\frac{\partial\phi}{\partial u}(q), \frac{\partial\phi}{\partial v}(q)\right) = \left\langle \frac{\partial^2\phi}{\partial v\partial u}(q), (N \circ \phi)(q) \right\rangle.$$

The same argument with the roles of u and v reversed tells us that

$$A\left(\frac{\partial\phi}{\partial v}(q), \frac{\partial\phi}{\partial u}(q)\right) = \left\langle \frac{\partial^2\phi}{\partial u\partial v}(q), (N \circ \phi)(q) \right\rangle.$$

Since $\frac{\partial^2\phi}{\partial v\partial u}(q) = \frac{\partial^2\phi}{\partial u\partial v}(q)$, we conclude that the two expressions are equal. $\square$

Remark 3.1.3 — The proof of proposition 3.1.2 made use of the fact that second partial derivatives commute, i.e., that the given chart $\phi: U \rightarrow \mathbb{R}^3$ at p satisfies $\frac{\partial^2\phi}{\partial v\partial u} = \frac{\partial^2\phi}{\partial u\partial v}$. Thus at a minimum our charts (which we have assumed to be smooth) need to have continuous second derivatives from now on.

Since the second fundamental form

$$A(X, Y) = \langle X, -dN_p(Y) \rangle$$

is a symmetric bilinear form on T_pS , it can be represented in any basis of T_pS by a symmetric matrix. The spectral theorem from linear algebra tells us that this matrix is then diagonalizable, with real eigenvalues whose corresponding eigenspaces are orthogonal to each other. We extract an orthonormal basis of T_pS consisting of eigenvectors, and label them as follows.

Definition 3.1.4. Given a regular surface $S \subset \mathbb{R}^3$ and a point $p \in S$, there are an orthonormal basis (X_1, X_2) of T_pS and scalars $\lambda_1, \lambda_2 \in \mathbb{R}$ such that

$$dN_p(X_1) = -\lambda_1 X_1, \quad dN_p(X_2) = -\lambda_2 X_2.$$

Then X_1, X_2 are called the **principal directions** at p , and λ_1, λ_2 the **principal curvatures** at p .

Note the minus signs in the definition: the second fundamental form is represented by the matrix $-dN_p$, not by dN_p itself. We compute the second fundamental form in terms of the principal directions at p as follows:

$$\left. \begin{aligned} A(X_1, X_1) &= \langle X_1, -dN_p(X_1) \rangle = \langle X_1, \lambda_1 X_1 \rangle \\ A(X_1, X_2) &= \langle X_1, -dN_p(X_2) \rangle = \langle X_1, \lambda_2 X_2 \rangle \\ A(X_2, X_2) &= \langle X_2, -dN_p(X_2) \rangle = \langle X_2, \lambda_2 X_2 \rangle \end{aligned} \right\} \implies \begin{aligned} A(X_1, X_1) &= \lambda_1 \\ A(X_1, X_2) &= 0 \\ A(X_2, X_2) &= \lambda_2 \end{aligned}.$$

Therefore, in this basis, A has the matrix

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

Lemma 3.1.5

If S is a regular surface with principal directions X_1, X_2 at p and has principal curvatures $\lambda_1(p) \leq \lambda_2(p)$, then

$$\begin{aligned}\lambda_1(p) &= \min\{A(X, X) \mid X \in T_p S, |X| = 1\}, \\ \lambda_2(p) &= \max\{A(X, X) \mid X \in T_p S, |X| = 1\}.\end{aligned}$$

Proof. Since the principal directions X_1, X_2 form an orthonormal basis of $T_p S$, any other unit vector $X \in T_p S$ can be written as

$$X = c_1 X_1 + c_2 X_2, \quad c_1^2 + c_2^2 = 1.$$

Then, as A is symmetric and bilinear, we have that

$$\begin{aligned}A(X, X) &= A(c_1 X_1 + c_2 X_2, c_1 X_1 + c_2 X_2) \\ &= c_1^2 A(X_1, X_1) + 2c_1 c_2 A(X_1, X_2) + c_2^2 A(X_2, X_2) \\ &= c_1^2 \lambda_1 + c_2^2 \lambda_2.\end{aligned}$$

Now, since $\lambda_2 \geq \lambda_1$, we have

$$A(X, X) = c_1^2 \lambda_1 + c_2^2 \lambda_2 \geq (c_1^2 + c_2^2) \lambda_1 = \lambda_1.$$

Similarly,

$$A(X, X) = c_1^2 \lambda_1 + c_2^2 \lambda_2 \leq (c_1^2 + c_2^2) \lambda_2 = \lambda_2.$$

We achieve equality when $(c_1, c_2) = (\pm 1, 0)$ and $(c_1, c_2) = (0, \pm 1)$, respectively. $\square$

Example 3.1.6

If $S \subset \mathbb{R}^3$ is a sphere of radius $r > 0$, and N is the outward pointing Gauss map, then we computed in example 2.4.7 that $N(p) = \frac{1}{r} \cdot p$ and

$$dN_p = \frac{1}{r} \cdot Id: T_p S \rightarrow T_p S.$$

The second fundamental form on S is then given by

$$A(X, Y) = \langle X, -\frac{1}{r} Y \rangle = -\frac{\langle X, Y \rangle}{r},$$

and both principal curvatures are $-\frac{1}{r}$. If, instead, we had chosen N to be the inward pointing Gauss map $N(p) = -\frac{1}{r} \cdot p$, then we would have had $-dN_p = -\frac{1}{r} \cdot id$, and

$$A(X, Y) = +\frac{\langle X, Y \rangle}{r},$$

with principal curvatures $+\frac{1}{r}$. Therefore, the sign of the principal curvatures depends on a choice of orientation of S .

For a plane, the computation is simpler.

Example 3.1.7

If $P \subset \mathbb{R}^3$ is a plane, then we computed in example 2.4.6 that $dN_p = 0$ for all $p \in P$, so the principal curvatures are both zero regardless of orientation.

There is a converse to example 3.1.7:

Proposition 3.1.8

If S is a connected surface and its principal curvatures are $\lambda_1 = \lambda_2 = 0$ on all of S , then S is contained in a plane.

Proof. The eigenvalues λ_1, λ_2 of $-dN_p$ are zero, so dN_p is zero for all p , and then since S is connected the Gauss map $N: S \rightarrow \mathbb{S}^2$ must be constant. We write $N(p) = \vec{v}$ for all p , where $\vec{v}$ is a fixed vector in $\mathbb{S}^2$.

Fix $p \in S$, and let $\phi: U \rightarrow \mathbb{R}^3$ be a chart for S at p . We can assume that U is connected by deleting any components that do not contain $\pi^{-1}(p)$. Given any other point $p' \in \phi(U)$, we can use the connectedness of S to find a regular path

$$\gamma: [0, 1] \rightarrow U$$

with $\gamma(0) = \phi^{-1}(p)$ and $\gamma(1) = \phi^{-1}(p')$. Each tangent vector $(\phi \circ \gamma)'(t) \in T_{\phi(\gamma(t))}S$ is orthogonal to $N(\phi(\gamma(t))) = \vec{v}$, so we have

$$\frac{d}{dt} \langle \phi(\gamma(t)), \vec{v} \rangle = \langle (\phi \circ \gamma)'(t), \vec{v} \rangle = 0.$$

$\langle \phi(\gamma(t)), \vec{v} \rangle$ is constant, so we evaluate it at $t = 1$ and $t = 0$ to attain that $\langle p', \vec{v} \rangle = \langle p, \vec{v} \rangle$. This holds for all $p' \in \phi(U)$, since p' was chosen arbitrarily. We've shown for our given chart $\phi: U \rightarrow \mathbb{R}^3$ at p that the entire image $\phi(U) \subset S$ lies in the plane

$$P = \{x \in \mathbb{R}^3 \mid \langle x - p, \vec{v} \rangle = 0\}.$$

However, since S is connected, we can apply the same argument to see that S is contained in the same plane P : given any other $p' \in S$, we take a regular path in S from p to p' , cover this path by finitely many charts (note that this path is compact even if S is not), and argue as above that each of them must be contained in P . $\square$

Remark 3.1.9 — Note that the statement above is not true if we only ask for $k \equiv 0$, where k is the Gaussian curvature.

We can interpret the second fundamental form at $p \in S$ geometrically in terms of curves in S that pass through p .

Definition 3.1.10. Let $C \subset S$ be a regular curve, with geodesic curvature k at a point $p \in C$. Let $N(p)$ be the unit normal vector to S (at p) and let $n(p)$ be the principal normal vector to C (at p). Then, the **normal curvature** to C at p is

$$k_n = k \cos(\theta),$$

where θ is the angle between $N(p)$ and $n(p)$. In other words, k_n is the length of the projection of $k \cdot n(p)$ onto the line through $N(p)$.

Remark 3.1.11 — If $\alpha(t)$ is a regular curve in S parametrized by arclength, we have $k_n(\alpha(t)) = \langle k(t)n(t), N(\alpha(t)) \rangle$, as

$$\langle k(t)n(t), N(\alpha(t)) \rangle = |n(t)||N(\alpha(t))| \cos(\theta) = \cos(\theta).$$

Remark 3.1.12 — It is useful to recall that we have the following inner product and cross product rules:

$$\begin{aligned} \langle v, w \rangle &= |v||w| \cos(\theta) \\ |v \times w| &= |v||w| \sin(\theta) \end{aligned}$$

Proposition 3.1.13

If C is a curve in S with unit tangent vector v at p , then

$$A(v, v) = k_n(p).$$

Proof. Let $\alpha: (-\epsilon, \epsilon) \rightarrow S$ be a parametrization by arclength of part of C , with $\alpha(0) = p$ and $\alpha'(0) = v$. We differentiate the relations

$$\langle \alpha'(t), N(\alpha(t)) \rangle = 0,$$

which holds because $\alpha'(t)$ is tangent to S at $\alpha(t)$ whereas $N(\alpha(t))$ is normal to it, to get

$$\langle \alpha''(t), N(\alpha(t)) \rangle + \langle \alpha'(t), dN_{\alpha(t)}(\alpha'(t)) \rangle = 0.$$

Setting $t = 0$, we attain that

$$\langle k \cdot n(p), N(p) \rangle + \langle v, dN_p(v) \rangle = 0,$$

so then

$$A(v, v) = -\langle v, dN_p(v) \rangle = \langle k \cdot n(p), N(p) \rangle = k \cos(\theta) = k_n(p).$$

□

We can combine this with lemma 3.1.5 to conclude the following:

Proposition 3.1.14

The principal curvatures to S at p are equal to the minimum and maximum values of the normal curvatures at p , taken over all regular curves in S that pass through p .

When the principal curvatures at p are equal, this gives a strong constraint on the geometry of S near p . We formalize this with the following definition.

Definition 3.1.15. A point $p \in S$ is **umbilical** if the principal curvatures λ_1 and λ_2 at p are equal.

Example 3.1.16

Every point on a plane is umbilical with $\lambda_1 = \lambda_2 = 0$, for all points. We also have the converse; if S has every point is umbilical with $\lambda_1 = \lambda_2 = 0$, then S is contained in a plane.

Theorem 3.1.17

If S is connected and all points of S are umbilical, then S is contained in either a plane ($\lambda_1 = \lambda_2 = 0$) or a sphere.

Proof. Define $\lambda: S \rightarrow \mathbb{R}$ by setting $\lambda(p) = \lambda_1(p) = \lambda_2(p)$ for all $p \in S$. Then since $\lambda_1(p)$ and $\lambda_2(p)$ are the eigenvalues of $-dN_p$, which is diagonalizable, we have

$$dN_p = -\lambda(p) \cdot Id_{T_p S},$$

for all $p \in S$. Fix a point $p \in S$ and a chart $\phi: U \rightarrow \mathbb{R}^3$ for S at p whose domain U is connected. According to proposition 2.4.8, we have

$$\frac{\partial(N \circ \phi)}{\partial u}(q) = dN_{\phi(q)} \left(\frac{\partial \phi}{\partial u}(q) \right) = -\lambda(\phi(q)) \cdot \frac{\partial \phi}{\partial u}(q)$$

for all $q \in U$, and similarly for the v -derivative: we write these relations as

$$\frac{\partial(N \circ \phi)}{\partial u} = -(\lambda \circ \phi) \frac{\partial \phi}{\partial u}, \quad \frac{\partial(N \circ \phi)}{\partial v} = -(\lambda \circ \phi) \frac{\partial \phi}{\partial v}.$$

Differentiating the first of these with respect to v give

$$\frac{\partial^2(N \circ \phi)}{\partial v \partial u} = -\frac{\partial(\lambda \circ \phi)}{\partial v} \cdot \frac{\partial \phi}{\partial u} - (\lambda \circ \phi) \frac{\partial^2 \phi}{\partial v \partial u},$$

while differentiating the second with respect to u gives

$$\frac{\partial^2(N \circ \phi)}{\partial u \partial v} = -\frac{\partial(\lambda \circ \phi)}{\partial u} \cdot \frac{\partial \phi}{\partial v} - (\lambda \circ \phi) \frac{\partial^2 \phi}{\partial u \partial v}.$$

Combining these equations with the facts that the second partial derivatives of $N \circ \phi$ and of ϕ commute, we see that

$$\begin{aligned} -\frac{\partial(\lambda \circ \phi)}{\partial v} \cdot \frac{\partial \phi}{\partial u} &= \frac{\partial^2(N \circ \phi)}{\partial v \partial u} + (\lambda \circ \phi) \frac{\partial^2 \phi}{\partial v \partial u} \\ &= \frac{\partial^2(N \circ \phi)}{\partial u \partial v} + (\lambda \circ \phi) \frac{\partial^2 \phi}{\partial u \partial v} \\ &= -\frac{\partial(\lambda \circ \phi)}{\partial u} \cdot \frac{\partial \phi}{\partial v}. \end{aligned}$$

However the vectors $\frac{\partial \phi}{\partial u}$ and $\frac{\partial \phi}{\partial v}$ are linearly independent since they span the tangent planes to S , so their coefficients must be zero, meaning that

$$\frac{\partial(\lambda \circ \phi)}{\partial u} = \frac{\partial(\lambda \circ \phi)}{\partial v} = 0.$$

In other words, the composition $\lambda \circ \phi: U \rightarrow \mathbb{R}$ is constant.

Now since S is connected and covered by charts $\phi: U \rightarrow \mathbb{R}^3$ as above, with λ constant on each $\phi(U)$, we conclude that λ is constant on S , say $\lambda(p) = \lambda_0$ for all $p \in S$. If $\lambda_0 = 0$ then we saw in proposition 3.1.8 that S is contained in a plane, so from now on we will assume that $\lambda_0 \neq 0$. If λ_0 is negative then reversing the orientation of S will make it positive, so we may as well assume that $\lambda_0 > 0$. On the charts $\phi: U \rightarrow \mathbb{R}^3$ we studied above, we showed that

$$\frac{\partial(N \circ \phi)}{\partial u} = -\lambda_0 \frac{\partial \phi}{\partial u}, \quad \frac{\partial(N \circ \phi)}{\partial v} = -\lambda_0 \frac{\partial \phi}{\partial v}.$$

This implies that

$$\frac{\partial}{\partial u} ((N \circ \phi) + \lambda_0 \phi) + \frac{\partial}{\partial v} ((N \circ \phi) + \lambda_0 \phi) = 0.$$

This means that the map

$$(N \circ \phi) + \lambda_0 \phi: U \rightarrow \mathbb{R}^3$$

is constant, so we let $\vec{c}$ denote its common value on U , and let $\vec{c}_0 = \frac{1}{\lambda_0} \vec{c}$. Then,

$$|\phi - \vec{c}_0| = \frac{|\lambda_0 \phi - \vec{c}|}{\lambda_0} = \frac{|-(N \circ \phi)|}{\lambda_0} = \frac{1}{\lambda_0},$$

so $\phi(U)$ is contained in a sphere of radius $\frac{1}{\lambda_0}$ centered at $\vec{c}_0$. By connectedness the same is true of all of S , and this completes the proof. $\square$

§3.2 Gaussian Curvature and Mean Curvature

The principal curvatures λ_1, λ_2 at a point p on a surface S depend on the choice of orientation: if we replace the Gauss map $N: S \rightarrow \mathbb{S}^2$ with $-N$, then the principal curvatures change sign to $-\lambda_1$ and $-\lambda_2$. However, their product does not change, and hence gives us a quantity that doesn't depend on orientation at all.

Definition 3.2.1. Let $p \in S$ be a point on a regular surface, whose principal curvatures are λ_1 and λ_2 . Then the **Gaussian curvature** of S at p is defined by

$$K(p) = \lambda_1 \lambda_2 = \det(-dN_p) = \det(dN_p),$$

and the **mean curvature** at p is

$$H(p) = \frac{\lambda_1 + \lambda_2}{2} = -\frac{1}{2} \text{Tr}(dN_p).$$

Remark 3.2.2 — We note that unlike Gaussian curvature, the mean curvature depends on the orientation of S (up to sign). (On the other hand, the points where $H = 0$ do not depend on orientation.)

Example 3.2.3

By example 2.4.6, the Gauss map of a plane satisfies $dN_p = 0$ everywhere, so its Gaussian and mean curvatures are given by $K = H = 0$.

Example 3.2.4

In example 2.4.7, we computed that a sphere S of radius r can be oriented so its Gauss map satisfies

$$dN_p = -\frac{1}{r} \cdot Id,$$

with principal curvatures $\lambda_1 = \lambda_2 = \frac{1}{r}$, as seen in example 3.1.6. Thus we have

$$K = \frac{1}{r^2}, \quad H = \frac{1}{r}$$

at all points of S . If we choose the opposite orientation, then we have $K = \frac{1}{r^2}$ and $H = -\frac{1}{r}$ instead.

We can interpret the Gaussian curvature in terms of a chart $\phi: U \rightarrow \mathbb{R}^3$ for S at the point p , but this will first require us to compute the second fundamental form in terms of ϕ . We arrange the chart so that $q = \phi^{-1}(p)$ has coordinates $(0, 0)$, and define a path by $\gamma(t) = \phi(at, bt)$ near $t = 0$. Then just as in the proof of proposition 3.1.13, we can differentiate

$$\langle \gamma'(t), N(\gamma(t)) \rangle = 0$$

to attain

$$\langle \gamma''(t), N(\gamma(t)) \rangle + \langle \gamma'(t), dN_{\gamma(t)}(\gamma'(t)) \rangle = 0.$$

This can be rewritten in terms of the second fundamental form as

$$A(\gamma'(t), \gamma'(t)) = \langle \gamma''(t), N(\gamma(t)) \rangle.$$

At $t = 0$, we have

$$\begin{aligned} \gamma'(0) &= a \frac{\partial \phi}{\partial u}(q) + b \frac{\partial \phi}{\partial v}(q) \\ \gamma''(0) &= a^2 \frac{\partial^2 \phi}{\partial u^2}(q) + 2ab \frac{\partial^2 \phi}{\partial u \partial v}(q) + b^2 \frac{\partial^2 \phi}{\partial v^2}(q), \end{aligned}$$

which we can substitute into the above to get

$$\begin{aligned} A \left(a \frac{\partial \phi}{\partial u}(q) + b \frac{\partial \phi}{\partial v}(q), a \frac{\partial \phi}{\partial u}(q) + b \frac{\partial \phi}{\partial v}(q) \right) \\ = \left\langle a^2 \frac{\partial^2 \phi}{\partial u^2}(q) + 2ab \frac{\partial^2 \phi}{\partial u \partial v}(q) + b^2 \frac{\partial^2 \phi}{\partial v^2}(q), N(p) \right\rangle. \end{aligned}$$

Proposition 3.2.5

Let p be a point in a regular surface S . If the Gaussian curvature at p satisfies $K(p) > 0$, then there is a neighborhood of p on which all points of S lie on the same side of the tangent plane $T_p S$. If instead $K(p) < 0$, then any neighborhood of p has points of S on both sides of $T_p S$.

Proof. Fix a chart $\phi: U \rightarrow \mathbb{R}^3$ for S at p , with $q = (0, 0) = \phi^{-1}(p)$ as above, and consider the second-order Taylor series for ϕ at p . This has the form

$$\begin{aligned} \phi(u, v) = p + \underbrace{\left(\frac{\partial \phi}{\partial u}(q) \cdot u + \frac{\partial \phi}{\partial v}(q) \cdot v \right)}_{\in T_p S} \\ + \frac{1}{2} \left(\frac{\partial^2 \phi}{\partial u^2}(q) \cdot u^2 + 2 \frac{\partial^2 \phi}{\partial u \partial v}(q) \cdot u \cdot v + \frac{\partial^2 \phi}{\partial v^2}(q) \cdot v^2 \right) + R(u, v), \end{aligned}$$

where the error term $R(u, v)$ satisfies

$$\lim_{(u,v) \rightarrow (0,0)} \frac{R(u, v)}{u^2 + v^2} = 0.$$

Since $T_p S$ is orthogonal to $N(p)$, it follows that

$$\begin{aligned} \langle \phi(u, v) - p, N(p) \rangle = \frac{1}{2} \left\langle u^2 \frac{\partial^2 \phi}{\partial u^2}(q) + 2uv \frac{\partial^2 \phi}{\partial u \partial v}(q) + v^2 \frac{\partial^2 \phi}{\partial v^2}(q), N(p) \right\rangle \\ + \langle R(u, v), N(p) \rangle, \end{aligned}$$

and we recognize the first term on the right from our above computation of A :

$$\langle \phi(u, v) - p, N(p) \rangle = \frac{1}{2} A(w, w) + \langle R(u, v), N(p) \rangle,$$

where $w = u \frac{\partial \phi}{\partial u}(q) + v \frac{\partial \phi}{\partial v}(q) \in T_p S$. The error term on the right satisfies

$$\left| \frac{\langle R(u, v), N(p) \rangle}{u^2 + v^2} \right| \leq \frac{|R(u, v)|}{u^2 + v^2},$$

so $\left| \frac{\langle R(u, v), N(p) \rangle}{u^2 + v^2} \right| \rightarrow 0$ as $(u, v) \rightarrow (0, 0)$.

Now suppose S has principal curvatures $\lambda_1 \leq \lambda_2$ at p , with corresponding principal directions X_1 and X_2 . If $K(p) > 0$ then either $\lambda_2 \geq \lambda_1 > 0$, and by lemma 3.1.5 we have

$$A(w, w) \geq \lambda_1 |w|^2 > 0,$$

for all $w \neq 0$; or $\lambda_1 \leq \lambda_2 < 0$, and then

$$A(w, w) \leq \lambda_2 |w|^2 < 0,$$

for all $w \neq 0$. Note here, we used that for nonzero w , $A(w, w) = |w|^2 A(\frac{w}{|w|}, \frac{w}{|w|})$. In either case, we have $|A(w, w)| > |\langle R(u, v), N(p) \rangle|$ for all sufficiently small nonzero (u, v) , and so $\phi(u, v) - p$ lies entirely above or entirely below the plane $(N(p))^\perp = T_p S$ near p .

In the remaining case, where $K(p) < 0$, we must have $\lambda_1 < 0 < \lambda_2$, so by the same reasoning the quantity $\langle \phi(u, v) - p, N(p) \rangle$ is negative for small multiples of X_1 and positive for small multiples of X_2 . This means that in any neighborhood of p there are points of S on either side of $T_p S$, as claimed. $\square$

The computation above also suggests the following interpretation of the principal curvatures at a point. Suppose we have a chart $\phi: U \rightarrow \mathbb{R}^3$ for S at p , and we apply a

rigid motion to $\mathbb{R}^3$ and a translation to U to arrange that $p = (0, 0, 0)$ is the image of $q = (0, 0) \in U$, and that the principal directions at $p = (0, 0, 0)$ are

$$X_1 = \frac{\partial \phi}{\partial u}(0, 0) = (1, 0, 0), \quad X_2 = \frac{\partial \phi}{\partial v}(0, 0) = (0, 1, 0),$$

with corresponding principal curvatures λ_1 and λ_2 . Then the Taylor series for ϕ has the form

$$\phi(u, v) = (u, v, 0) + \underbrace{\frac{1}{2} \left(u^2 \frac{\partial^2 \phi}{\partial u^2}(q) + 2uv \frac{\partial^2 \phi}{\partial u \partial v}(q) + v^2 \frac{\partial^2 \phi}{\partial v^2}(q) \right)}_{=Q(u, v)} + \dots,$$

and we have seen from above that the quadratic term $Q(u, v)$ satisfies

$$\begin{aligned} \langle Q(u, v), N(p) \rangle &= A(uX_1 + vX_2, uX_1 + vX_2) \\ &= u^2 \lambda_1 + v^2 \lambda_2. \end{aligned}$$

Now we observe that the unit normal at $p = (0, 0, 0)$ is given in terms of ϕ by

$$N(p) = \frac{\frac{\partial \phi}{\partial u}(q) \times \frac{\partial \phi}{\partial v}(q)}{\left| \frac{\partial \phi}{\partial u}(q) \times \frac{\partial \phi}{\partial v}(q) \right|} = (0, 0, 1),$$

which tells us that $\langle Q(u, v), N(p) \rangle$ is the z -coordinate of $Q(u, v)$ and we have

$$\phi(u, v) = \left(u, v, \frac{1}{2}(\lambda_1 u^2 + \lambda_2 v^2) \right) + \dots,$$

where the omitted terms are of strictly higher order in each coordinate. In other words, the surface S is approximated near p by the graph of $z = \frac{1}{2}(\lambda_1 x^2 + \lambda_2 y^2)$. This approximation suggests that we should divide points $p \in S$ into several types based on the principal curvatures at p , or equivalently on the Gaussian and mean curvatures at p .

Definition 3.2.6. A point p on a regular surface S is:

- **elliptic** if $K(p) > 0$ (λ_1, λ_2 have the same sign).
- **hyperbolic** if $K(p) < 0$ (λ_1, λ_2 have opposite signs).
- **parabolic** if $K(p) = 0$ and $H(p) \neq 0$ (exactly one λ_i is zero).
- **planar** if $K(p) = H(p) = 0$ ($\lambda_1 = \lambda_2 = 0$).

For each type of point, we can find a surface whose points all have that type. But if we restrict our attention to compact surfaces, the only way that every point can have the same type is if all of them are elliptic.

Proposition 3.2.7

Let $S \subset \mathbb{R}^3$ be a compact surface. Then S has an elliptic point.

Proof. Since S is compact, the distance-squared function $F(x, y, z) = x^2 + y^2 + z^2$ on S achieves its maximum at some point $p \in S$. We will show that p is elliptic.

First, if $\alpha: (-\epsilon, \epsilon) \rightarrow S$ is any smooth curve with $\alpha(0) = p$, then the function

$$f(t) = F(\alpha(t)) = \langle \alpha(t), \alpha(t) \rangle$$

has a critical point at $t = 0$, hence

$$0 = f'(0) = 2\langle \alpha(0), \alpha'(0) \rangle = 2\langle p, \alpha'(0) \rangle.$$

We can arrange for $\alpha'(0)$ to be any element of $T_p S$, so $T_p S$ is orthogonal to p . In particular this means that the unit normal $N(p)$ is a multiple of p , so up to changing the Gauss map by an overall sign we can write $N(p) = \frac{p}{|p|}$. Now we pick a unit vector $v \in T_p S$ and suppose that α is parametrized by arclength, with $\alpha'(0) = v$. The critical point of $f(t)$ at $t = 0$ is a local maximum, so $f''(0) \leq 0$. We compute that

$$f''(0) = 2 \underbrace{\langle \alpha'(0), \alpha'(0) \rangle}_{=\langle v, v \rangle=1} + 2 \underbrace{\langle \alpha(0), \alpha''(0) \rangle}_{=p},$$

and since this is at most zero we have $\langle p, \alpha''(0) \rangle \leq -1$, or

$$\langle N(p), \alpha''(0) \rangle \leq -\frac{1}{|p|} < 0.$$

The left side is precisely the normal curvature $k_n(p)$ of α at p , which by proposition 3.1.13 is equal to the specialization $A(v, v)$ of the second fundamental form, so we have shown that $A(v, v) < 0$. Since this is true for any unit vector $v \in T_p S$, we conclude by lemma 3.1.5 that both of the principal curvatures $\lambda_1(p), \lambda_2(p)$ are strictly negative, and so

$$K(p) = \lambda_1 \cdot \lambda_2 > 0.$$

□

§3.3 Computing the Gaussian and Mean Curvatures

Computing the curvatures of a surface at a point requires us to unpack a lot of definitions. Fortunately, there are some fairly concise formulas that make this easier.

Notation 3.3.1. From now on if we fix $\phi: U \rightarrow \mathbb{R}^3$ for S at p , we will write $q = \phi^{-1}(p)$ and

$$\phi_u = \frac{\partial \phi}{\partial u}, \quad \phi_v = \frac{\partial \phi}{\partial v}.$$

Theorem 3.3.2

Let $\phi: U \rightarrow S$ be a chart for S at p , with $q = \phi^{-1}(p)$, and define matrices

$$g = \begin{pmatrix} \langle \phi_u(q), \phi_u(q) \rangle & \langle \phi_u(q), \phi_v(q) \rangle \\ \langle \phi_v(q), \phi_u(q) \rangle & \langle \phi_v(q), \phi_v(q) \rangle \end{pmatrix}$$

and

$$A = \begin{pmatrix} A(\phi_u(q), \phi_u(q)) & A(\phi_u(q), \phi_v(q)) \\ A(\phi_v(q), \phi_u(q)) & A(\phi_v(q), \phi_v(q)) \end{pmatrix}.$$

If we set $\sigma = g^{-1}A$, then the Gaussian and mean curvatures at p satisfy

$$K(p) = \det(\sigma) = \frac{\det(A)}{\det(g)}, \quad H(p) = \frac{1}{2} \text{Tr}(\sigma).$$

Proof. By definition, we have $K(p) = \det(-dN_p)$ and $H(p) = \frac{1}{2}\text{Tr}(-dN_p)$, so it will suffice to compute these in terms of any basis of T_pS . We fix an orthonormal basis of T_pS , and express $dN_p: T_pS \rightarrow T_pS$ and the tangent vectors $\phi_u(q)$ and $\phi_v(q)$ in these coordinates. Then, the inner product of two column vectors in T_pS can be written as a matrix product $\langle X, Y \rangle = X^T Y$, and the 2×2 matrix

$$B = \begin{pmatrix} | & | \\ \phi_u(q) & \phi_v(q) \\ | & | \end{pmatrix}$$

is invertible because $\phi_u(q)$ and $\phi_v(q)$ form another basis for T_pS , so

$$g = \begin{pmatrix} - & \phi_u(q) & - \\ - & \phi_v(q) & - \end{pmatrix} \begin{pmatrix} | & | \\ \phi_u(q) & \phi_v(q) \\ | & | \end{pmatrix} = B^T B$$

is invertible as well. Therefore, we can write A as a product of 2×2 matrices:

$$\begin{aligned} A &= \begin{pmatrix} - & \phi_u(q) & - \\ - & \phi_v(q) & - \end{pmatrix} \begin{pmatrix} | & | \\ -dN_p(\phi_u) & -dN_p(\phi_v) \\ | & | \end{pmatrix} \\ &= \begin{pmatrix} - & \phi_u(q) & - \\ - & \phi_v(q) & - \end{pmatrix} \cdot (-dN_p) \cdot \begin{pmatrix} | & | \\ \phi_u(q) & \phi_v(q) \\ | & | \end{pmatrix} \\ &= B^T (-dN_p) B \\ &= B^T B \cdot B^{-1} (-dN_p) B. \end{aligned}$$

Since $g = B^T B$ is invertible, we can rewrite this to get

$$\sigma = g^{-1} A = B^{-1} (-dN_p) B.$$

Therefore, $-dN_p$ is conjugate to σ , so they have determinant and trace, and this means that

$$\begin{aligned} K(p) &= \det(-dN_p) = \det(\sigma) \\ H(p) &= \frac{1}{2}\text{Tr}(-dN_p) = \frac{1}{2}\text{Tr}(\sigma) \end{aligned}$$

as claimed. □

Remark 3.3.3 — We note that $\det(g) = |\phi_u(q)|^2 \cdot |\phi_v(q)|^2 - \langle \phi_u(q), \phi_v(q) \rangle^2$. By the Cauchy-Schwarz inequality, this tells us that this is nonzero when one is not a scalar multiple of the other.

Remark 3.3.4 — If we choose any two variables $x, y \in \{u, v\}$, then we know that $\frac{\partial \phi}{\partial y}(q) \in T_pS$ is orthogonal to $N(p) = N(\phi(q))$, so we differentiate

$$\left\langle \frac{\partial \phi}{\partial y}(q), N(\phi(q)) \right\rangle = 0$$

with respect to x to get

$$\left\langle \frac{\partial^2 \phi}{\partial x \partial y}, N(p) \right\rangle + \left\langle \frac{\partial \phi}{\partial y}(q), \underbrace{\frac{\partial(N \circ \phi)}{\partial x}(q)}_{=dN_p(\frac{\partial \phi}{\partial x})} \right\rangle = 0,$$

or

$$\left\langle \frac{\partial^2 \phi}{\partial x \partial y}, N(p) \right\rangle = A \left(\frac{\partial \phi}{\partial y}(q), \frac{\partial \phi}{\partial x}(q) \right).$$

Thus, we can rewrite the matrix A in theorem 3.3.2 in the form

$$A = \begin{pmatrix} \langle \phi_{uu}(q), N(p) \rangle & \langle \phi_{uv}(q), N(p) \rangle \\ \langle \phi_{vu}(q), N(p) \rangle & \langle \phi_{vv}(q), N(p) \rangle \end{pmatrix},$$

where the double subscripts denote the second partial derivatives.

Example 3.3.5

We use theorem 3.3.2 to compute the curvature of the saddle $S = \{z = x^2 - y^2\}$, using the chart $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by

$$\phi(u, v) = (u, v, u^2 - v^2).$$

Solution. The first partial derivatives of the chart ϕ are given by $\phi_u = (1, 0, 2u)$ and $\phi_v = (0, 1, -2v)$, so we have

$$g = \begin{pmatrix} \langle \phi_u(q), \phi_u(q) \rangle & \langle \phi_u(q), \phi_v(q) \rangle \\ \langle \phi_v(q), \phi_u(q) \rangle & \langle \phi_v(q), \phi_v(q) \rangle \end{pmatrix} = \begin{pmatrix} 1 + 4u^2 & -4uv \\ -4uv & 1 + 4v^2 \end{pmatrix}.$$

These vectors have cross product

$$\phi_u \times \phi_v = \det \begin{pmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2u \\ 0 & 1 & -2v \end{pmatrix} = (-2u, 2v, 1)$$

at $\phi(u, v)$, so the Gauss map is

$$N(\phi(u, v)) = \frac{(-2u, 2v, 1)}{|(-2u, 2v, 1)|} = \frac{(-2u, 2v, 1)}{\sqrt{1 + 4u^2 + 4v^2}}.$$

The second partial derivatives of ϕ are

$$\phi_{uu} = (0, 0, 2), \quad \phi_{uv} = \phi_{vu} = (0, 0, 0), \quad \phi_{vv} = (0, 0, -2),$$

so according to remark 3.3.4, we have

$$A = \begin{pmatrix} \langle \phi_{uu}, N \rangle & \langle \phi_{uv}, N \rangle \\ \langle \phi_{vu}, N \rangle & \langle \phi_{vv}, N \rangle \end{pmatrix} = \frac{1}{\sqrt{1 + 4u^2 + 4v^2}} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}.$$

Thus, we compute

$$\begin{aligned} \sigma &= g^{-1}A = \frac{1}{1 + 4u^2 + 4v^2} \begin{pmatrix} 1 + 4v^2 & 4uv \\ 4uv & 1 + 4u^2 \end{pmatrix} \cdot \frac{1}{\sqrt{1 + 4u^2 + 4v^2}} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} \\ &= \frac{2}{(1 + 4u^2 + 4v^2)^{\frac{3}{2}}} \begin{pmatrix} 1 + 4v^2 & -4uv \\ 4uv & -(1 + 4u^2) \end{pmatrix}. \end{aligned}$$

Thus, the Gaussian and mean curvatures at $p = \phi(u, v) = (u, v, u^2 - v^2)$ are

$$\begin{aligned} K(p) = \det(\sigma) &= \frac{4}{(1 + 4u^2 + 4v^2)^3} \left(-(1 + 4u^2)(1 + 4v^2) + 16u^2v^2 \right) \\ &= -\frac{4}{(1 + 4u^2 + 4v^2)^2} \end{aligned}$$

and

$$H(p) = \frac{1}{2} \text{Tr}(\sigma) = \frac{4(v^2 - u^2)}{(1 + 4u^2 + 4v^2)^{\frac{3}{2}}}.$$

□

Remark 3.3.6 — Note that we don't have any elliptic points, as the saddle is not compact.

Example 3.3.7

A **helicoid** is the surface obtained from the helix C of example 1.3.8 by drawing for each point $p = (x, y, z) \in C$ the horizontal line $\{(tx, ty, z) \mid t \in \mathbb{R}\}$ from p to the point $(0, 0, z)$ at the same height on the z -axis. It has parametrization given by

$$\begin{aligned}\phi: \mathbb{R}^2 &\rightarrow \mathbb{R}^3 \\ (u, v) &\mapsto (v \cos(u), v \sin(u), \alpha u)\end{aligned}$$

for some constant $\alpha > 0$, and we compute:

$$\begin{aligned}\phi_u &= (-v \sin(u), v \cos(u), \alpha) \\ \phi_v &= (\cos(u), \sin(u), 0) \\ \phi_u \times \phi_v &= (-\alpha \sin(u), \alpha \cos(u), -v)\end{aligned}$$

so the Gauss map is given by

$$N(\phi(u, v)) = \frac{\phi_u \times \phi_v}{|\phi_u \times \phi_v|} = \frac{(-\alpha \sin(u), \alpha \cos(u), -v)}{\sqrt{v^2 + \alpha^2}}.$$

The second partial derivatives of ϕ are given by $\phi_{vv} = (0, 0, 0)$ and

$$\phi_{uu} = (-v \cos(u), -v \sin(u), 0), \quad \phi_{uv} = (-\sin(u), \cos(u), 0),$$

so we have

$$\begin{aligned}g &= \begin{pmatrix} \langle \phi_u, \phi_u \rangle & \langle \phi_u, \phi_v \rangle \\ \langle \phi_v, \phi_u \rangle & \langle \phi_v, \phi_v \rangle \end{pmatrix} = \begin{pmatrix} v^2 + \alpha^2 & 0 \\ 0 & 1 \end{pmatrix} \\ A &= \begin{pmatrix} \langle \phi_{uu}, N \rangle & \langle \phi_{uv}, N \rangle \\ \langle \phi_{vu}, N \rangle & \langle \phi_{vv}, N \rangle \end{pmatrix} = \frac{1}{\sqrt{v^2 + \alpha^2}} \begin{pmatrix} 0 & \alpha \\ \alpha & 0 \end{pmatrix} \\ \sigma &= g^{-1}A = \frac{1}{\sqrt{v^2 + \alpha^2}} \begin{pmatrix} 0 & \frac{\alpha}{v^2 + \alpha^2} \\ \alpha & 0 \end{pmatrix}.\end{aligned}$$

Thus, at $p = \phi(u, v)$ the Gaussian and mean curvatures are given by

$$K(p) = \det(\sigma) = -\frac{\alpha^2}{(v^2 + \alpha^2)^2}, \quad H(p) = \frac{1}{2}\text{Tr}(\sigma) = 0.$$

This tells us that S is a *minimal surface*.

§3.4 The First Fundamental Form

We have seen that the second fundamental form of a surface S determines the curvature of S . In this section we introduce another fundamental form that governs quantities such as arclength.

Definition 3.4.1. Let $S \subset \mathbb{R}^3$ be a regular surface. The **first fundamental form**, or **metric**, at a point $p \in S$ is the symmetric bilinear form

$$g: T_p S \times T_p S \rightarrow \mathbb{R}$$

defined by $g(v, w) = \langle v, w \rangle$.

If $\phi: U \rightarrow S$ is a chart for S at p , with $q = \phi^{-1}(p)$, then theorem 2.2.6 says that $p \in T_p S$ is spanned by $\phi_u(q)$ and $\phi_v(q)$. In this basis the first fundamental form can be written

$$\begin{aligned} g(a\phi_u + b\phi_v, c\phi_u + d\phi_v) &= ac\langle\phi_u, \phi_u\rangle + (ad + bc)\langle\phi_u, \phi_v\rangle + bd\langle\phi_v, \phi_v\rangle \\ &= \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} \langle\phi_u, \phi_u\rangle & \langle\phi_u, \phi_v\rangle \\ \langle\phi_v, \phi_u\rangle & \langle\phi_v, \phi_v\rangle \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix}, \end{aligned}$$

so that it is represented by the matrix

$$\begin{pmatrix} \langle\phi_u, \phi_u\rangle & \langle\phi_u, \phi_v\rangle \\ \langle\phi_v, \phi_u\rangle & \langle\phi_v, \phi_v\rangle \end{pmatrix}$$

that we called g in theorem 3.3.2 to compute the Gaussian and mean curvatures of S in terms of $\sigma = g^{-1}A$. To see that the metric of S determines the arclength of curves in S , we can take a curve $\alpha: [a, b] \rightarrow S$ contained in the image $\phi(U)$ of some chart; following lemma 2.2.5, we can suppose that it has the form

$$\alpha(t) = \phi(u(t), v(t))$$

for some smooth functions $u, v: [a, b] \rightarrow \mathbb{R}$. Viewed as a curve in $\mathbb{R}^3$, we know that α has arclength

$$\begin{aligned} L(\alpha) &= \int_a^b |\alpha'(t)| dt = \int_a^b |\phi_u(\alpha(t))u'(t) + \phi_v(\alpha(t))v'(t)| dt \\ &= \int_a^b \sqrt{\begin{pmatrix} u'(t) & v'(t) \end{pmatrix} g \begin{pmatrix} u'(t) \\ v'(t) \end{pmatrix}} dt, \end{aligned}$$

where g is the matrix from above. This means that if a map between surfaces preserves the first fundamental form, it will also preserve the lengths of curves in these surfaces.

Definition 3.4.2. A smooth map $F: S_1 \rightarrow S_2$ between surfaces is a **local isometry** if it preserves the first fundamental form, meaning that

$$\langle dF_p(X), dF_p(Y) \rangle = \langle X, Y \rangle$$

for all $p \in S_1$ and $X, Y \in T_p S_1$. If F is also bijective, then we call it an **isometry**.

We note that if F is a local isometry, then each $dF_p: T_p S_1 \rightarrow T_{F(p)} S_2$ must be injective: if $dF_p(X) = 0$ for some tangent vector $X \in T_p S_1$ then

$$\langle X, X \rangle = \langle dF_p(X), dF_p(X) \rangle = \langle 0, 0 \rangle = 0,$$

so X must be zero as well. Now since dF_p is an injective map between two-dimensional vector spaces, it must actually be an isomorphism, and so proposition 2.3.9 tells us that F must be a diffeomorphism from some neighborhood of $p \in S_1$ to some neighborhood of $F(p) \in S_2$. In other words, a local isometry is necessarily a local diffeomorphism.

Example 3.4.3

Let $S \subset \mathbb{R}^3$ be a regular surface, and fix a matrix $M \in SO(3)$, so that $M^T M = I$. We define

$$S' = M \cdot S = \{M \cdot p \mid p \in S\},$$

and a map $F: S \rightarrow S'$ be $F(p) = M \cdot p$. Then for any $p \in S$ and $v \in T_p S$, we have $dF_p(v) = M \cdot v$, and so

$$\begin{aligned} \langle dF_p(X), dF_p(Y) \rangle &= \langle M \cdot X, M \cdot Y \rangle \\ &= (M \cdot X)^T (M \cdot Y) \\ &= X^T (M^T M) Y \\ &= X^T Y \\ &= \langle X, Y \rangle. \end{aligned}$$

This means that F is a local isometry, and in fact it is an isometry since it has inverse

$$F^{-1}(p) = B^{-1} \cdot p = B^T \cdot p.$$

Lemma 3.4.4

A map $F: S_1 \rightarrow S_2$ is a local isometry if and only if it preserves the lengths of tangent vectors, meaning that

$$|dF_p(X)| = |X|$$

for all $p \in S_1$ and $X \in T_p(S_1)$.

Proof. Certainly, if F is a local isometry, then it preserves lengths, because

$$|dF_p(X)| = \sqrt{\langle dF_p(X), dF_p(X) \rangle} = \sqrt{\langle X, X \rangle} = |X|,$$

so we only need to prove the converse. To do so, we use a trick to write the usual inner product on $\mathbb{R}^3$ in terms of lengths: we have

$$\langle X + Y, X + Y \rangle = \langle X, X \rangle + 2\langle X, Y \rangle + \langle Y, Y \rangle,$$

which upon rearranging becomes

$$\langle X, Y \rangle = \frac{\langle X + Y, X + Y \rangle - \langle X, X \rangle - \langle Y, Y \rangle}{2},$$

or simply

$$\langle X, Y \rangle = \frac{1}{2}(|X + Y|^2 - |X|^2 - |Y|^2).$$

Now, if dF_p preserves lengths for all p , then this immediately implies that it preserves inner products $\langle X, Y \rangle$, so F is a local isometry as claimed. $\square$

Example 3.4.5

Let $S_1 = \{z = 0\}$ be the xy -plane, and let $S_2 = \{x^2 + y^2 = 1\}$ be a cylinder. We define a map $F: S_1 \rightarrow S_2$ by

$$F(u, v, 0) = (\cos(u), \sin(u), v).$$

At $p = (u, v, 0) \in S_1$, a tangent vector $X = (a, b, 0)$ can be written as $X = \alpha'(0)$ where

$$\alpha(t) = (u + ta, v + tb, 0).$$

This has length squared equal to $\langle X, X \rangle = a^2 + b^2$. We apply dF_p to X , computing that

$$\begin{aligned} dF_p(X) &= (F \circ \alpha)'(0) = \left. \frac{d}{dt}(\cos(u + ta), \sin(u + ta), v + tb) \right|_{t=0} \\ &= (-a \sin(u + ta), a \cos(u + ta), b) \big|_{t=0} \\ &= (-a \sin(u), a \cos(u), b) \end{aligned}$$

and so

$$\begin{aligned} \langle dF_p(X), dF_p(X) \rangle &= a^2(\sin^2(u) + \cos^2(u)) + b^2 \\ &= a^2 + b^2 \\ &= \langle X, X \rangle. \end{aligned}$$

Since F preserves lengths of tangent vectors, lemma 3.4.4 says that it is a local isometry. It is not an isometry, however, because it is not injective: we have $F(2\pi k, 0, 0) = (1, 0, 0)$ for all $k \in \mathbb{Z}$.

Proposition 3.4.6

A smooth map $F: S_1 \rightarrow S_2$ is a local isometry if and only if it preserves the lengths of curves, meaning that for all curves $\alpha: [a, b] \rightarrow S_1$, we have

$$L(\alpha) = L(F \circ \alpha): [a, b] \rightarrow S_2.$$

Proof. Suppose that F is a local isometry. Then $(F \circ \alpha)'(t) = dF_{\alpha(t)}(\alpha'(t))$, so

$$\begin{aligned} L(F \circ \alpha) &= \int_a^b |(F \circ \alpha)'(t)| dt \\ &= \int_a^b |dF_{\alpha(t)}(\alpha'(t))| dt \\ &= \int_a^b \sqrt{\langle dF_{\alpha(t)}(\alpha'(t)), dF_{\alpha(t)}(\alpha'(t)) \rangle} dt \\ &= \int_a^b \sqrt{\langle \alpha'(t), \alpha'(t) \rangle} dt \\ &= \int_a^b |\alpha'(t)| dt = L(\alpha), \end{aligned}$$

where we have used the fact that F is a local isometry to get from the third line to the

fourth.

For the converse, we suppose that F satisfies $L(\alpha) = L(F \circ \alpha)$ for all curves α . Given a curve $\alpha: (-\epsilon, \epsilon) \rightarrow S_1$ with $\alpha(0) = p$ and $\alpha'(0) = v$, we fix some $\delta \in (0, \epsilon)$ and differentiate the relation

$$L(\alpha|_{[-\delta, t]}) = L((F \circ \alpha)|_{[-\delta, t]})$$

at $t = 0$ to attain the following:

$$\begin{aligned} |\alpha'(0)| &= \left. \frac{d}{dt} \int_{[-\delta]}^t |\alpha'(s)| ds \right|_{t=0} \\ &= \left. \frac{d}{dt} \int_{[-\delta]}^t |(F \circ \alpha)'(s)| ds \right|_{t=0} \\ &= |(F \circ \alpha)'(0)| \\ &= |dF_{\alpha(0)}(\alpha'(0))|, \end{aligned}$$

or $|v| = |dF_p(v)|$. Since we can find such a curve α for every $p \in S_1$ and every $v \in T_p S_1$, we conclude that dF_p preserves the lengths of all tangent vectors, so by lemma 3.4.4 it must be a local isometry. $\square$

§3.5 Christoffel Symbols

If we fix a chart $\phi: U \rightarrow \mathbb{R}^3$ for a surface S , then we can use this to compute both the first and second fundamental forms. Letting x_1, x_2 denote the standard coordinates on $U \subset \mathbb{R}^2$, these forms are given by the 2×2 matrices

$$g = \left(\left\langle \frac{\partial \phi}{\partial x_i}, \frac{\partial \phi}{\partial x_j} \right\rangle \right)_{i,j=1,2}, \quad A = \left(\left\langle N, \frac{\partial^2 \phi}{\partial x_i \partial x_j} \right\rangle \right)_{i,j=1,2},$$

where N is the Gauss map. Note that this was not the original definition of A , but we showed it to be equivalent in remark 3.3.4. At any point $p \in \phi(U) \subset S$, theorem 2.2.6 says that $\frac{\partial \phi}{\partial x_1}$ and $\frac{\partial \phi}{\partial x_2}$ form a basis of $T_p S$, and then these vectors plus the unit normal $N(p)$ give a basis of all of $\mathbb{R}^3$. We can thus express any vector at all as a linear combination of these three, and the above expression for A suggests that it would be useful to try this with the second derivatives $\frac{\partial^2 \phi}{\partial x_i \partial x_j}$.

Definition 3.5.1. Let $\phi: U \rightarrow \mathbb{R}^3$ be a chart for the regular surface S . The **Christoffel symbols** Γ_{ij}^k are the coefficients in $\frac{\partial \phi}{\partial x_k}$ in

$$\frac{\partial^2 \phi}{\partial x_i \partial x_j} = \Gamma_{ij}^1 \cdot \frac{\partial \phi}{\partial x_1} + \Gamma_{ij}^2 \cdot \frac{\partial \phi}{\partial x_2} + A_{ij} \cdot N,$$

for each $i, j, k \in \{1, 2\}$.

Remark 3.5.2 — The identity $\frac{\partial^2 \phi}{\partial x_i \partial x_j} = \frac{\partial^2 \phi}{\partial x_j \partial x_i}$ tells us that the Christoffel symbols are symmetric in i and j , meaning that $\Gamma_{ij}^k = \Gamma_{ji}^k$.

Remark 3.5.3 — Since the unit normal N is orthogonal to both $\frac{\partial\phi}{\partial x_1}$ and $\frac{\partial\phi}{\partial x_2}$, taking inner products with N in the definition above provides

$$\left\langle N, \frac{\partial^2\phi}{\partial x_i \partial x_j} \right\rangle = \langle A_{ij}N, N \rangle = A_{ij},$$

so that $A_{ij} = A\left(\frac{\partial\phi}{\partial x_i}, \frac{\partial\phi}{\partial x_j}\right)$ as suggested by the notation.

Example 3.5.4

Let $P = \{z = 0\}$ be the xy -plane, with the chart $\phi(u, v) = (u, v, 0)$. This has normal vector

$$N = \frac{\phi_u \times \phi_v}{|\phi_u \times \phi_v|} = (0, 0, 1),$$

and the second derivatives of ϕ are

$$\frac{\partial^2\phi}{\partial u^2} = \frac{\partial^2\phi}{\partial u \partial v} = \frac{\partial^2\phi}{\partial v^2} = 0,$$

so all the Christoffel symbols for ϕ are zero.

Remark 3.5.5 — A useful way to compute the Christoffel symbols is via taking the inner product with $\frac{\partial\phi}{\partial x_k}$ (for $k = 1$ or 2):

$$\left\langle \frac{\partial^2\phi}{\partial x_i \partial x_j}, \frac{\partial\phi}{\partial x_k} \right\rangle = \Gamma_{ij}^1 g_{1k} + \Gamma_{ij}^2 g_{2k}.$$

This then gives the following matrix equation:

$$\begin{pmatrix} \langle \phi_{x_i x_j}, \phi_{x_1} \rangle \\ \langle \phi_{x_i x_j}, \phi_{x_2} \rangle \end{pmatrix} = g \begin{pmatrix} \Gamma_{ij}^1 \\ \Gamma_{ij}^2 \end{pmatrix}.$$

Proposition 3.5.6

The Christoffel symbols Γ_{ij}^k are determined completely by the first fundamental form g .

Proof. We write $g_{ij} = g\left(\frac{\partial\phi}{\partial x_i}, \frac{\partial\phi}{\partial x_j}\right)$ for readability and let

$$g = \begin{pmatrix} g_{11} & g_{12} \\ g_{12} & g_{22} \end{pmatrix},$$

which is invertible. We then take the defining equation from the definition and compute the inner product of both sides with $\frac{\partial\phi}{\partial x_k}$ (for $k = 1$ or 2):

$$\left\langle \frac{\partial^2\phi}{\partial x_i \partial x_j}, \frac{\partial\phi}{\partial x_k} \right\rangle = \Gamma_{ij}^1 g_{1k} + \Gamma_{ij}^2 g_{2k}.$$

We compute the derivatives of the various g_{jk} as

$$\frac{\partial g_{jk}}{\partial x_i} = \frac{\partial}{\partial x_i} \left\langle \frac{\partial \phi}{\partial x_j}, \frac{\partial \phi}{\partial x_j} \right\rangle = \left\langle \frac{\partial^2 \phi}{\partial x_i \partial x_j}, \frac{\partial \phi}{\partial x_j} \right\rangle + \left\langle \frac{\partial \phi}{\partial x_j}, \frac{\partial^2 \phi}{\partial x_i \partial x_k} \right\rangle,$$

which we can use the above to rewrite as

$$\frac{\partial g_{jk}}{\partial x_i} = (\Gamma_{ij}^1 g_{1k} + \Gamma_{ij}^2 g_{2k}) + (\Gamma_{ik}^1 g_{1j} + \Gamma_{ik}^2 g_{2j}).$$

In particular, if $j = k$, then we get

$$\frac{\partial g_{jj}}{\partial x_i} = 2 (\Gamma_{ij}^1 g_{1j} + \Gamma_{ij}^2 g_{2j}).$$

If, instead $j \neq k$, then we have $\{j, k\} = \{1, 2\}$, so in the case where $i = j$ (hence $i \neq k$), we have

$$\begin{aligned} \frac{\partial g_{ik}}{\partial x_i} &= (\Gamma_{ii}^1 g_{1k} + \Gamma_{ii}^2 g_{2k}) + (\Gamma_{ik}^1 g_{1i} + \Gamma_{ik}^2 g_{2i}) \\ &= (\Gamma_{ii}^1 g_{1k} + \Gamma_{ii}^2 g_{2k}) + \frac{1}{2} \frac{\partial g_{ii}}{\partial x_k}, \end{aligned}$$

or

$$\Gamma_{ii}^1 g_{1k} + \Gamma_{ii}^2 g_{2k} = \frac{\partial g_{ik}}{\partial x_i} - \frac{1}{2} \frac{\partial g_{ii}}{\partial x_i} \quad (i \neq k).$$

We combine the above equations in three cases now. If $i = j = 1$ and $k = 1$ or $k = 2$, then we get

$$\begin{cases} \Gamma_{11}^1 g_{11} + \Gamma_{11}^2 g_{21} = \frac{1}{2} \frac{\partial g_{11}}{\partial x_1} \\ \Gamma_{11}^1 g_{11} + \Gamma_{11}^2 g_{21} = \frac{\partial g_{12}}{\partial x_1} - \frac{1}{2} \frac{\partial g_{11}}{\partial x_2} \end{cases} \implies g \begin{pmatrix} \Gamma_{11}^1 \\ \Gamma_{11}^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \frac{\partial g_{11}}{\partial x_1} \\ \frac{\partial g_{12}}{\partial x_1} - \frac{1}{2} \frac{\partial g_{11}}{\partial x_2} \end{pmatrix};$$

if instead $(i, j, k) = (2, 1, 1)$ or $(i, j, k) = (1, 2, 2)$, then since $\Gamma_{12}^k = \Gamma_{21}^k$ we get

$$\begin{cases} \Gamma_{21}^1 g_{11} + \Gamma_{21}^2 g_{21} = \frac{1}{2} \frac{\partial g_{11}}{\partial x_1} \\ \Gamma_{12}^1 g_{12} + \Gamma_{12}^2 g_{22} = \frac{\partial g_{12}}{\partial x_1} - \frac{1}{2} \frac{\partial g_{11}}{\partial x_2} \end{cases} \implies g \begin{pmatrix} \Gamma_{12}^1 \\ \Gamma_{12}^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \frac{\partial g_{11}}{\partial x_2} \\ \frac{1}{2} \frac{\partial g_{22}}{\partial x_1} \end{pmatrix};$$

and finally, if $i = j = 2$ and either $k = 1$ or $k = 2$, then we get

$$\begin{cases} \Gamma_{22}^1 g_{11} + \Gamma_{22}^2 g_{21} = \frac{\partial g_{21}}{\partial x_2} - \frac{1}{2} \frac{\partial g_{22}}{\partial x_1} \\ \Gamma_{22}^1 g_{12} + \Gamma_{22}^2 g_{22} = \frac{1}{2} \frac{\partial g_{22}}{\partial x_2} \end{cases} \implies g \begin{pmatrix} \Gamma_{22}^1 \\ \Gamma_{22}^2 \end{pmatrix} = \begin{pmatrix} \frac{\partial g_{21}}{\partial x_2} - \frac{1}{2} \frac{\partial g_{22}}{\partial x_1} \\ \frac{1}{2} \frac{\partial g_{22}}{\partial x_2} \end{pmatrix}.$$

In every case, since g is invertible we can solve these matrix equations for the corresponding pairs of Christoffel symbols $(\Gamma_{ij}^1, \Gamma_{ij}^2)$, and the solutions depend only on the entries of g and their partial derivatives $\square$

Example 3.5.7

We parametrize the cylinder $C = \{x^2 + y^2 = 1\}$ by $\psi(u, v) = (\cos(u), \sin(u), v)$. Letting the plane $P = \{z = 0\}$ have the chart $\phi(u, v) = (u, v, 0)$ as in example 3.5.4, we showed that the Christoffel symbols for ϕ are identically zero. We also saw in example 3.4.5 that the map $F: P \rightarrow C$ defined by

$$F(x, y, 0) = (\cos(x), \sin(x), y)$$

is a local isometry, so it preserves the first fundamental form. Since $\psi = F \circ \phi$, we conclude by proposition 3.5.6 that the Christoffel symbols for ψ are zero as well.

§3.6 Gauss's Theorema Egregium

Suppose that we have a chart $\phi: U \rightarrow \mathbb{R}^3$ for a surface S . Then by the definition of Christoffel symbols, we have

$$\begin{aligned}\frac{\partial^2 \phi}{\partial x_1 \partial x_1} &= \Gamma_{11}^1 \frac{\partial \phi}{\partial x_1} + \Gamma_{11}^2 \frac{\partial \phi}{\partial x_2} + A_{11}N \\ \frac{\partial^2 \phi}{\partial x_1 \partial x_2} &= \Gamma_{12}^1 \frac{\partial \phi}{\partial x_1} + \Gamma_{12}^2 \frac{\partial \phi}{\partial x_2} + A_{12}N,\end{aligned}$$

where we are using ' N ' as a shorthand for the composition $N \circ \phi: U \rightarrow \mathbb{S}^2$. We use the fact that partial derivatives commute to write

$$\frac{\partial}{\partial x_2} \left(\frac{\partial^2 \phi}{\partial x_1 \partial x_1} \right) = \frac{\partial}{\partial x_1} \left(\frac{\partial^2 \phi}{\partial x_1 \partial x_2} \right),$$

which we'll use to find some interesting relations among the Christoffel symbols; note that this sort of computation was already useful in proposition 3.1.2 (to prove that the second fundamental form is symmetric) and theorem 3.1.17 (to prove that a surface whose points are all umbilical must lie in a plane or a sphere), but this is the first time we've tried to compare third partial derivatives. The computation will be a bit long a tedious, but worth it in the end.

The above equality of third derivatives can be written in terms of the basis $\left(\frac{\partial \phi}{\partial x_1}, \frac{\partial \phi}{\partial x_2}, N\right)$ of $\mathbb{R}^3$ as

$$\frac{\partial}{\partial x_2} \left(\Gamma_{11}^1 \frac{\partial \phi}{\partial x_1} + \Gamma_{11}^2 \frac{\partial \phi}{\partial x_2} + A_{11}N \right) = \frac{\partial}{\partial x_1} \left(\Gamma_{12}^1 \frac{\partial \phi}{\partial x_1} + \Gamma_{12}^2 \frac{\partial \phi}{\partial x_2} + A_{12}N \right).$$

Expanding this and writing $(u, v) = (x_1, x_2)$, we get the long equation

$$\begin{aligned}(\Gamma_{11}^1 \phi_{vu} + (\Gamma_{11}^1)_v \phi_u) + (\Gamma_{11}^2 \phi_{vv} + (\Gamma_{11}^2)_v \phi_v) + (A_{11}N_v + (A_{11})_v N) \\ = (\Gamma_{12}^1 \phi_{uu} + (\Gamma_{12}^1)_u \phi_u) + (\Gamma_{12}^2 \phi_{uv} + (\Gamma_{12}^2)_u \phi_v) + (A_{12}N_u + (A_{12})_u N).\end{aligned}$$

We will try to extract the ϕ_u -component of this equation. Remembering that in this basis we have

$$\phi_{uu} = \Gamma_{11}^1 \phi_u + \dots, \quad \phi_{uv} = \Gamma_{12}^1 \phi_u + \dots, \quad \phi_{vv} = \Gamma_{22}^1 \phi_u + \dots$$

Lemma 3.6.1

The partial derivatives N_u and N_v of the map $N \circ \phi$ satisfy

$$N_u = a\phi_u + b\phi_v, \quad N_v = c\phi_u + d\phi_v,$$

where the coefficients a, b, c, d are determined by

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} = -g^{-1}A = -\begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}^{-1} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}.$$

Proof. We first check that in the basis ϕ_u, ϕ_v, N of $\mathbb{R}^3$, both N_u and N_v have zero N -coordinate: we have

$$\langle N_u, N \rangle = \frac{1}{2} \frac{\partial}{\partial u} \langle N, N \rangle = \frac{1}{2} \frac{\partial}{\partial u} (1) = 0$$

and likewise $\langle N_v, N \rangle = 0$. Since N is orthogonal to the other basis vectors, this proves that N_u and N_v both lie in $T_p S = \text{Span}(\phi_u, \phi_v)$. Now the partial derivatives of N (by which we really mean $N \circ \phi$) satisfy

$$\begin{aligned} N_u = \frac{\partial(N \circ \phi)}{\partial u} = dN_p(\phi_u) &\implies \begin{cases} A_{11} = \langle \phi_u, -dN_p(\phi_u) \rangle = -\langle \phi_u, N_u \rangle \\ A_{21} = \langle \phi_v, -dN_p(\phi_u) \rangle = -\langle \phi_v, N_u \rangle \end{cases} \\ N_v = \frac{\partial(N \circ \phi)}{\partial v} = dN_p(\phi_v) &\implies \begin{cases} A_{12} = \langle \phi_u, -dN_p(\phi_v) \rangle = -\langle \phi_u, N_v \rangle \\ A_{22} = \langle \phi_v, -dN_p(\phi_v) \rangle = -\langle \phi_v, N_v \rangle \end{cases} \end{aligned}$$

Taking the inner product of ϕ_u with each of $N_u = a\phi_u + b\phi_v$ and $N_v = c\phi_u + d\phi_v$ in turn, and likewise for ϕ_v , we have

$$\begin{aligned} -A_{11} = \langle \phi_u, N_u \rangle &= ag_{11} + bg_{12} & -A_{21} = \langle \phi_v, N_u \rangle &= ag_{21} + bg_{22} \\ -A_{21} = \langle \phi_u, N_v \rangle &= cg_{11} + dg_{12} & -A_{22} = \langle \phi_v, N_v \rangle &= cg_{21} + dg_{22} \end{aligned}$$

and these can be packaged into a single matrix equation

$$-\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix}.$$

The matrix g is invertible, so we can now solve for a, b, c, d as claimed. $\square$

Lemma 3.6.1 says that the ϕ_u -coefficients of N_u and N_v are given by the top row of

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \frac{-1}{g_{11}g_{22} - g_{12}^2} \begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

meaning that

$$a = \frac{g_{12}A_{21} - g_{22}A_{11}}{g_{11}g_{22} - g_{12}^2}, \quad c = \frac{g_{12}A_{22} - g_{22}A_{12}}{g_{11}g_{22} - g_{12}^2}.$$

With this at hand, we can finally extract the ϕ_u -coefficients term by term from the boxed equation above to get

$$\Gamma_{11}^1 \Gamma_{21}^1 + (\Gamma_{11}^1)_v + \Gamma_{11}^2 \Gamma_{22}^1 + A_{11}c = \Gamma_{12}^1 \Gamma_{11}^1 + (\Gamma_{12}^1)_u + \Gamma_{12}^2 \Gamma_{12}^1 + A_{12}a,$$

which after some rearranging (and applying $\Gamma_{12}^1 = \Gamma_{21}^1$) becomes

$$\begin{aligned} (\Gamma_{11}^1)_v - (\Gamma_{12}^1)_u &= \Gamma_{12}^2 \Gamma_{12}^1 - \Gamma_{11}^2 \Gamma_{22}^1 + \frac{A_{12}(g_{12}A_{21} - g_{22}A_{11}) - A_{11}(g_{12}A_{22} - g_{22}A_{12})}{g_{11}g_{22} - g_{12}^2} \\ &= \Gamma_{12}^2 \Gamma_{12}^1 - \Gamma_{11}^2 \Gamma_{22}^1 + \frac{(A_{12}A_{21} - A_{11}A_{22})g_{12}}{g_{11}g_{22} - g_{12}^2} \\ &= \Gamma_{12}^2 \Gamma_{12}^1 - \Gamma_{11}^2 \Gamma_{22}^1 + \frac{\det(A)}{\det(g)} g_{12}. \end{aligned}$$

Now we recognize the Gaussian curvature $K = \frac{\det(A)}{\det(g)}$ from theorem 3.3.2, so we have shown that

$$K \cdot g_{12} = \Gamma_{12}^2 \Gamma_{12}^1 - \Gamma_{11}^2 \Gamma_{22}^1 + (\Gamma_{12}^1)_u - (\Gamma_{12}^1)_v.$$

If instead we extract the ϕ_v -coefficients from that same equation, then we get

$$\Gamma_{11}^1 \Gamma_{21}^2 + \Gamma_{11}^2 \Gamma_{22}^2 + (\Gamma_{11}^2)_v + A_{11}d = \Gamma_{12}^1 \Gamma_{11}^2 + \Gamma_{12}^2 \Gamma_{12}^2 + (\Gamma_{12}^2)_u + A_{12}b$$

and this rearranges to give

$$\begin{aligned} (\Gamma_{12}^2)_u - (\Gamma_{11}^2)_v &= \Gamma_{11}^1 \Gamma_{21}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{12}^1 \Gamma_{11}^2 - \Gamma_{12}^2 \Gamma_{12}^2 \\ &\quad + \frac{A_{11}(g_{21}A_{12} - g_{11}A_{22}) - A_{12}(g_{21}A_{11} - g_{11}A_{21})}{g_{11}g_{22} - g_{12}^2}, \end{aligned}$$

in which the numerator of the last fraction is equal to

$$(A_{12}A_{21} - A_{11}A_{22})g_{11},$$

or simply

$$-\det(A)g_{11},$$

and so the whole fraction is equal to

$$-\frac{\det(A)}{\det(g)}g_{11} = -K \cdot g_{11}.$$

We have thus derived the *Gauss equations*.

Proposition 3.6.2 (Gauss Equations)

The Gaussian curvature of a surface satisfies

$$\begin{aligned} K \cdot g_{11} &= \Gamma_{11}^1 \Gamma_{21}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{12}^1 \Gamma_{11}^2 - \Gamma_{12}^2 \Gamma_{12}^2 + (\Gamma_{11}^2)_v - (\Gamma_{12}^2)_u \\ K \cdot g_{12} &= \Gamma_{12}^2 \Gamma_{12}^1 - \Gamma_{11}^2 \Gamma_{22}^1 + (\Gamma_{12}^1)_u - (\Gamma_{11}^1)_v, \end{aligned}$$

in terms of the first fundamental form and Christoffel symbols for a given chart.

Remark 3.6.3 — Don't expect to remember these equations for the exam.

Theorem 3.6.4 (Theorema Egregium (Gauss))

The Gaussian curvature K of a surface S depends only on the first fundamental form of S . In other words, it is an *intrinsic* invariant of S .

Proof. By the Gauss equations (proposition 3.6.2), we can write

$$K \cdot g_{11} = \alpha, \quad K \cdot g_{12} = \beta,$$

where α and β are expressions that only depend on the Christoffel symbols. Proposition 3.5.6 says in turn that the Christoffel symbols are determined entirely by the first fundamental form, hence the same is true of α and β . We combine these equations to write

$$K \underbrace{(g_{11}g_{22} - g_{12}^2)}_{=\det(g) \neq 0} = \alpha g_{22} - \beta g_{12},$$

or equivalently

$$K = \frac{\alpha g_{22} - \beta g_{12}}{g_{11}g_{22} - g_{12}^2}.$$

Now every term on the right side is completely determined by the first fundamental form, so K must be as well. $\square$

An important corollary of the Theorema Egregium is that local isometries preserve Gaussian curvature.

Corollary 3.6.5

If $F: S_1 \rightarrow S_2$ is a local isometry, then the Gaussian curvatures on S_1 and S_2 are related by $K_{S_1}(p) = K_{S_2}(F(p))$ of all $p \in S_1$.

We can deduce from this that it's impossible to gift-wrap a sphere without distorting the wrapping paper:

Corollary 3.6.6

There is no local isometry from a plane to a sphere.

Proof. Examples 3.2.3 and 3.2.4 tell us that a plane has curvature $K = 0$ everywhere, whereas a sphere of radius $r > 0$ has nonzero curvature $K = \frac{1}{r^2}$ everywhere. $\square$

§3.7 Surfaces of Constant Curvature

We have seen in example 3.2.4 that spheres have constant positive curvature. In fact, we will prove in this section that these are the only examples among compact surfaces in $\mathbb{R}^3$.

Theorem 3.7.1

Let $S \subset \mathbb{R}^3$ be a compact, connected surface of constant curvature $K = K_0$. Then $K_0 > 0$, and S is a sphere of radius $\frac{1}{\sqrt{K_0}}$.

Proof. Proposition 3.3.2 guarantees that S has an elliptic point p , so then $K_0 = K(p) > 0$. Supposing that every point of S is umbilical, theorem 3.1.17 then says that S is contained in a sphere S' . In fact, we must have $S = S'$, because S' is connected and S is both closed (by compactness) and open in S' (because every point of S is in the open image of some chart). By example 3.2.4, if $S = S'$ has radius r then $K_0 = \frac{1}{r^2}$, so this would complete the proof.

It remains to be shown that every point of S is umbilical. We write the principal curvatures at $x \in S$ as $\lambda_1(x) \leq \lambda_2(x)$, and then since S is compact, the continuous function $\lambda_1: S \rightarrow \mathbb{R}$ achieves its minimum at some point $p \in S$; we also have

$$\lambda_2 = \frac{K_0}{\lambda_1}: S \rightarrow \mathbb{R},$$

so λ_2 is also maximized at the same point p . Since for all $x \in S$ we have

$$\lambda_1(p) \leq \lambda_1(x) \leq \lambda_2(x) \leq \lambda_2(p),$$

it follows that all points $x \in S$ are umbilical if and only if p is umbilical, so we will suppose that $\lambda_1(p) < \lambda_2(p)$ and get a contradiction.

We can arrange a rigid motion of $\mathbb{R}^3$ that $p = (0, 0, 0)$; that the principal directions at p are $X_1 = (1, 0, 0)$ and $X_2 = (0, 1, 0)$ corresponding to $\lambda_1(p)$ and $\lambda_2(p)$, so $T_p S =$

$\text{Span}(X_1, X_2)$ is the xy -plane; and then that the normal vector at p is $N(p) = (0, 0, 1)$. Now recall that S near p can be written as a graph. Then, S cannot locally be a graph of the form $x = g(y, z)$ or $y = h(x, z)$ near p , because in those cases $N(p)$ would have nonzero x or y coordinate, so lemma 2.1.18 says that S is a graph $z = f(x, y)$ on some neighborhood of p , parametrized by a chart of the form

$$\begin{aligned}\phi: U &\rightarrow \mathbb{R}^3 \\ (u, v) &\mapsto (u, v, f(u, v)).\end{aligned}$$

At any point $q \in U$, we observe that

$$\begin{aligned}\phi_u(q) &= (1, 0, f_u(q)) \\ \phi_v(q) &= (0, 1, f_v(q)) \\ N(\phi(q)) &= \frac{\phi_u(q) \times \phi_v(q)}{|\phi_u(q) \times \phi_v(q)|} = \frac{(f_u(q), f_v(q), 1)}{\sqrt{1 + |\nabla f|^2}},\end{aligned}$$

and in particular $f(0, 0) = 0$ and $\nabla f(0, 0) = (f_u(0, 0), f_v(0, 0)) = (0, 0)$. The matrix A of the second fundamental form is determined by the second derivatives of ϕ , which are

$$\begin{aligned}\phi_{uu}(q) &= (0, 0, f_{uu}(q)) \\ \phi_{uv}(q) &= \phi_{vu}(q) = (0, 0, f_{uv}(q)) \\ \phi_{vv}(q) &= (0, 0, f_{vv}(q)),\end{aligned}$$

so following remark 3.3.4, we have

$$A \circ \phi = \begin{pmatrix} \langle \phi_{uu}, N \circ \phi \rangle & \langle \phi_{uv}, N \circ \phi \rangle \\ \langle \phi_{vu}, N \circ \phi \rangle & \langle \phi_{vv}, N \circ \phi \rangle \end{pmatrix} = \frac{1}{\sqrt{1 + |\nabla f|^2}} \begin{pmatrix} f_{uu} & f_{uv} \\ f_{vu} & f_{vv} \end{pmatrix}.$$

We will use this to estimate the second fundamental form near $p = \phi(0, 0)$, where

$$\begin{aligned}\lambda_1(p) &= A(X_1, X_1) = f_{uu}(0, 0) \\ \lambda_2(p) &= A(X_2, X_2) = f_{vv}(0, 0)\end{aligned}$$

We define a family of unit tangent vectors at points near p by

$$\begin{aligned}E_1(t) &= \frac{\phi_u(0, t)}{|\phi_u(0, t)|} = \frac{(1, 0, f_u(0, t))}{\sqrt{1 + (f_u(0, t))^2}} \in T_{\phi(0, t)}S \\ E_2(t) &= \frac{\phi_v(t, 0)}{|\phi_v(t, 0)|} = \frac{(0, 1, f_v(t, 0))}{\sqrt{1 + (f_v(t, 0))^2}} \in T_{\phi(t, 0)}S,\end{aligned}$$

and set $h_i(t) = A(E_i(t), E_i(t))$ for $i = 1, 2$; then, $E_i(0) = X_i \in T_p S$ and so

$$h_i(0) = A(X_i, X_i) = \lambda_i(p), \quad i = 1, 2.$$

Lemma 3.1.5 gives us a pair of inequalities

$$\begin{aligned}h_1(t) &\geq \lambda_1(\phi(0, t)) \geq \lambda_1(p) \\ h_2(t) &\leq \lambda_2(\phi(t, 0)) \leq \lambda_2(p),\end{aligned}$$

so $h_1(t)$ has a local minimum at $t = 0$ while $h_2(t)$ has a local maximum at $t = 0$. This means that $h_1''(t) \geq 0$ while $h_2''(t) \leq 0$, and combining this gives us an inequality

$$h_1''(t) - h_2''(t) \geq 0.$$

We now evaluate the left side carefully to get our contradiction. We now use the bilinearity of the second fundamental form, we evaluate

$$\begin{aligned} h_1(t) &= \frac{A(\phi_u(0, t), \phi_u(0, t))}{1 + (f_u(0, t))^2} = \frac{1}{1 + (f_u(0, t))^2} \frac{f_{uu}(0, t)}{\sqrt{1 + |\nabla f(0, t)|^2}} \\ h_2(t) &= \frac{A(\phi_v(t, 0), \phi_v(t, 0))}{1 + (f_v(t, 0))^2} = \frac{1}{1 + (f_v(t, 0))^2} \frac{f_{vv}(t, 0)}{\sqrt{1 + |\nabla f(t, 0)|^2}}. \end{aligned}$$

We can approximate these using the Taylor expansion of f . We have seen that this has the form

$$f(u, v) = \underbrace{f(u, v)}_{=0} + \frac{1}{2}(\lambda_1 u^2 + \lambda_2 v^2) + O(|u|^3 + |v|^3)$$

where $\lambda_1 = \lambda_1(p)$ and $\lambda_2 = \lambda_2(p)$, so then we can compute

$$\begin{aligned} f_u(u, v) &= \lambda_1 u + O(u^2 + v^2) \\ f_v(u, v) &= \lambda_2 v + O(u^2 + v^2) \\ |\nabla f(u, v)|^2 &= f_u^2 + f_v^2 = \lambda_1^2 u^2 + \lambda_2^2 v^2 + O(|u|^3 + |v|^3) \end{aligned}$$

and substitute these into the above expressions for $h_1(t)$ and $h_2(t)$, using $(1 + x)^r = 1 + rx + O(x^2)$ to simplify:

$$\begin{aligned} h_1(t) &= \frac{1}{1 + (O(t^2))^2} \frac{f_{uu}(0, t)}{(1 + \lambda_2^2 t^2 + O(t^3))^{\frac{1}{2}}} \\ &= f_{uu}(0, t) \cdot (1 - O(t^4)) \cdot (1 - \frac{1}{2}\lambda_2^2 t^2 + O(t^3)) \\ &= f_{uu}(0, t) \cdot (1 - \frac{1}{2}\lambda_2^2 t^2) + O(t^3), \end{aligned}$$

and similarly

$$h_2(t) = f_{vv}(t, 0) \cdot (1 - \frac{1}{2}\lambda_1^2 t^2) + O(t^3).$$

We evaluate the second derivatives of each $t = 0$:

$$\begin{aligned} h_1''(0) &= f_{vvuu}(0, t)(1 - \frac{1}{2}\lambda_2^2 t^2) + 2f_{vuu}(0, t)(-\lambda_2^2 t) + f_{uu}(0, t)(-\lambda_2^2) + O(t) \big|_{t=0} \\ &= f_{vvuu}(0, 0) - \lambda_2^2 f_{uu}(0, 0) \end{aligned}$$

and likewise

$$\begin{aligned} h_2''(0) &= f_{uuvv}(t, 0)(1 - \frac{1}{2}\lambda_1^2 t^2) + 2f_{uvv}(t, 0)(-\lambda_1^2 t) + f_{vv}(t, 0)(-\lambda_1^2) + O(t) \big|_{t=0} \\ &= f_{uuvv}(0, 0) - \lambda_1^2 f_{vv}(0, 0), \end{aligned}$$

so now since $f_{vvuu}(0) = f_{uuvv}(0)$ we conclude that

$$\begin{aligned} h_1''(0) - h_2''(0) &= \lambda_1^2 f_{vv}(0, 0) - \lambda_2^2 f_{uu}(0, 0) \\ &= \lambda_1^2 \lambda_2 - \lambda_2^2 \lambda_1 \\ &= \lambda_1 \lambda_2 \cdot (\lambda_1 - \lambda_2). \end{aligned}$$

Since $\lambda_1 \lambda_2 = K(p) > 0$ while $\lambda_1 - \lambda_2 < 0$, we conclude that $h_1''(0) - h_2''(0) < 0$, and this is a contradiction. $\square$

4 Area and Integration on Surfaces

§4.1 Surface Integrals

Suppose we have a surface $S \subset \mathbb{R}^3$, with a chart $\phi: U \rightarrow S$. A small rectangle $R \subset U$ with side $(\delta_u, 0)$ and $(0, \delta_v)$ has area $\delta_u \delta_v$. If the side lengths are small enough, then the image $\phi(R) \subset S$ is very close to a parallelogram with sides $\delta_u \frac{\partial \phi}{\partial u}$ and $\delta_v \frac{\partial \phi}{\partial v}$. Therefore, we can approximate its areas as

$$\text{area}(\phi(R)) \approx \left| \left(\delta_u \frac{\partial \phi}{\partial u} \right) \times \left(\delta_v \frac{\partial \phi}{\partial v} \right) \right| \cong \delta_u \delta_v \left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right|$$

This motivates the following definition.

Definition 4.1.1. Let $D \subset U$ be compact, where $\phi: U \rightarrow \mathbb{R}^3$ is a chart for S . Then,

$$\text{area}(\phi(D)) = \int_D \left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right| du dv.$$

Proposition 4.1.2

The definition of $\text{area}(\phi(D))$ does not depend on the choice of oriented coordinate chart ϕ .

Proof. Let $\phi: U \rightarrow S$ and $\psi: U' \rightarrow S$ be two different charts whose images both contain $\phi(D) = \psi(D')$. We can shrink U and U' as needed, keeping $\phi(D)$ and $\psi(D')$ in their images, to ensure that

$$f = \phi^{-1} \circ \psi: U' \xrightarrow{\sim} U$$

is homeomorphism. We will write f in coordinates as $f(u, v) = (x(u, v), y(u, v))$. Then, $\psi = \phi \circ f$, so

$$\begin{aligned} \frac{\partial \psi}{\partial u} \times \frac{\partial \psi}{\partial v} &= \left(\frac{\partial \phi}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial \phi}{\partial y} \frac{\partial y}{\partial u} \right) \times \left(\frac{\partial \phi}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial \phi}{\partial y} \frac{\partial y}{\partial v} \right) \\ &= \left(\frac{\partial \phi}{\partial x} \times \frac{\partial \phi}{\partial y} \right) \cdot \left(\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right) \\ &= \left(\frac{\partial \phi}{\partial x} \times \frac{\partial \phi}{\partial y} \right) \cdot \det \begin{pmatrix} x_u & y_u \\ x_v & y_v \end{pmatrix}. \end{aligned}$$

Assuming that f is orientation preserving, so that the matrix on the right has positive determinant, we have

$$\begin{aligned} \text{area}_\psi(\psi(D')) &= \int_{D'} \left| \frac{\partial \psi}{\partial u} \times \frac{\partial \psi}{\partial v} \right| du dv \\ &= \int_{D'} \left| \frac{\partial \phi}{\partial x} \times \frac{\partial \phi}{\partial y} \right| \cdot \det \begin{pmatrix} x_u & y_u \\ x_v & y_v \end{pmatrix} du dv \\ &= \int_D \left| \frac{\partial \phi}{\partial x} \times \frac{\partial \phi}{\partial y} \right| dx dy \\ &= \text{area}_\phi(\phi(D)). \end{aligned}$$

□

Remark 4.1.3 — If we choose a chart ψ whose orientation is opposite that of ϕ , then the only change to the proof of proposition 4.1.2 is that the determinant of $\begin{pmatrix} x_u & y_u \\ x_v & y_v \end{pmatrix}$ is negative, and so upon taking absolute values the integrand gains a factor of -1 ; in this case, we get

$$\text{area}_\psi(\psi(D')) = -\text{area}_\phi(\phi(D)).$$

We recall from theorem 3.3.2, the notation

$$g = \begin{pmatrix} \langle \phi_u, \phi_u \rangle & \langle \phi_u, \phi_v \rangle \\ \langle \phi_v, \phi_u \rangle & \langle \phi_v, \phi_v \rangle \end{pmatrix}$$

for the matrix representing the first fundamental form.

Lemma 4.1.4

With $\phi: U \rightarrow S$ and $D \subset U$ as in definition 4.1.1, we have

$$\text{area}(\phi(D)) = \int_D \sqrt{\det(g)} du dv.$$

Proof. We use the identity $|a \times b|^2 = |a|^2|b|^2 - \langle a, b \rangle^2$, which holds because if θ is the angle between a and b , then the two sides are equal to $|a|^2|b|^2 \sin^2(\theta)$ and $|a|^2|b|^2 \cos^2(\theta)$, respectively. In the case where $a = \frac{\partial \phi}{\partial u}$ and $b = \frac{\partial \phi}{\partial v}$, this becomes

$$\begin{aligned} \left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right|^2 &= \left| \frac{\partial \phi}{\partial u} \right|^2 \left| \frac{\partial \phi}{\partial v} \right|^2 - \left\langle \frac{\partial \phi}{\partial u}, \frac{\partial \phi}{\partial v} \right\rangle^2 \\ &= g_{11}g_{22} - g_{12}^2 \\ &= \det(g). \end{aligned}$$

Therefore, the integrand $\left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right|$ of definition 4.1.1 is equal to $\sqrt{\det(g)}$ as claimed. $\square$

We can define surface integrals over some compact portion of a chart by the same construction.

Definition 4.1.5. Let $f: S \rightarrow \mathbb{R}$ be a smooth function on S , and let $D \subset U$ be a compact subset of the domain of some chart $\phi: U \rightarrow \mathbb{R}^3$ for S . Then, we define

$$\int_{\phi(D)} f dA = \int_D (f \circ \phi) \left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right| du dv = \int_D (f \circ \phi) \sqrt{\det(g)} du dv.$$

Again, this definition doesn't depend on the particular chart ϕ ; the proof of this is exactly the same as in proposition 4.1.2. We can now use this to define integration of functions over compact surfaces as follows:

Definition 4.1.6. Let $S \subset \mathbb{R}^3$ be a compact surface, and $f: S \rightarrow \mathbb{R}$ a smooth function. Divide S into finitely many pieces as

$$S = S_1 \cup S_2 \cup \cdots \cup S_k,$$

where $i \neq j$ then $S_i \cap S_j \subset \partial S_i \cap \partial S_j$, and such that for each i there is a chart $\phi_i: U_i \rightarrow S$ such that $S_i = \phi_i(D_i)$ for some compact $D_i \subset U_i$. Then, we define

$$\int_S f dA = \sum_{i=1}^k \int_{S_i} f dA = \sum_{i=1}^k \int_{D_i} (f \circ \phi_i) \left| \frac{\partial \phi_i}{\partial u} \times \frac{\partial \phi_i}{\partial v} \right| du dv.$$

Remark 4.1.7 — Note that the definition of $\int_S f dA$ also doesn't depend on any choices. The finite decomposition always exists because S is compact; the individual summands $\int_{S_i} f dA$ are well-defined; and if we take a decomposition and split the various S_i into several pieces then the value does not change, so given two different decompositions we can show that the value associated to each is the same as the value associated to a common refinement.

Example 4.1.8

Let $S \subset \mathbb{R}^3$ be the unit sphere. Here, we compute the surface area of S .

Solution. This can be parametrized in spherical coordinates as

$$\phi(u, v) = (\cos(u) \sin(v), \sin(u) \sin(v), \cos(v)),$$

but ϕ is only a chart if we take $0 < v < \pi$. Here, we have

$$\begin{aligned}\phi_u &= (-\sin(u) \sin(v), \cos(u) \sin(v), 0) \\ \phi_v &= (\cos(u) \cos(v), \sin(u) \cos(v), -\sin(v))\end{aligned}$$

$$\begin{aligned}\sqrt{\det(g)} &= |\phi_u \times \phi_v| = |(-\cos(u) \sin^2(v), -\sin(u) \sin^2(v), -\sin(v) \cos(v))| \\ &= \sin(v)\end{aligned}$$

We take $\epsilon > 0$ small and set

$$\begin{aligned}D_1 &= \{0 \leq u \leq \pi, \epsilon \leq v \leq \pi - \epsilon\}, & S_1 &= \phi(D_1) \\ D_2 &= \{\pi \leq u \leq 2\pi, \epsilon \leq v \leq \pi - \epsilon\}, & S_2 &= \phi(D_2),\end{aligned}$$

and let $D_3 = S \cap \{z \geq \cos(\epsilon)\}$ and $D_4 = S \cap \{z \leq -\cos(\epsilon)\}$. Then,

$$\begin{aligned}\int_{S_1} dA &= \int_{D_1} \sqrt{\det(g)} dA = \int_{\epsilon}^{\pi-\epsilon} \int_0^{\pi} \sin(v) du dv \\ &= \pi \int_{\epsilon}^{\pi-\epsilon} \sin(v) dv \\ &= 2\pi \cos(\epsilon).\end{aligned}$$

The same computation gives $\int_{S_2} dA = 2\pi \cos(\epsilon)$ as well. As for S_3 , we may use the chart

$$\begin{aligned}\psi: \{u^2 + v^2 < 1\} &\rightarrow \mathbb{R}^3 \\ (u, v) &\mapsto (u, v, \sqrt{1 - u^2 - v^2}),\end{aligned}$$

with $S_3 = \psi(D_3)$, where $D_3 = \{u^2 + v^2 \leq \sin^2(\epsilon)\}$. Then,

$$\psi_u = \left(1, 0, \frac{-u}{\sqrt{1 - u^2 - v^2}}\right), \quad \psi_v = \left(0, 1, \frac{-v}{\sqrt{1 - u^2 - v^2}}\right)$$

and so

$$|\psi_u \times \psi_v| = \left| \left(\frac{u}{\sqrt{1 - u^2 - v^2}}, \frac{v}{\sqrt{1 - u^2 - v^2}}, 1 \right) \right| = \frac{1}{\sqrt{1 - u^2 - v^2}}$$

Thus, we can compute the area of S_3 by

$$\begin{aligned}\int_{S_3} dA &= \int_{D_3} \frac{1}{\sqrt{1 - u^2 - v^2}} du dv = \int_0^{2\pi} \int_0^{\sin(\epsilon)} \frac{1}{\sqrt{1 - r^2}} r dr d\theta \\ &= 2\pi \left[-(1 - r^2)^{\frac{1}{2}} \right]_{r=0}^{r=\sin(\epsilon)} \\ &= 2\pi(1 - \cos(\epsilon)).\end{aligned}$$

The computation for S_4 is the same, so S has area

$$\begin{aligned}\int_S dA &= \sum_{i=1}^4 \int_{S_i} dA \\ &= 2\pi \cos(\epsilon) + 2\pi \cos(\epsilon) + 2\pi(1 - \cos(\epsilon)) + 2\pi(1 - \cos(\epsilon)) \\ &= 4\pi,\end{aligned}$$

which is what we expect. $\square$

Remark 4.1.9 — We may also argue using limits after we've found the $2\pi \cos(\epsilon)$ and say that $\epsilon \rightarrow 0$ makes the area tend to 4π .

§4.2 Geodesics

Let $\alpha: [a, b] \rightarrow S$ be a regular curve in an oriented surface $S \subset \mathbb{R}^3$, which has Gauss map N . Then, $\alpha'(t)$ is tangent to S and $N(\alpha(t))$ is orthogonal to it, so if α is parametrized by arclength, then the triple $(\alpha', N \times \alpha', N)$ is an orthonormal basis of $\mathbb{R}^3$. The curvature vector $\vec{k}(t) = \alpha''(t)$ is orthogonal to $\alpha'(t)$, because

$$\langle \alpha'(t), \alpha''(t) \rangle = \frac{1}{2} \frac{d}{dt} \underbrace{\langle \alpha'(t), \alpha'(t) \rangle}_{=1} = 0,$$

so we can write it in terms of the rest of the basis as

$$\vec{k} = k_n N + k_g (N \times \alpha'),$$

where $k_n = \langle \vec{k}, N \rangle$ is the normal curvature, while $k_g = \langle \vec{k}, N \times \alpha' \rangle$.

Definition 4.2.1. Let $S \subset \mathbb{R}^3$ be an oriented surface with normal vector N , and let $\alpha: [a, b] \rightarrow S$ be a curve parametrized by arclength. Then, the quantity

$$k_g(t) = \langle \vec{k}(t), N(\alpha(t)) \times \alpha'(t) \rangle$$

is called the **geodesic curvature** of α .

Remark 4.2.2 — Note that this is not the same as our original notion of curvature $k(t) = |\vec{k}(t)| = |\alpha''(t)|$. In fact, we have

$$\begin{aligned}k(t)^2 &= \langle \vec{k}(t), \vec{k}(t) \rangle \\ &= \langle k_n N + k_g (N \times \alpha'), k_n N + k_g (N \times \alpha') \rangle \\ &= k_n^2 \underbrace{\langle N, N \rangle}_{=1} + 2k_n k_g \underbrace{\langle N, N \times \alpha' \rangle}_{=0} + k_g^2 \underbrace{\langle N \times \alpha', N \times \alpha' \rangle}_{=1},\end{aligned}$$

or simply

$$k^2 = k_n^2 + k_g^2.$$

If $k_g = 0$, then we have that $\vec{k} = k_n N$, meaning that $\alpha(t)$ only accelerates in directions orthogonal to S .

Definition 4.2.3. A curve $\gamma: [a, b] \rightarrow S$ is called a **geodesic** if it has zero geodesic curvature, meaning that $k_g = 0$.

Example 4.2.4

Let $S = \{z = 0\}$ be the xy -plane, with constant Gauss map $N = (0, 0, 1)$, and fix a curve $\alpha: [a, b] \rightarrow S$ defined by

$$\alpha(t) = (x(t), y(t), 0)$$

parametrized by arclength, so $(x'(t))^2 + (y'(t))^2 = 1$. This has normal curvature

$$k_n(t) = \langle \alpha''(t), N(\alpha(t)) \rangle = \langle (x''(t), y''(t), 0), (0, 0, 1) \rangle = 0.$$

so the curvature vector satisfies

$$(x''(t), y''(t), 0) = \vec{k}(t) = k_g(t) \cdot (N \times \alpha'(t)) = k_g(t) \cdot (-y'(t), x'(t), 0).$$

Since $(-y', x', 0)$ has unit length, we have $k_g(t) = 0$ if and only if $x''(t) = y''(t) = 0$. Thus a curve in S is a geodesic if and only if it is a straight line.

Example 4.2.5

Let $S = \{x^2 + y^2 + z^2 = 1\}$ be the unit sphere in $\mathbb{R}^3$, fix $r \in [0, 1)$, and let

$$\alpha(t) = \left(r \cos\left(\frac{t}{r}\right), r \sin\left(\frac{t}{r}\right), \sqrt{1-r^2} \right)$$

be a circle in S of radius r , parametrized by arclength. Then,

$$k_g(t) = \langle \alpha''(t), N(\alpha(t)) \times \alpha'(t) \rangle,$$

so we compute that

$$\alpha''(t) = \left(-\frac{1}{r} \cos\left(\frac{t}{r}\right), -\frac{1}{r} \sin\left(\frac{t}{r}\right), 0 \right)$$

while

$$\begin{aligned} N(\alpha(t)) \times \alpha'(t) &= \left(r \cos\left(\frac{t}{r}\right), r \sin\left(\frac{t}{r}\right), \sqrt{1-r^2} \right) \times \left(-\sin\left(\frac{t}{r}\right), \cos\left(\frac{t}{r}\right), 0 \right) \\ &= \left(-\sqrt{1-r^2} \cos\left(\frac{t}{r}\right), -\sqrt{1-r^2} \sin\left(\frac{t}{r}\right), r \right), \end{aligned}$$

and then their inner product is

$$k_g(t) = \langle \alpha''(t), N(\alpha(t)) \times \alpha'(t) \rangle = \frac{\sqrt{1-r^2}}{r}.$$

This is zero if and only if $r = 1$, so the circle $\alpha(t)$ is a geodesic precisely when γ is a *great circle* (such as the equator).

Proposition 4.2.6

If $F: S \rightarrow S'$ is a local isometry, then it sends geodesics to geodesics.

Proof. Let $\gamma: [a, b] \rightarrow S$ be a geodesic parametrized by arclength, and fix a chart

$\phi: U \rightarrow \mathbb{R}^3$ for S at some point $\gamma(t)$. From the equation

$$\gamma''(t) = \vec{k}(t) = k_n(t) \cdot N(\gamma(t)) + k_g(t) \cdot (N(\gamma(t)) \times \gamma'(t)),$$

we see that γ is a geodesic if and only if $\gamma''(t)$ is a multiple of $N(\gamma(t))$, hence if and only if $\gamma''(t)$ is orthogonal to the tangent vectors ϕ_u and ϕ_v at $\gamma(t)$. We now use lemma 2.2.5 to write $\gamma(t) = \phi(u(t), v(t))$ in local coordinates, and then γ has derivatives

$$\begin{aligned}\gamma'(t) &= \phi_u(t) \cdot u'(t) + \phi_v(t) \cdot v'(t) \\ \gamma''(t) &= (\phi_{uu}u' + \phi_{vu}v') \cdot u'(t) + (\phi_{uv}u' + \phi_{vv}v') \cdot v'(t).\end{aligned}$$

We know that $\langle \gamma''(t), \phi_u \rangle = 0$, and in these coordinates we can expand $\gamma''(t)$ to get

$$\begin{aligned}\langle \gamma''(t), \phi_u \rangle &= \langle \phi_{uu}, \phi_u \rangle u'(t)^2 + 2 \langle \phi_{uv}, \phi_u \rangle u'(t)v'(t) + \langle \phi_{vv}, \phi_u \rangle v'(t)^2 \\ &= \left(\frac{1}{2} \frac{\partial}{\partial u} \langle \phi_u, \phi_u \rangle \right) u'(t)^2 + \left(\frac{\partial}{\partial v} \langle \phi_u, \phi_u \rangle \right) u'(t)v'(t) \\ &\quad + \left(\frac{\partial}{\partial v} \langle \phi_u, \phi_v \rangle - \frac{1}{2} \frac{\partial}{\partial u} \langle \phi_v, \phi_v \rangle \right) v'(t)^2.\end{aligned}$$

in other words, the equation $\langle \gamma''(t), \phi_u \rangle = 0$ is equivalent to

$$\frac{1}{2}(g_{11})_u \cdot u'(t)^2 + (g_{11})_v \cdot u'(t)v'(t) + ((g_{12})_v - \frac{1}{2}(g_{22})_u) \cdot v'(t)^2 = 0.$$

We get a similar equation by expanding $\langle \gamma''(t), \phi_v \rangle = 0$. In each case, these two equations are determined completely by $u(t), v(t)$, and the first fundamental form g . Now since G is an isometry, it preserves g with respect to the chart $\psi = F \circ \phi$, and then in local coordinates we have

$$F(\gamma(t)) = \psi(u(t), v(t)),$$

so the fact that $\langle \gamma''(t), \phi_u \rangle = \langle \gamma''(t), \phi_v \rangle = 0$ immediately implies that

$$\langle (F \circ \gamma)''(t), \psi_u \rangle = \langle (F \circ \gamma)''(t), \psi_v \rangle = 0.$$

This means as above that $F \circ \gamma$ is a geodesic as well. $\square$

It is usually very hard to describe geodesics explicitly, but proposition 4.2.6 can help:

Example 4.2.7

Let $S = \{z = 0\}$ and $S' = \{x^2 + y^2 = 1\}$ and consider the isometry $F: S \rightarrow S'$ defined by

$$F(x, y, 0) = (\cos(x), \sin(x), y).$$

We know that $\gamma(t) = (at + b, ct + d, 0)$ with $a^2 + c^2 = 1$ is a geodesic in S . This says that

$$(F \circ \gamma)(t) = (\cos(at + b), \sin(at + b), ct + d)$$

is a geodesic in S' .

Remark 4.2.8 — Every geodesic in S' is a helix or a vertical line or a great circle.

Proof. F is an isometry, so we can invert F locally near $\gamma(t)$, a point on a curve in a geodesic of S' . If we let $\gamma(t) = (\cos(\theta), \sin(\theta), z)$, then we can have

$$F^{-1}: (\cos(t), \sin(t), s) \mapsto (t, s, 0).$$

Since F is a local isometry, its inverse is also an isometry. Thus, $F^{-1} \circ \gamma$ is a segment of a straight line. $\square$

Exercise 4.2.9. Use the pair of equations

$$\begin{aligned}\langle \gamma''(t), \phi_u \rangle &= \frac{d}{dt} \langle \gamma'(t), \phi_u(u(t), v(t)) \rangle - \left\langle \gamma'(t), \frac{d}{dt} \phi_u(u(t), v(t)) \right\rangle \\ \langle \gamma''(t), \phi_v \rangle &= \frac{d}{dt} \langle \gamma'(t), \phi_v(u(t), v(t)) \rangle - \left\langle \gamma'(t), \frac{d}{dt} \phi_v(u(t), v(t)) \right\rangle\end{aligned}$$

to show that any geodesic $\gamma(t) = \phi(u(t), v(t))$ satisfies the **geodesic equations**

$$\begin{aligned}\frac{d}{dt}(g_{11}u' + g_{12}v') &= \frac{1}{2}((g_{11})_u(u'(t))^2 + 2(g_{12})_u(u'(t)v'(t)) + (g_{22})_u(v'(t))^2) \\ \frac{d}{dt}(g_{21}u' + g_{22}v') &= \frac{1}{2}((g_{11})_v(u'(t))^2 + 2(g_{12})_v(u'(t)v'(t)) + (g_{22})_v(v'(t))^2)\end{aligned}$$

Exercise 4.2.10. Use local isometries to show that conversely, every geodesic on the cylinder C is the image $F(l)$ for some straight line l in P .

We will show that geodesics minimize arclength locally, in the sense that they cannot be perturbed to become shorter.

Definition 4.2.11. A **variation** of the curve $\alpha: [0, L] \rightarrow S$ is a smooth map

$$\begin{aligned}[0, L] \times [-\epsilon, \epsilon] &\rightarrow S \\ (t, s) &\mapsto \alpha_s(t),\end{aligned}$$

with $\alpha_0(t) = \alpha(t)$, satisfying $\alpha_s(0) = \alpha(0)$ and $\alpha_s(L) = \alpha(L)$ for all s .

Proposition 4.2.12

Let $\alpha: [0, L] \rightarrow S$ be a curve parametrized by arclength. If α minimizes arclength from $\alpha(0)$ to $\alpha(L)$, in the sense that for any variation $\alpha_s(t)$ the length $L(\alpha_s)$ is locally minimized at $s = 0$, then α is a geodesic.

Proof. Let $\alpha_s(t)$ be a variation of α . The fact that the function $s \mapsto L(\alpha_s)$ is locally minimized at $s = 0$ means that its derivative at $s = 0$ is zero. We attempt to evaluate this derivative as follows:

$$\begin{aligned}\left. \frac{d}{ds} L(\alpha_s) \right|_{s=0} &= \left. \frac{d}{ds} \int_0^L |\alpha'_s(t)| dt \right|_{s=0} \\ &= \left. \frac{d}{ds} \int_0^L \sqrt{\langle \alpha'_s(t), \alpha'_s(t) \rangle} dt \right|_{s=0} \\ &= \int_0^L \left. \frac{d}{ds} \sqrt{\langle \alpha'_s(t), \alpha'_s(t) \rangle} \right|_{s=0} dt,\end{aligned}$$

in which we are allowed to exchange differentiation because the integrand $\sqrt{\langle \alpha'_s(t), \alpha'_s(t) \rangle}$ is smooth near $s = 0$. Note that $|\alpha'_0(t)| = 1$ implies that $|\alpha'_s(t)|$ will be nonzero for s sufficiently close to zero, which is what we require for smoothness. Via the chain rule, we evaluate the integrand as

$$\left. \frac{d}{ds} \sqrt{\langle \alpha'_s(t), \alpha'_s(t) \rangle} \right|_{s=0} = \frac{\left\langle \left. \frac{d}{ds} \alpha'_s(t) \right|_{s=0}, \alpha'_0(t) \right\rangle}{\sqrt{\langle \alpha'_0(t), \alpha'_0(t) \rangle}} = \left\langle \left. \frac{d}{ds} \alpha'_s(t) \right|_{s=0}, \alpha'_0(t) \right\rangle,$$

again because $|\alpha'_0(t)| = 1$ for all t , and we know that the first input on the right satisfies

$$\frac{d}{ds}\alpha'_s(t)\Big|_{s=0} = \frac{d}{dt}\left(\frac{d\alpha_s}{dt}(t)\Big|_{s=0}\right),$$

by the smoothness of the map $(s, t) \mapsto \alpha_s(t)$, so we have shown that

$$\frac{d}{ds}L(\alpha_s)\Big|_{s=0} = \int_0^L \left\langle \frac{d}{dt}\left(\frac{d\alpha_s}{ds}\Big|_{s=0}\right), \alpha'_0(t) \right\rangle dt.$$

Integrating by parts, we now have

$$\frac{d}{ds}L(\alpha_s)\Big|_{s=0} = \left[\frac{d\alpha_s}{ds}\Big|_{s=0} \alpha'_0(t) \right]_{t=0}^{t=L} - \int_0^L \left\langle \frac{d\alpha_s}{ds}\Big|_{s=0}, \alpha''_0(t) \right\rangle dt.$$

Now, $\alpha_s(0) = \alpha(0)$ and $\alpha_s(L) = \alpha(L)$ are both constant in s , so $\frac{d\alpha_s}{ds}$ vanishes at $t = 0$ and at $t = L$. Writing

$$\alpha''_0(t) = \alpha''(t) = k_n(t)N(\alpha(t)) + k_g(t)(N(\alpha(t)) \times \alpha'(t)),$$

the last equation becomes

$$\frac{d}{ds}L(\alpha_s)\Big|_{s=0} = - \int_0^L \left\langle \frac{d\alpha_s}{ds}\Big|_{s=0}, k_n(t)N(\alpha(t)) + k_g(t)(N(\alpha(t)) \times \alpha'(t)) \right\rangle dt.$$

However, $s \mapsto \alpha_s(t)$ is a path in S for each fixed t , so its derivative $\frac{d\alpha_s}{ds}\Big|_{s=0}$ is tangent to S and therefore orthogonal to $N(\alpha(t))$, leaving us with

$$\frac{d}{ds}L(\alpha_s)\Big|_{s=0} = - \int_0^L k_g(t) \left\langle \frac{d\alpha_s}{ds}\Big|_{s=0}, N(\alpha(t)) \times \alpha'(t) \right\rangle dt.$$

Now, if $\alpha(t)$ is not a geodesic, then we must have $k_g(t_0) \neq 0$ for some $t_0 \in [0, L]$, hence $k_g \neq 0$ on a neighborhood V of $t = t_0$ by continuity. In this case, we can find a variation $\alpha_s(t)$ or which $\frac{d\alpha_s}{ds}\Big|_{s=0}$ vanishes outside V , and is nonzero in the direction of $N \times \alpha'$ on an even smaller neighborhood of $t = t_0$. This particular variation satisfies $\frac{d}{ds}L(\alpha_s)\Big|_{s=0} \neq 0$, so then $\alpha(t)$ cannot be length-minimizing after all, and we conclude that any length-minimizing curve must be a geodesic. $\square$

We now consider what it means for geodesics to minimize arclength.

Example 4.2.13 (Geodesics do not need to globally minimize arclength)

On the unit sphere, we pick two points p and q on the equator and let γ be a curve from p to q that wraps around the equator several times. This is a geodesic (see example 4.2.5), so we can't make it shorter by perturbing it, but it is not the shortest curve from p to q .

Example 4.2.14

There may not be a geodesic between a given pair of points.

Let $S = \{z = 0\} \setminus \{(0, 0, 0)\}$ be the xy -plane minus the origin, and let $p = (-1, 0, 0)$ and $q = (1, 0, 0)$. Any geodesic in S is also a geodesic in the xy -plane, hence example 4.2.4 says it must be a straight line. But a straight line from p to q must pass through the origin, so S does not have a geodesic from p to q .

Example 4.2.15

Geodesics which do minimize length globally may not be unique.

Let S be the unit sphere, with p and q the north and south poles. Then any arc of constant longitude ($\theta = c$ in spherical coordinates) is a globally length-minimizing path from p to q , of length π , but there is one such arc for every possible value of θ .

§4.3 A Local Version of the Gauss–Bonnet Theorem

Definition 4.3.1. A surface with **boundary** $S \subset \mathbb{R}^3$ is the same as a regular surface, except that some points $p \in S$ have charts

$$\begin{aligned}\phi: U &\rightarrow \mathbb{R}^3 \\ (0,0) &\mapsto p,\end{aligned}$$

where $U \subset \mathbb{R}^2$ is not open in $\mathbb{R}^2$, but is rather the intersection of some open neighborhood of $(0,0)$ with the upper half-plane $\{v \geq 0\}$. The **boundary** of S , denoted ∂S , is the union of all points with such charts; it is a union of curves in $\mathbb{R}^3$.

If S is an oriented surface with boundary, then we will say that ∂S is **positively oriented** if it admits a parametrization

$$\alpha: [a, b] \rightarrow \partial S$$

for which the tangent vector $N(\alpha(t)) \times \alpha'(t)$ points *inwards* into S along ∂S ; the picture to keep in mind is that as we walk along ∂S , the interior of S should be to the left.

We can extend the definition of integration to compact surfaces with boundary without any trouble. Our goal for now is to prove the following theorem, relating an integral over a surface with boundary to the total geodesic curvature of its boundary. These surfaces must be contained in charts, but eventually we will generalize this to all compact surfaces with boundary.

Theorem 4.3.2 (Local Gauss–Bonnet Theorem)

Let $\phi: U \rightarrow \mathbb{R}^3$ be a chart that extends smoothly to the closure $\bar{U}$, and such that $S = \phi(\bar{U})$ is a surface with boundary $\partial S = \phi(\partial \bar{U})$. If $\bar{U}$ is diffeomorphic to a closed disk, and S is oriented compatibly with ϕ (meaning that $N = \frac{\phi_u \times \phi_v}{|\phi_u \times \phi_v|}$), then

$$\int_{\partial S} k_g ds + \int_S K dA = 2\pi.$$

where ∂S is positively oriented with respect to S and the integral over ∂S is taken with respect to arclength.

By the definition of a surface integral, we know that

$$\int_S K dA = \int_{\bar{U}} K \sqrt{\det g} du dv = \int_{\bar{U}} (K \circ \phi) \left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right| du dv.$$

Our strategy will be to find a pair of functions M and L so that this expression is equal to

$$\int_{\bar{U}} \left| \frac{\partial M}{\partial u} - \frac{\partial L}{\partial v} \right| du dv,$$

and then we will apply Green's theorem to convert this into $\int_{\partial\bar{U}} Ldu + Mdv$, which we will evaluate explicitly.

Proof of Theorem 4.3.2. We start by fixing an orthonormal basis for T_pS , of the form

$$E_1 = \frac{\phi_u}{|\phi_u|}, \quad E_2 = N \times E_1$$

so that $N = E_1 \times E_2$. We parametrize ∂S by arclength, oriented positively, as $\alpha: [0, L] \rightarrow \partial S$. Then, since $|\alpha'(t)| = 1$, there is a smooth map $\theta: [0, L] \rightarrow \mathbb{R}$ such that

$$\alpha'(t) = \cos(\theta(t))E_1(\alpha(t)) + \sin(\theta(t))E_2(\alpha(t)).$$

Claim 4.3.3. $k_g(\alpha(t)) = \theta'(t) - \langle (E_1 \circ \alpha)(t), (E_2 \circ \alpha)'(t) \rangle$.

Proof. We compute from the above expression for $\alpha'(t)$ that

$$\begin{aligned} \alpha''(t) &= (-\sin(\theta)E_1 + \cos(\theta)E_2)\theta'(t) + \cos(\theta)E_1' + \sin(\theta)E_2' \\ N(\alpha(t)) \times \alpha'(t) &= \cos(\theta)E_2 - \sin(\theta)E_1, \end{aligned}$$

where for the sake of readability we write θ and θ' without the (t) , and use E_1 and E_2 to stand for $(E_1 \circ \alpha)(t)$ and $(E_2 \circ \alpha)(t)$. Then,

$$\begin{aligned} k_g(t) &= \langle \alpha''(t), N(\alpha(t)) \times \alpha'(t) \rangle \\ &= \theta' + \langle \cos(\theta)E_1' + \sin(\theta)E_2', -\sin(\theta)E_1 + \cos(\theta)E_2 \rangle. \end{aligned}$$

Now, we can differentiate each of the relations

$$\langle E_1, E_1 \rangle = 1, \quad \langle E_2, E_2 \rangle = 1, \quad \langle E_1, E_2 \rangle = 0$$

to get

$$\langle E_1', E_1 \rangle = \langle E_2', E_2 \rangle = 0, \quad \langle E_1', E_2 \rangle + \langle E_1, E_2' \rangle = 0,$$

and this allows us to simplify the above expression for $k_g(t)$ to

$$\begin{aligned} k_g &= \theta' + \cos^2(\theta) \langle E_1', E_2 \rangle - \sin^2(\theta) \langle E_2', E_1 \rangle \\ &= \theta' - (\cos^2(\theta) + \sin^2(\theta)) \langle E_1, E_2' \rangle \\ &= \theta' - \langle E_1, E_2' \rangle \end{aligned}$$

as claimed. □

Claim 4.3.4. We have

$$\left\langle \frac{\partial E_1}{\partial u}, \frac{\partial E_2}{\partial v} \right\rangle - \left\langle \frac{\partial E_2}{\partial u}, \frac{\partial E_1}{\partial v} \right\rangle = (K \circ \phi) \left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right|.$$

Proof. We note that for each $j = 1, 2$, we have

$$\left\langle E_j, \frac{\partial E_j}{\partial u} \right\rangle = \frac{1}{2} \frac{\partial}{\partial u} \underbrace{\langle E_j, E_j \rangle}_{=1} = 0, \quad \left\langle E_j, \frac{\partial E_j}{\partial v} \right\rangle = \frac{1}{2} \frac{\partial}{\partial v} \underbrace{\langle E_j, E_j \rangle}_{=1} = 0.$$

Recalling that $\frac{\partial N}{\partial u}$ really means $\frac{\partial(N \circ \phi)}{\partial u} = dN(\frac{\partial \phi}{\partial u})$, we can similarly show that

$$\left\langle \frac{\partial E_j}{\partial u}, N \right\rangle = \frac{\partial}{\partial u} \underbrace{\langle E_j, N \rangle}_{=0} - \left\langle E_j, \frac{\partial N}{\partial u} \right\rangle = A(E_j, \frac{\partial \phi}{\partial u}),$$

and likewise with v in place of u . Since (E_1, E_2, N) is an orthonormal basis of $\mathbb{R}^3$, it follows that $\frac{\partial E_1}{\partial u}$ has E_1 -coordinate zero in this basis, so we can write

$$\frac{\partial E_1}{\partial u} = aE_2 + \left\langle \frac{\partial E_1}{\partial u}, N \right\rangle N = aE_2 + A(E_1, \phi_u)N$$

for $a = \left\langle \frac{\partial E_1}{\partial u}, E_2 \right\rangle$. The same argument says that

$$\begin{aligned} \frac{\partial E_1}{\partial v} &= bE_2 + A(E_1, \phi_v)N, \\ \frac{\partial E_2}{\partial u} &= cE_1 + A(E_2, \phi_u)N, \\ \frac{\partial E_2}{\partial v} &= dE_1 + A(E_2, \phi_v)N \end{aligned}$$

for suitable b, c, d . With each of these expressions at hand, we now have

$$\left\langle \frac{\partial E_1}{\partial u}, \frac{\partial E_2}{\partial v} \right\rangle - \left\langle \frac{\partial E_2}{\partial u}, \frac{\partial E_1}{\partial v} \right\rangle = A(E_1, \phi_u)A(E_2, \phi_v) - A(E_2, \phi_u)A(E_1, \phi_v).$$

Now, if we write the tangent vectors ϕ_u, ϕ_v in terms of the basis E_1, E_2 of $T_p S$ as

$$\begin{aligned} \phi_u &= c_{11}E_1 + c_{12}E_2 \\ \phi_v &= c_{21}E_1 + c_{22}E_2, \end{aligned}$$

then $\phi_u \times \phi_v = (c_{11}c_{22} - c_{12}c_{21})(E_1 \times E_2) = (c_{11}c_{22} - c_{12}c_{21})N$, and the RHS of the above equation equals

$$A(E_1, c_{11}E_1 + c_{12}E_2)A(E_2, c_{21}E_1 + c_{22}E_2) - A(E_2, c_{11}E_1 + c_{12}E_2)A(E_1, c_{21}E_1 + c_{22}E_2),$$

which can be rearranged (using the fact that A is symmetric and bilinear) to get

$$\underbrace{(c_{11}c_{22} - c_{12}c_{21})}_{=|\phi_u \times \phi_v|} \cdot \underbrace{(A(E_1, E_1)A(E_2, E_2) - A(E_1, E_2)A(E_2, E_1))}_{=\det(A)}.$$

Since $(K \circ \phi) = \frac{\det(A)}{\det(g)}$, where $g = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, this proves the claim. $\square$

Claim 4.3.5. $\int_0^L \langle (E_1 \circ \alpha)(t), (E_2 \circ \alpha)'(t) \rangle dt = \int_{\bar{U}} (K \circ \phi) \left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right| dudv.$

Proof. We use lemma 2.2.5 to write $\alpha(t)$ locally in the form $\alpha(t) = \phi(u(t), v(t))$. In these coordinates, we have

$$(E_2 \circ \alpha)'(t) = \frac{\partial E_2}{\partial u} u'(t) + \frac{\partial E_2}{\partial v} v'(t).$$

Writing E_1 and E_2 as shorthand for $E_1 \circ \alpha$ and $E_2 \circ \alpha$, we now have

$$\begin{aligned} \int_0^L \langle E_1, E_2' \rangle dt &= \int_0^L \left\langle E_1, \frac{\partial E_2}{\partial u} u'(t) \right\rangle dt + \int_0^L \left\langle E_1, \frac{\partial E_2}{\partial v} v'(t) \right\rangle dt \\ &= \int_{\bar{U}} \left(\left\langle E_1, \frac{\partial E_2}{\partial u} \right\rangle du + \left\langle E_1, \frac{\partial E_2}{\partial v} \right\rangle dv \right) \\ &= \int_{\bar{U}} \left(\frac{\partial}{\partial u} \left\langle E_1, \frac{\partial E_2}{\partial v} \right\rangle - \frac{\partial}{\partial v} \left\langle E_1, \frac{\partial E_2}{\partial u} \right\rangle \right) dudv, \end{aligned}$$

where at the last step, we've applied Green's theorem. The new integrand equals the following:

$$\begin{aligned} & \left(\left\langle \frac{\partial E_1}{\partial u}, \frac{\partial E_2}{\partial v} \right\rangle + \left\langle E_1, \frac{\partial^2 E_2}{\partial u \partial v} \right\rangle \right) - \left(\left\langle \frac{\partial E_1}{\partial v}, \frac{\partial E_2}{\partial u} \right\rangle + \left\langle E_1, \frac{\partial^2 E_2}{\partial v \partial u} \right\rangle \right) \\ &= \left\langle \frac{\partial E_1}{\partial u}, \frac{\partial E_2}{\partial v} \right\rangle - \left\langle \frac{\partial E_1}{\partial v}, \frac{\partial E_2}{\partial u} \right\rangle \\ &= (K \circ \phi) \left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right| \end{aligned}$$

by an application of claim 4.3.4, so we conclude that

$$\int_0^L \langle E_1, E_2' \rangle dt = \int_{\bar{U}} (K \circ \phi) \left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right| dudv,$$

as claimed. $\square$

Having established the above claims, we now integrated the identity of claim 4.3.3 and then apply claim 4.3.5 to get

$$\begin{aligned} \int_0^L k_g(\alpha(t)) dt &= \int_0^L (\theta'(t) - \langle (E_1 \circ \alpha)(t), (E_2 \circ \alpha)'(t) \rangle) dt \\ &= \int_0^L \theta'(t) dt - \int_{\bar{U}} (K \circ \phi) \left| \frac{\partial \phi}{\partial u} \times \frac{\partial \phi}{\partial v} \right| dudv, \end{aligned}$$

which is equivalent to

$$\int_{\partial S} k_g ds + \int_S K dA = \int_0^L \theta'(t) dt = \theta(L) - \theta(0).$$

We recall that $\theta(t)$ measures the angle that $\alpha'(t)$ makes with the tangent vector $E_1 = \frac{\partial \phi}{\partial u}$ and $\theta(L) - \theta(0)$ is a multiple of 2π since $\alpha'(L) = \alpha'(0)$, so the ratio

$$\frac{\theta(L) - \theta(0)}{2\pi} \in \mathbb{Z}$$

is equal to the total number of rotations that $(\phi^{-1} \circ \alpha)'(t)$ makes with respect to the fixed vector $(1, 0) \in \mathbb{R}^2$ in the u -direction. Now, $\phi^{-1} \circ \alpha$ is a parametrization of $\partial \bar{U}$, so in fact this number of rotations is the index of the plane curve $\partial \bar{U}$, so we will be done if we can prove that

$$\text{Ind}(\partial \bar{U}) = 1.$$

This last claim that $\text{Ind}(\partial \bar{U}) = 1$ requires a bit of algebraic topology, but we sketch the proof here. We take a diffeomorphism

$$f: \bar{U} \xrightarrow{\sim} \mathbb{D}^2 = \{z \in \mathbb{C} \mid |z| \leq 1\},$$

and gradually shrink the disk on the right to define regular homotopy

$$\gamma_s(t) = f^{-1} \left((1 - (1 - \epsilon)s)e^{it} \right)$$

for some very small $\epsilon > 0$. This means that we have a smooth family of closed, regular plane curve γ_s in $\bar{U} \subset \mathbb{R}^2$, where γ_0 is the original boundary $\partial \bar{U}$, and γ_1 is the inverse image under f of a tiny radius ϵ circle in $\mathbb{D}^2$. We recall from theorem 1.5.5 that the index of a closed plane curve doesn't change under this regular homotopy, so

$$\text{Ind}(\partial \bar{U}) = \text{Ind}(\gamma_0) = \text{Ind}(\gamma_1).$$

If ϵ is very small, then $d(f^{-1})$ is nearly constant on the disk of radius ϵ centered at $(0, 0)$, so γ_1 will nearly be an ellipse. Particularly, γ_1 will be convex, and we can 'inflate' it by another regular homotopy, similar to blowing up a balloon, to turn it into a circle. This establishes that $\text{Ind}(\gamma_1) = 1$, proving the theorem. $\square$

§4.4 Triangles and Triangulations of Surfaces

Having proved theorem 4.3.2, which applies to closed, smooth disks in a coordinate chart, we now generalize it slightly to triangles. The advantage of this is that we can triangulate a surface S , partitioning it into triangles, and add up the contributions from each to understand something global about S .

Definition 4.4.1. A **curvilinear triangle** in $\mathbb{R}^2$ is a continuous map $\beta: \mathbb{R} \rightarrow \mathbb{R}^2$ that is 3-periodic, meaning $\beta(t+3) = \beta(t)$ for all t , with the following property. For some t_0, t_1, t_2 with $0 \leq t_0 < t_1 < t_2 < t_0 + 3$, we have as follows:

1. β is smooth on each of the open intervals $(t_0, t_1), (t_1, t_2), (t_2, t_3)$, where $t_3 = t_0 + 3$.
2. The restriction of β to $[0, 3)$ is injective.
3. The one-sided derivatives $\beta'_-(t_i)$ and $\beta'_+(t_i)$ exist and meet at an angle θ_i that is not a multiple of π .

Now given a chart $\phi: U \rightarrow \mathbb{R}^3$ for S , and a curvilinear triangle $T \subset U$, we consider the image $\phi(T) \subset S$. At each vertex of $\phi(T)$, we can parametrize the incoming and outgoing edges by

$$\begin{aligned}\alpha_{\text{in}}: (-\epsilon, 0] &\rightarrow S \\ \alpha_{\text{out}}: [0, \epsilon) &\rightarrow S,\end{aligned}$$

and then their tangent vectors meet at some angle $\theta \in (-\pi, \pi)$ satisfying

$$\cos(\theta) = \frac{\langle \alpha'_{\text{in}}(0), \alpha'_{\text{out}}(0) \rangle}{|\alpha'_{\text{in}}(0)| \cdot |\alpha'_{\text{out}}(0)|}.$$

Here we take $\theta > 0$ if

$$(\alpha'_{\text{in}}(0), \alpha'_{\text{out}}(0), N(\alpha(0)))$$

is a positive basis of $\mathbb{R}^3$, and $\theta < 0$ otherwise. Then θ and $\pi - \theta$ are called the exterior angle and the interior angle of $\phi(T)$ at this vertex, respectively.

Theorem 4.4.2 (Gauss-Bonnet Theorem for Triangles)

Let $D \subset U \subset \mathbb{R}^2$ be compact and homeomorphic to a disk, with boundary ∂D a curvilinear triangle, and let $\phi(D) \subset S$ be its image under a chart $\phi: U \rightarrow \mathbb{R}^3$ for S . If $\phi(\partial D)$ is oriented positively with respect to $\phi(D)$, and if it has exterior angles $\theta_1, \theta_2, \theta_3$, and edges $\alpha_1, \alpha_2, \alpha_3$ parametrized by arclength, then

$$\left(\sum_{i=1}^3 \int_{\alpha_i} k_g ds \right) + \left(\sum_{i=1}^3 \theta_i \right) + \int_{\phi(D)} K dA = 2\pi.$$

Proof. This is essentially the same as the proof of theorem 4.3.2. We choose the same orthonormal basis $E_1 = \frac{\phi_u}{|\phi_u|}$ and $E_2 = N \times E_1$, and use the parametrization of $\phi(\partial D)$ by arclength to define the angle function $\theta(t)$ as before. Then the identities

$$k_g(\alpha(t)) = \theta'(t) - \langle (E_1 \circ \alpha)(t), (E_2 \circ \alpha)'(t) \rangle$$

and

$$\int_0^L \langle (E_1 \circ \alpha)(t), (E_2 \circ \alpha)'(t) \rangle dt = \int_U (K \circ \phi) \left| \frac{\partial \phi}{\partial u} \frac{\partial \phi}{\partial v} \right| du dv$$

are still true, and Green's theorem still applies to $\phi(D)$, so we integrate k_g to get

$$\left(\sum_{i=1}^3 \int_{\alpha_i} k_g ds \right) + \int_{\phi(D)} K dA = \int_{\phi(\partial D)} \theta'(t) dt.$$

But now $\theta(t)$ is only piecewise continuous: it increases by 2π as we travel around the boundary $\phi(\partial D)$, but it is discontinuous at each vertex, where it jumps by the corresponding exterior angle θ_i . Thus we now have

$$\int_{\phi(\partial D)} \theta'(t) dt = 2\pi - \sum_{i=1}^3 \theta_i$$

and substituting this into the above expression completes the proof. $\square$

Definition 4.4.3. From now on we will call a compact region $T \subset S$ a **triangle** if it is homeomorphic to a disk, and its boundary ∂T is piecewise smooth with exactly three vertices. In particular, the image $\phi(D)$ in theorem 4.4.2 is a triangle.

Remark 4.4.4 — Let T be a triangle in a surface S whose sides are all geodesics, and whose interior angles are $\alpha_i = \pi - \theta_i$. If the edges of T are geodesics, then the integral $\int_{\partial T} k_g ds$ in theorem 4.4.2 is zero, so the theorem says that $\sum(\pi - \alpha_i) + \int_T K dA = 2\pi$, or

$$\left(\sum_{i=1}^3 \alpha_i \right) - \pi = \int_T K dA.$$

This leads to several interesting special cases:

- If S is a plane then $K = 0$, and we have $\sum \alpha_i = \pi$.
- If S is the unit sphere then $K = 1$, and so $\sum \alpha_i - \pi = \text{area}(T)$.
- If S is negatively curved then $\int_T K dA < 0$ and $\sum \alpha_i < \pi$.

All of this is still true if T is not in the image of a single chart, though at the moment we haven't quite proved it in general.

Definition 4.4.5. Let S be a compact regular surface, possibly with boundary. A **triangulation** of S is a partition of S into curvilinear triangles, say

$$S = \bigcup_{i=1}^n T_i,$$

where:

- if $i \neq j$, then $T_i \cap T_j$ is either empty, a vertex of each triangle, an edge of each triangle.
- each edge belongs to exactly two triangles if it lies in the interior of S , or to one triangle if it lies on ∂S .

Claim 4.4.6. Any compact surface with boundary admits a triangulation.

Remark 4.4.7 — Claim 4.4.6 is beyond the scope of this module, so we will just assume it to be true in what follows, and use it to define the following numerical invariant of a surface.

Definition 4.4.8. Let S be a compact surface, possibly with boundary. If S admits a triangulation with V vertices, E edges, and F faces, then we define the **Euler characteristic** of S to be

$$\chi(S) = V - E + F.$$

Claim 4.4.9. The Euler characteristic of S is an invariant of S : it does not depend on the choice of triangulation.

Eventually this claim will follow from the Gauss-Bonnet theorem, but for now we will secretly assume whenever we write ' $\chi(S)$ ' that we have a fixed triangulation in mind.

Example 4.4.10

The sphere $\mathbb{S}^2$ has a triangulation shown (in the notes), constructed by inflating a tetrahedron until it is round. This has $V = 4$ vertices, $E = 6$ edges, and $F = 4$ faces, so $\chi = 4 - 6 + 4 = 2$.

Example 4.4.11

We can triangulate a torus T by first triangulating a square, and then gluing opposite sides together in pairs to form T : The triangulation shown in the notes has $V = 9$ vertices, $E = 27$ edges, and $F = 18$ faces; note that the vertices and edges along the left side are the same as those on the right side, so we must be careful not to double-count them, and likewise for the top and bottom, and so

$$\chi(T) = V - E + F = 9 - 27 + 18 = 0.$$

Example 4.4.12

Let Σ_g be a regular surface of genus g , built by gluing g tori together: for example, Σ_0 is a sphere and Σ_1 is a torus. We can inductively build a triangulation of Σ_g by taking triangulations of Σ_{g-1} and Σ_1 , removing a triangular face from each, and gluing the results along their boundaries. If the triangulation of Σ_{g-1} had V_{g-1} vertices, E_{g-1} edges, and F_{g-1} faces, and the triangulation of Σ_1 had V_1 vertices, E_1 edges, and F_1 faces, then we have built a triangulation of Σ_g whose vertex, edge, and face counts are

$$\begin{aligned} V &= V_{g-1} + V_1 - 3 \\ E &= E_{g-1} + E_1 - 3 \\ F &= (F_{g-1} - 1) + (F_1 - 1) \end{aligned}$$

so that

$$\begin{aligned} \chi(\Sigma_g) &= V - E + F = (V_{g-1} + V_1 - 3) - (E_{g-1} + E_1 - 3) + ((F_{g-1} - 1) + (F_1 - 1)) \\ &= (V_{g-1} - E_{g-1} + F_{g-1}) + (V_1 - E_1 + F_1) - 2 \\ &= \chi(\Sigma_{g-1}) + \chi(\Sigma_1) - 2 \end{aligned}$$

Now since $\chi(\Sigma_1) = 0$, it follows by induction that $\chi(\Sigma_g) = 2 - 2g$ for all $g \geq 0$.

The classification of surfaces says that under mild hypotheses, the Euler characteristic determines the topology of S :

Theorem 4.4.13

Let $S \subset \mathbb{R}^3$ be a **closed oriented surface**, meaning that S is compact with empty boundary. Then $\chi(S) = 2 - 2g$ for some integer $g \neq 0$, and moreover S is diffeomorphic to the genus- g surface Σ_g .

Remark 4.4.14 — Note that this is not true if $\partial S \neq \emptyset$. We should get that $\chi(S \setminus \mathbb{D}^2) = \chi(S) - 1$. This is because we lose one vertex, three edges, and three faces.

Theorem 4.4.15

If S is oriented and compact with boundary, then S is homeomorphic to a genus g surface with n disks removed for some g, n , and $\chi(S) = 2 - 2g - n$.

§4.5 Gauss-Bonnet Theorem

We are finally ready to prove the main theorem of this module, which says that the total curvature of a surface is determined by purely topological information.

Theorem 4.5.1 (Gauss-Bonnet Theorem)

Let $S \subset \mathbb{R}^3$ be a compact oriented surface, whose boundary is positively oriented if it is nonempty. Then,

$$\int_{\partial S} k_g ds + \int_S K dA = 2\pi \cdot \chi(S).$$

Proof. We take a triangulation of S and repeatedly subdivide it, dividing each triangle in half by including another triangle inside the preexisting triangle, so that every triangle is contained in a chart. (note that this does not change the Euler characteristic) We will suppose that this triangulation has V vertices, E edges, and F faces, and that exactly V_∂ of the V vertices lie on ∂S . Writing $S = T_1 \cup \dots \cup T_n$ with $n = F$, and writing the exterior angles of T_i as θ_{ij} and the interior angles as $\alpha_{ij} = \pi - \theta_{ij}$ for $j = 1, 2, 3$, we apply Gauss-Bonnet for triangles (theorem 4.4.2) to say that

$$\int_{\partial T_i} k_g ds + \int_{T_i} K dA = 2\pi - \sum_{j=1}^3 \theta_{ij} = \left(\sum_{j=1}^3 \alpha_{ij} \right) - \pi$$

We sum over all $i = 1, \dots, n$, and observe that all the contributions from interior edges of the triangulation cancel in pairs: each interior edge belongs to exactly two triangles, and is oriented differently for each of them. This leaves only the contributions from boundary edges, so

$$\sum_{i=1}^n \int_{\partial T_i} k_g ds = \int_{\partial S} k_g ds.$$

Meanwhile, on the RHS, the sum of interior angles α_{ij} contains a total of 2π at each interior vertex and π at each boundary vertex, so ultimately we have

$$\begin{aligned} \int_{\partial S} k_g ds + \int_S K dA &= 2\pi(V - V_\partial) + \pi V_\partial - \pi F \\ &= 2\pi V - \pi V_\partial - \pi F. \end{aligned}$$

To simplify the right side, we count the total number of edges over all faces of the triangulation: each face contributes three edges, but most of these edges are counted twice since they belong to two different faces, so

$$\begin{aligned} 3F &= 2(\# \text{ interior edges}) + (\# \text{ boundary edges}) \\ &= 2E - (\# \text{ boundary edges}) \\ &= 2E - (\# \text{ boundary vertices}) \\ &= 2E - V_\partial. \end{aligned}$$

Here the numbers of boundary edges and of boundary vertices are equal, because vertices and edges of the triangulation necessarily alternate along each component of ∂S . Now we have

$$\begin{aligned} \int_{\partial S} k_g ds + \int_S K dA &= 2\pi V - \pi V_\partial - \pi F \\ &= 2\pi V - \pi(2E - 3F) - \pi F \\ &= 2\pi V - 2\pi E + 2\pi F \\ &= 2\pi \chi(S). \end{aligned}$$

and this concludes the proof. $\square$

We remark that we can now prove the invariance of Euler characteristic, which was claim 4.4.9: given two different triangulations of S , say T and T' , we apply the Gauss–Bonnet theorem to each to get

$$\chi_T(S) = \int_{\partial S} k_g ds + \int_S K dA = \chi_{T'}(S).$$

Therefore, the Euler characteristic of S is the same with respect to either triangulation. We conclude with some applications of the Gauss–Bonnet theorem. We will use the following result from algebraic topology without proof:

Theorem 4.5.2 (Jordan Curve theorem)

Let $C \subset \mathbb{R}^2$ be a simple closed curve. Then the complement $\mathbb{R}^2 \setminus C$ consists of two connected components; one of these is unbounded, and the other has compact closure that is homeomorphic to a disk.

Definition 4.5.3. We say that a closed curve $\alpha: [a, b] \rightarrow \mathbb{R}^n$ is **simple** if it is injective on $[a, b)$, with $\alpha(a) = \alpha(b)$; in other words, it has no points of self-intersection.

Proposition 4.5.4

Let $S \subset \mathbb{R}^3$ be a closed, orientable surface. If $K \geq 0$ on all of S , then S is diffeomorphic to a sphere.

Proof. Since ∂S is empty, theorem 4.5.1 simply says that

$$\chi(S) = \frac{1}{2\pi} \int_S K dA \geq 0,$$

with equality if and only if $K = 0$ everywhere. But S has an elliptic point by proposition 3.2.7, so we must have $\chi(S) > 0$. Thus, if we look at the genus, we have $2 - 2g > 0$ occurs only if $g = 0$. Thus, $S \cong \mathbb{S}^2$ by theorem 4.4.13. $\square$

Proposition 4.5.5

Let $S \subset \mathbb{R}^3$ be a closed, orientable surface. If $K > 0$ at all points of S , then any two simple closed geodesics on S must intersect.

Proof. By proposition 3.2.7 we know that S is diffeomorphic to a sphere. If γ_1, γ_2 are disjoint simple closed geodesics, then the Jordan curve theorem (theorem 4.5.2) says that each γ_i divides $\mathbb{S}^2$ into a pair of disks, so we can write

$$S \cong (\text{disk}) \cup_{\gamma_1} \Sigma \cup_{\gamma_2} (\text{disk}),$$

where $\Sigma \subset S$ is a surface with boundary $\gamma_1 \cup \gamma_2$ and Euler characteristic $\chi(\Sigma) = 0$. Now theorem 4.4.13 tells us that

$$0 = 2\pi \cdot \chi(\Sigma) = \int_{\partial \Sigma} k_g ds + \int_{\Sigma} K dA = \int_{\Sigma} K dA > 0,$$

and we have a contradiction. Note that $\partial \Sigma = \gamma_1 \cup \gamma_2$ is a geodesic, so its integral is 0. $\square$

Proposition 4.5.6

Let $S \subset \mathbb{R}^3$ be diffeomorphic to a disk, and suppose that $K \leq 0$ on S . Let γ_1 and γ_2 be two geodesics in S which are not contained in a single geodesic. Then γ_1 and γ_2 intersect in at most one point.

Proof. Supposing otherwise, we parametrize γ_1 and γ_2 so that

$$\gamma_1(0) = \gamma_2(0) = p$$

and let $t_1 > 0$ be the smallest positive number for which $\gamma_1(t_1)$ also lies on γ_2 , say

$$\gamma_1(t_1) = \gamma_2(t_2) = q$$

for some t_2 . Then $\gamma_1([0, t_1])$ and $\gamma_2([0, t_2])$ bound some region T that must be homeomorphic to a disk, say with exterior angles p and q (by the Jordan curve theorem). We can view T as a degenerate triangle in S - we pick any point other than p and q as the third vertex, with exterior angle $\theta_3 = 0$ - and we apply the Gauss-Bonnet theorem for triangles, theorem 4.4.2, to say that

$$\int_{\partial R} k_g ds + (\theta_1 + \theta_2 + \theta_3) + \int_R K dA = 2\pi.$$

Now we have $k_g = 0$ on all of ∂R , and $\theta_3 = 0$, so then

$$2\pi - \theta_1 - \theta_2 = \int_R K dA \leq 0$$

since K is never positive. In other words, we have $\theta_1 + \theta_2 \geq 2\pi$. But by definition each of the θ_i is at most π , and we cannot have $\theta_i = \pi$ since γ_1 and γ_2 are not contained in a single geodesic, so in fact

$$\theta_1 + \theta_2 < \pi + \pi = 2\pi.$$

This is a contradiction. □